BIRATIONAL BOUNDEDNESS OF VARIETIES WITH LARGE SESHADRI CONSTANTS

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Abstract. We prove that, in each dimension at least two, the normal projective varieties whose anti-canonical divisor has moving Seshadri constant at least the dimension minus one are birationally bounded. Thus we answer Question 8.6 in Zhuang's paper on Fano varieties with large Seshadri constants.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Seshadri constants, introduced by Demailly [Dem92], measure the local positivity of a nef divisor at a point. For a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on a projective variety X and a smooth point $x \in X$, we denote by $\epsilon(L, x)$ the Seshadri constant of L at x, and by $\epsilon(L)$ the supremum of $\epsilon(L, x)$ over the smooth points x of X (see Definition 2.7). When L is an ample Cartier divisor, $\epsilon(L, x)$ is the asymptotic order of the jets at x generated by the global sections of the multiples of L [Laz04, Theorem 5.1.17].

For a Fano variety X of dimension d, a large value of $\epsilon(-K_X)$ characterizes the projective space. If X is smooth and $\epsilon(-K_X) > d$, then $X \cong \mathbb{P}^d$ [BS09, Theorem 1.7]. A projective variety X is $\mathbb{Q}$-*Fano* if X has klt singularities and $-K_X$ is an ample $\mathbb{Q}$-Cartier divisor [LZ18, p. 26]. If X is a $\mathbb{Q}$-Fano variety and $\epsilon(-K_X) > d$, then also $X \cong \mathbb{P}^d$ [LZ18, Theorem 2], and the $\mathbb{Q}$-Fano varieties X of dimension d with $\epsilon(-K_X, x) = d$ at a smooth point x are classified in [LZ18, Theorem 3].

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Zhuang [Zhu18] studied the thresholds d and $d-1$ for normal projective varieties X of dimension d, with no assumption on the singularities of X and no positivity assumption on $-K_X$. Since $-K_X$ need not be $\mathbb{Q}$-Cartier or nef, Zhuang replaces $\epsilon(-K_X)$ by the moving Seshadri constant, which is defined through jets of global sections. Let D be a $\mathbb{Q}$-divisor on X, $x \in X$ a smooth point, and m a positive integer such that mD is a Weil divisor (for $D = -K_X$, any $m \geq 1$). We denote by $s(mD, x)$ the largest integer $s \geq 0$ such that $H^0(X, \mathcal{O}_X(mD))$ generates s-jets at x, and we set $s(mD, x) = -1$ if $H^0(X, \mathcal{O}_X(mD))$ does not generate $\mathcal{O}_X(mD)$ at x. The moving Seshadri constant of D at x is

$$\epsilon_{\mathrm{mov}}(D, x) = \limsup_{m \to +\infty} \frac{s(mD, x)}{m},$$

and $\epsilon_{\mathrm{mov}}(D)$ is the supremum of $\epsilon_{\mathrm{mov}}(D, x)$ over the smooth points x of X (see Definition 2.4). When D is $\mathbb{Q}$-Cartier, $\epsilon_{\mathrm{mov}}(D, x)$ coincides with the moving Seshadri constant defined by Nakamaye [Nak03], by [ELM+09, Proposition 6.6] (see [Zhu18, Section 1, after equation (1)]). For positive characteristic, see [Zhu22], and for moving Seshadri constants over arbitrary fields, see [Mur24, Section 3].

Let X be a normal projective variety of dimension d. Zhuang proved the following results.

(1) For any real number $t > 0$, there exists a number $M(d, t)$ depending only on d and t such that if $\epsilon_{\mathrm{mov}}(-K_X) > d - 1 + t$, then $\mathrm{vol}(-K_X) \leq M(d, t)$ [Zhu18, Theorem 3.6].
(2) If $\epsilon_{\mathrm{mov}}(-K_X) > d-1$, then X is birational to a terminal Fano variety Y with $\epsilon(-K_Y) \geq \epsilon_{\mathrm{mov}}(-K_X)$ [Zhu18, Theorem 3.7]. Terminal Fano varieties of dimension d form a bounded set (see Definition 2.14) by [Bir21, Theorem 1.1]. Thus the set of normal projective varieties X of dimension d with $\epsilon_{\mathrm{mov}}(-K_X) > d - 1$ is birationally bounded.
(3) Let $B \geq 0$ be a $\mathbb{Q}$-divisor on X such that (X, B) is a pair, $L = -(K_X + B)$ is nef, and $\epsilon(L) \geq d - 1$. Then X is birational to a canonical Fano variety Y with $\epsilon(-K_Y) \geq d - 1$, or to $E \times \mathbb{P}^{d-1}$ for an elliptic curve E [Zhu18, Theorem 1.6].
(4) If $\epsilon_{\mathrm{mov}}(-K_X) > d$, then $X \cong \mathbb{P}^d$ [Zhu18, Theorem 1.7]. If $\epsilon_{\mathrm{mov}}(-K_X) = d$, and either $\mathrm{vol}(-K_X) > d^d$, or $d = 2$, or there exists a $\mathbb{Q}$-divisor $B \geq 0$ such that $L = -(K_X + B)$ is nef and $\epsilon(L) \geq d$, then X is of Fano type, that is, there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ such that (X, Δ) is klt and $-(K_X + \Delta)$ is ample [Zhu18, Theorem 1.8].

The number t in (1) cannot be taken to be 0. Indeed, for $d \geq 2$, the weighted projective space $X = \mathbb{P}(1, a_1, \ldots, a_d)$ with $a_1 = 1$ and $a_2 = \cdots = a_d = a$ satisfies $\epsilon(-K_X) = d - 1 + 2/a > d - 1$ [Zhu18, Example 1.3], and $\epsilon_{\mathrm{mov}}(-K_X) = \epsilon(-K_X)$ since $-K_X$ is ample (Lemma 2.9(3)), while

$$\mathrm{vol}(-K_X) = \frac{(2 + (d-1)a)^d}{a^{d-1}} \to +\infty \quad \text{as } a \to +\infty.$$

The second alternative in (3) occurs: the cubic hypersurface $X = (x_0^3 + x_1^3 + x_2^3 = 0) \subset \mathbb{P}^{d+1}$ satisfies $\epsilon(-K_X) = d - 1$ and is birational to $E \times \mathbb{P}^{d-1}$ for an elliptic curve E, so X is not rationally connected [Zhu18, Section 1, paragraph after Theorem 1.5].

In (1), (2), and (4), the hypotheses are on $\epsilon_{\mathrm{mov}}(-K_X)$. Zhuang notes that, except for [Zhu18, Theorem 1.6], the theorems of [Zhu18] are proved in such a moving form [Zhu18, Section 1, paragraph before Theorem 1.7]. The moving form of (3) is the following question of Zhuang.

Question 1.1 ([Zhu18, Question 8.6])**.** Is the set of normal projective varieties X of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$ birationally bounded?

For $d = 2$, the threshold $d-1$ in Question 1.1 cannot be lowered [Zhu18, Example 8.7]. Let C be a smooth projective curve of genus g, D a divisor on C with $\deg D \gg 0$, $X = \mathbb{P}_C(\mathcal{O}_C \oplus \mathcal{O}_C(-D))$, $C_0 \subset X$ the negative section, and F a fiber of $X \to C$. By [Zhu18, Example 8.7], the Zariski decomposition [Laz04, Theorem 2.3.19] of $-K_X$ is

$$-K_X = \Big(1 + \frac{2g-2}{\deg D}\Big) C_0 + \Big(1 - \frac{2g-2}{\deg D}\Big)(C_0 + (\deg D) F),$$

and $\epsilon_{\mathrm{mov}}(-K_X) = 1 - \frac{2g-2}{\deg D}$. Varying g and $\deg D$, Zhuang observes that for any real number $c < 1$, the set of surfaces X with $\epsilon_{\mathrm{mov}}(-K_X) > c$ is not birationally bounded, and he notes that $\mathbb{P}^{d-1}$-bundles over C yield such examples in dimension d [Zhu18, Example 8.7]. When $g = 1$, we have $\epsilon_{\mathrm{mov}}(-K_X) = 1 = d - 1$, and since $(C_0^2) = -\deg D$,

$$\mathrm{vol}(-K_X) = \big((C_0 + (\deg D)F)^2\big) = \deg D$$

by [Laz04, Corollary 2.3.22]. Thus for $g = 1$ the curve $C = E$ can be fixed while $\deg D \to +\infty$, so $\mathrm{vol}(-K_X)$ is unbounded, and no bound on $\mathrm{vol}(-K_X)$ as in [Zhu18, Theorem 3.6] holds at the threshold. In particular, even over a fixed base curve E, the surfaces X with $\epsilon_{\mathrm{mov}}(-K_X) = 1$ do not form a bounded set (see Definition 2.14), since the members X of a bounded set of smooth projective surfaces have bounded $\mathrm{vol}(-K_X)$: in a bounded family one can choose a very ample A with (A^2) and $(-K_X \cdot A)$ bounded, and $\mathrm{vol}(-K_X) \le (-K_X \cdot A)^2/(A^2)$.

We now explain why the methods of [Zhu18] do not reach the case $\epsilon_{\mathrm{mov}}(-K_X) = d - 1$. In this case [Zhu18, Theorem 3.7] cannot be used, since it needs the strict inequality. Moreover, [Zhu18, Theorem 1.6] can be used only when a $\mathbb{Q}$-divisor $B \ge 0$ such that $-(K_X + B)$ is nef and $\epsilon(-(K_X + B)) \ge d - 1$ is given, and the hypothesis of Question 1.1 does not provide such a $\mathbb{Q}$-divisor B.

Zhuang proves [Zhu18, Theorem 3.7] as follows. First, X is replaced by a higher model $\varphi : X' \to X$ such that X' is $\mathbb{Q}$-factorial and terminal and $K_{X'}$ is φ-nef, and

$$\epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X)$$

by [Zhu18, Lemma 3.4]. Next, since $-K_{X'}$ is big by Lemma 2.6(4), a $K_{X'}$-MMP [Zhu18, Theorem 2.2] ends with a Mori fiber space $f : Y \to Z$, and

$$\epsilon_{\mathrm{mov}}(-K_Y) \ge \epsilon_{\mathrm{mov}}(-K_{X'}) > d - 1$$

by [Zhu18, Corollary 3.3]. If $\dim Z = 0$, then Y is a terminal Fano variety. If $\dim Z > 0$, let F be a general fiber of f. The strict inequality passes to F by [Zhu18, Lemma 3.1], so $F \cong \mathbb{P}^{\dim F}$ by [Zhu18, Theorem 1.7], and $\epsilon(-K_{\mathbb{P}^r}) = r + 1$ [Laz04, Examples 5.1.4 and 5.1.7] forces $\dim F = d - 1$. By [Zhu18, Lemma 5.3], applied with $\Gamma = -\delta K_Y$ for $0 < \delta \ll 1$, $F \cong \mathbb{P}^{d-1}$, and either $Z \cong \mathbb{P}^1$, or $\epsilon_{\mathrm{mov}}(-K_Y) = d - 1$ and Z is an elliptic curve. The strict inequality excludes the second case. Finally, the generic fiber of f becomes isomorphic to $\mathbb{P}^{d-1}$ over an algebraic closure of $\mathbb{C}(Z)$. Since the Brauer group of the function field of a curve is zero [Ser79, Chapter X, §7], the generic fiber of f is isomorphic to $\mathbb{P}^{d-1}_{\mathbb{C}(Z)}$. Thus X is rational, and in particular X is birational to the terminal Fano variety $\mathbb{P}^d$.

At the threshold $\epsilon_{\mathrm{mov}}(-K_X) = d - 1$, the argument breaks in three places. In the examples below, $-K_X$ is an ample Cartier divisor, so $\epsilon_{\mathrm{mov}}(-K_X) = \epsilon(-K_X)$ by [Laz04, Theorem 5.1.17]. We use that $\epsilon(M, x) \le (M \cdot \gamma)/\operatorname{mult}_x \gamma$ for any irreducible curve $\gamma \ni x$, and that

$$\epsilon(M + M', x) \ge \epsilon(M, x) + \epsilon(M', x)$$

for nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors M and M', both by the formula for $\epsilon(M, x)$ as an infimum over curves [Laz04, Proposition 5.1.5].

(a) If $\dim Z = 1$, the strict inequality $\epsilon_{\mathrm{mov}}(-K_F) > \dim F$ is no longer available, and F need not be a projective space. For example, let $d = 3$, let $h : X \to \mathbb{P}^3$ be the blow-up along a smooth complete intersection C of two quadrics, and let H be a plane in $\mathbb{P}^3$. The pencil of quadrics containing C induces a morphism $X \to \mathbb{P}^1$ whose general fiber is $\mathbb{P}^1 \times \mathbb{P}^1$, and $2h^*H - h^{-1}(C)$ is the pullback of a point of $\mathbb{P}^1$. Thus X embeds in $\mathbb{P}^3 \times \mathbb{P}^1$, and

$$-K_X = 2h^*H + (2h^*H - h^{-1}(C))$$

is the restriction of $\mathcal{O}(2, 1)$. Moreover, $X \to \mathbb{P}^1$ is a Mori fiber space since $\rho(X) = 2$. Any point $x \in X$ is contained in a line l in a fiber of $X \to \mathbb{P}^1$, and $(-K_X \cdot l) = 2(h^*H \cdot l) = 2$, so $\epsilon(-K_X, x) \le 2$. If $x \notin h^{-1}(C)$, then $\epsilon(h^*H, x) \ge 1$, because for any irreducible curve

$\gamma \ni x$ a plane through $h(x)$ not containing $h(\gamma)$ shows $(h^*H \cdot \gamma) \geq \mathrm{mult}_x \gamma$ (as in [Laz04, Example 5.1.18]). Hence $\epsilon(-K_X, x) \geq 2\epsilon(h^*H, x) \geq 2$, and $\epsilon(-K_X) = 2 = d - 1$.

(b) The curve Z can be an elliptic curve: the conclusion of [Zhu18, Lemma 5.3] allows this case when $\epsilon_{\mathrm{mov}}(-K_Y) = d - 1$, and by the cubic hypersurface above, X need not be rationally connected at the threshold. The blow-up X' of the cubic hypersurface X above along the vertex $(x_0 = x_1 = x_2 = 0)$ is a $\mathbb{P}^{d-1}$-bundle over the plane cubic curve $E = (x_0^3 + x_1^3 + x_2^3 = 0) \subset \mathbb{P}^2$ (as in [Har77, Example V.2.11.4], where the vertex is a point), hence a Mori fiber space over E, and

$$\epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X) = d - 1$$

by [Zhu18, Lemma 3.4], since X' is smooth and a direct calculation shows that $K_{X'}$ is ample over X, so $X' \to X$ is a terminal modification in the sense of [Zhu18, Definition 2.3].

(c) The base Z can be a surface. Since $\epsilon(-K_{\mathbb{P}^{d-2}}) = d - 1$, the strict inequality excludes $\dim F = d - 2$, but the equality does not. For example, let $d = 3$, $X = \mathbb{P}^2 \times \mathbb{P}^1$, and $f : X \to \mathbb{P}^2$ the first projection. Then $-K_X = \mathcal{O}(3, 2)$, and a fiber l of f satisfies $(-K_X \cdot l) = 2$. Moreover, $-K_X = 2\mathcal{O}(1, 1) + \mathcal{O}(1, 0)$, where $\mathcal{O}(1, 1)$ is very ample and $\mathcal{O}(1, 0)$ is nef, so $\epsilon(-K_X, x) \geq 2$ at any point x by [Laz04, Example 5.1.6]. Hence $\epsilon(-K_X) = 2 = d - 1$.

The last step of the argument for [Zhu18, Theorem 3.7] uses that the Brauer group of the function field of a curve is zero [Ser79, Chapter X, §7]. No analogue holds over a surface: the conic $xu^2 + yv^2 = w^2$ over $\mathbb{C}(x, y)$ has no rational point, as one sees by comparing the parities of the orders of vanishing of xu^2, yv^2, and w^2 along $x = 0$ and using that y is not a square in $\mathbb{C}(y)$. Thus the rationality argument of Zhuang does not extend to Mori fiber spaces over surfaces, and we prove birational boundedness directly.

In this paper, we answer Question 1.1 affirmatively for $d \geq 2$.

Theorem 1.2. *Let $d \geq 2$ be an integer. Then the set of normal projective varieties X of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$ (see Definition 2.4) is birationally bounded (see Definition 2.14).*

Remark 1.3. In Theorem 1.2, we assume that $d \geq 2$. If $d = 1$ and X is a smooth projective curve of genus $g \geq 2$, then

$$\deg(-mK_X) = m(2 - 2g) < 0$$

for any integer $m \geq 1$, hence $H^0(X, \mathcal{O}_X(-mK_X)) = 0$ and $s(-mK_X, x) = -1$ for any $x \in X$. Thus

$$\epsilon_{\mathrm{mov}}(-K_X) = \limsup_{m \to +\infty} \frac{-1}{m} = 0 = d - 1.$$

In particular, for $d = 1$, any smooth projective curve of genus at least 2 satisfies the condition $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$ of Theorem 1.2.

Sketch of the proof. By Construction-Theorem 3.1, we may assume that X is terminal, $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$, and there exists a Mori fiber space $f : X \to Z$. Its proof restricts jets to the geometric generic fiber $X_{\bar{\eta}}$ of f: if $\dim Z > 0$, then

$$d - 1 \leq \epsilon_{\mathrm{mov}}(-K_{X_{\bar{\eta}}}) \leq d - \dim Z + 1,$$

and

$$\epsilon_{\mathrm{mov}}(-K_{X_{\bar{\eta}}}) > \dim X_{\bar{\eta}}$$

only for projective space [Zhu18, Theorem 1.7], so $\dim Z \leq 2$, and $X_{\bar{\eta}} \cong \mathbb{P}^{d-2}$ if $\dim Z = 2$. We distinguish three cases.

(I) $\epsilon_{\mathrm{mov}}(-K_X) > d - 1$ or $\dim Z = 0$. Then X is birational to a terminal Fano variety, and Theorem 2.16 concludes.

(II) $\dim Z = 1$ (Theorem 5.1). Then Z has genus at most one (Proposition 5.2). If a general fiber F of f satisfies $((-K_F)^{d-1}) = (d-1)^{d-1}$, then $-K_X$ is nef and big by [Zhu18, Lemma 5.4] (Proposition 5.2(1)). Otherwise X is birational to $Z \times \mathbb{P}^{d-1}$ by Lemma 4.1 and Proposition 4.2.

(III) $\dim Z = 2$ (Theorem 12.1). Then the geometric generic fiber of f is $\mathbb{P}^{d-2}$, Z is δ_d-lc [Bir23, Corollary 1.2], and a K_Z-MMP ends with a Mori fiber space onto a curve (Theorem 11.1) or with a surface Z'' with ample $-K_{Z''}$ of bounded volume [Bir21, Theorem 2.11]. In the second case, Proposition 12.2 bounds $\mathrm{vol}(-K_X)$, and Lemma 7.6 concludes.

The key new step is Proposition 12.2, which bounds $\mathrm{vol}(-K_X)$ whether or not f has a rational section. For a very ample H on Z, restricting $-K_X$ to $T = f^{-1}(C)$ for a very general $C \in |H|$ yields

$$\mathrm{vol}(-K_X) \leq 8d(d-1)^d \frac{(-K_Z \cdot H)^2}{(H^2)}.$$

Only (I) and the case $((-K_F)^{d-1}) = (d-1)^{d-1}$ of (II) follow from [Zhu18, Theorem 3.7 and Lemma 5.4] and Theorem 2.16. The other cases need four new ideas.

(a) Fibers that are not projective spaces. Over a curve, Zhuang's arguments need the general fiber to be $\mathbb{P}^{d-1}$, to construct, through lines in F, klt boundaries that bound the genus of Z [Zhu18, proof of Lemma 5.1], and to prove rationality via $\mathrm{Br}(\mathbb{C}(Z)) = 0$ [Zhu18, proof of Theorem 3.7]. At equality, F need not be a projective space. If $((-K_F)^{d-1}) > (d-1)^{d-1}$, then we take a very general point $p \in F$, a $\mathbb{Q}$-divisor $G \sim_{\mathbb{Q}} -K_F$ with $\mathrm{mult}_p G > d-1$, and $A := \frac{1}{m} M|_F$ for a member M of $|-mK_X|$ given by the generation of jets at p. For small $\delta > 0$, a multiple of $(1-\delta)A + \delta G$ makes p an isolated non-klt center of a pair (F, Δ) with $-(K_F + \Delta)$ ample. A small log canonical threshold of the restricted base ideals would force another non-klt center, contradicting the connectedness of non-klt loci [Zhu18, Lemma 2.12] (Proposition 5.2).

For rationality, Lemma 4.1 finds by weak approximation [HT06, Theorem 2] a $\mathbb{C}(Z)$-rational point x of the generic fiber at which $|-mK_X|$ generates as many jets as at a very general point, and Proposition 4.2 descends the classification of [LZ18, Lemma 13] by blowing up x, whose exceptional divisor is isomorphic to $\mathbb{P}^{d-2}_{\mathbb{C}(Z)}$.

(b) Volume over surface bases. The length argument for the nefness of $-K_X$ in [Zhu18, proof of Lemma 5.4] needs the non-lc loci of the pairs (X, D_q), where $D_q \sim_{\mathbb{Q}} -qK_X$ and $q \gg 0$, to be contained in fibers of f. Over a surface, they may contain curves dominating curves of Z, and $-K_X$ is only f-ample. On T, the non-lc loci are contained in finitely many fibers of $T \to C$, but the length argument fails for $-K_X|_T$, since adjunction adds the pullback of the ample divisor $H|_C$ (Remark 7.1).

Instead, Lemma 7.2 passes to a double cover of C of genus one if $C \cong \mathbb{P}^1$, after which any rational curve is vertical, and uses the cone theorem for non-lc pairs [Fuj11, Theorem 1.1] with the D_q of Theorem 2.19, which exist for any $q \geq 2$ because $\epsilon_{\mathrm{mov}}(-K_X) = d-1$. Then $\mathrm{vol}(-K_X|_T) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H)$ (Theorem 10.2), where Λ (Definition 10.1) is a positive multiple of the CM class of f, and $(\Lambda \cdot H) \geq 0$ by [CP21, Theorem 7.1] and the K-semistability of $\mathbb{P}^{d-2}$ (Theorem 10.3). Proposition 12.2 then follows from Lemma 7.4, which bounds $\mathrm{vol}(-K_X)$ by $d\,\mathrm{vol}(-K_X|_T)$ times the supremum of the rational numbers $s \geq 0$ such that $-K_X - sT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor ≥ 0, and from Lemma 7.3, which bounds any such s using the canonical bundle formula.

(c) Ruled bases. If the K_Z-MMP ends with a Mori fiber space onto a curve B (the curve C of Theorem 11.1), it yields no polarization H with $(-K_Z \cdot H)^2/(H^2)$ bounded in terms of d. We regard the composition $h : X \to B$ as a fibration whose very general fibers X_b are terminal weak Fano varieties. If $((-K_{X_b})^{d-1}) > (d-1)^{d-1}$, then, as in (II), Lemma 4.1 and Proposition 4.2 applied to h show that X is birational over B to $B \times \mathbb{P}^{d-1}$, and B has genus at most one by (d).

If $((-K_{X_b})^{d-1}) = (d-1)^{d-1}$, the argument for the nefness of $-K_X$ in [Zhu18, proof of Lemma 5.4] uses $\rho(X) = 2$, while $\rho(X) \geq 3$ here (Remark 11.2). Instead, the generation of jets

forces the restrictions of $|-mK_X|$ to X_b to have the volume of $-K_{X_b}$, so the graded sequence of their base ideals has zero asymptotic order along any divisor over X_b [FKL16, Theorem A] and infinite log canonical threshold [Blu21, Proposition 2.6(1)]. This yields, for any $q > 0$, a $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -qK_X$ and (X, D) klt over an open subset of B. With a lower bound $\alpha > 0$, depending only on d, of log canonical thresholds of anti-canonical $\mathbb{Q}$-divisors on X_b [Bir21, Theorem 1.6], Lemmas 7.3 and 7.4 imply $\mathrm{vol}(-K_X) \leq 8d(d-1)^{d-1}/\alpha$ (Theorem 11.1).

(d) The base surface. By Theorem 9.2, $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$. Thus $-K_Z$ is big, so the K_Z-MMP in (III) ends with a Mori fiber space, and the curve B in (c) has genus at most one by Proposition 6.1.

Even when f has a section σ, restricting jets to σ as in [Zhu18, Lemma 3.1] only controls $-K_X|_\sigma = -K_Z + (-K_{X/Z})|_\sigma$, and $(-K_{X/Z})|_\sigma$ has no sign. Instead, Construction-Theorem 8.1 takes determinants of vertical derivatives of several sections of $-mK_X$. Since the jets of $|-mK_X|$ have order close to $m(d-1)$, the maximum on the fibers $\mathbb{P}^{d-2}$, the canonical class of the vertical foliation cancels $K_{X/Z}$ in the limit, and the pullback of $-K_Z$ to any smooth projective surface with a morphism to X that is an embedding near a point t over a very general point of X and is étale over Z at t has moving Seshadri constant at least one at t (Lemma 9.5). For any curve through a very general point of Z, Theorem 2.22 yields a section of f over an affine open subset of the curve, and we choose the surface so that the image of the surface in X contains this section, so that no degree is lost.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce moving Seshadri constants of $\mathbb{Q}$-divisors on normal projective varieties, recall Brauer–Severi schemes, prove a consequence of the canonical bundle formula, and recall results of Zhuang, Birkar, and Belkale–Fakhruddin that we use repeatedly.

In Section 3 we reduce Theorem 1.2 to terminal Mori fiber spaces (Construction-Theorem 3.1). In Section 4 we study rational points on the generic fiber of a contraction onto a smooth projective curve, and the rationality of the generic fiber over the function field of the curve, which are used in Sections 5 and 11. In Section 5 we treat Mori fiber spaces over curves (Theorem 5.1).

Sections 6–12 treat the case when the base of the Mori fiber space is a surface, and in Section 6 we study normal projective surfaces S such that $\epsilon_{\mathrm{mov}}(-K_S) \geq 1$. In Section 7 we prove four lemmas on nefness, volume, and birationality, which are used in Sections 10–12. In Section 8 we construct relative Wronskians.

In Section 9 we show that the base surface Z of a Mori fiber space in the surface case of Construction-Theorem 3.1 satisfies $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$ (Theorem 9.2). In Section 10 we study curve slices of Mori fiber spaces over surfaces and the CM divisor. In Section 11 we treat Mori fiber spaces over surfaces with a ruling. In Section 12 we treat Mori fiber spaces over arbitrary surfaces (Theorem 12.1). In Section 13 we prove Theorem 1.2.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let δ be a positive real number. A pair (X, B) is *δ-lc* if $a(E, X, B) \geq \delta$ for any prime divisor E over X.

Definition 2.3. Let (X, B) be a pair. The *non-klt locus* $\mathrm{Nklt}(X, B)$ (resp. the *non-lc locus* $\mathrm{Nlc}(X, B)$) is the union of the centers on X of the prime divisors E over X such that $a(E, X, B) \leq 0$ (resp. $a(E, X, B) < 0$). Assume that (X, B) is klt. For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on X, the *log canonical threshold* of D with respect to (X, B) is

$$\mathrm{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\}.$$

For a non-zero coherent ideal $\mathfrak{a} \subset \mathcal{O}_X$, we define

$$\mathrm{lct}(X, B; \mathfrak{a}) := \inf_E \frac{a(E, X, B)}{\mathrm{mult}_E \mathfrak{a}},$$

where E runs over the prime divisors over X such that $\mathrm{mult}_E \mathfrak{a} > 0$. For a point $x \in X$, we define $\mathrm{lct}(X \ni x, B; D) := \mathrm{lct}(U, B|_U; D|_U)$ and

$$\mathrm{lct}(X \ni x, B; \mathfrak{a}) := \mathrm{lct}(U, B|_U; \mathfrak{a}\mathcal{O}_U),$$

where U is a sufficiently small open neighborhood of x. If $B = 0$, then we omit B and write $\mathrm{lct}(X; D)$, $\mathrm{lct}(X; \mathfrak{a})$, $\mathrm{lct}(X \ni x; D)$, and $\mathrm{lct}(X \ni x; \mathfrak{a})$.

2.2. **Moving Seshadri constants.**

Definition 2.4. Let X be a normal projective variety, $x \in X$ a smooth point, D a Weil divisor on X, and $V \subset H^0(X, \mathcal{O}_X(D))$ a linear subspace. We denote by $s(V, x)$ the largest integer $s \geq 0$ such that the natural map

$$V \to \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1}$$

is surjective, and we set $s(V, x) := -1$ if

$$V \to \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x$$

is not surjective. We say that V *generates s-jets* at x if $s(V, x) \geq s$, and we write $s(D, x) := s(H^0(X, \mathcal{O}_X(D)), x)$. For a $\mathbb{Q}$-divisor D on X, the *moving Seshadri constant* of D at x is

$$\epsilon_{\mathrm{mov}}(D, x) := \limsup_{m \to +\infty} \frac{s(mD, x)}{m},$$

where m runs over the positive integers such that mD is a Weil divisor, and the *moving Seshadri constant* of D is

$$\epsilon_{\mathrm{mov}}(D) := \sup_x \epsilon_{\mathrm{mov}}(D, x),$$

where x runs over the smooth points of X.

Remark 2.5. Definition 2.4 agrees with [Zhu18, equation (1)], where ϵ_{mov} is denoted by ϵ_m. We write ϵ_{mov} instead, since the letter m denotes a degree in Definition 2.4. For $D = -K_X$, the definition uses the reflexive sheaves $\mathcal{O}_X(-mK_X)$ for all integers $m \geq 1$, also for the integers m such that $-mK_X$ is not Cartier. Indeed, X is only assumed to be normal, so K_X need not be $\mathbb{Q}$-Cartier, and there may be no integer $m \geq 1$ such that $-mK_X$ is Cartier. Lemma 2.12 below moves $\epsilon_{\mathrm{mov}}(-K_X)$ to a terminal modification X' of X. Since $K_{X'}$ is $\mathbb{Q}$-Cartier, Lemma 2.6(2) then allows us to use only the degrees m such that $-mK_{X'}$ is Cartier.

Lemma 2.6. *Let X be a normal projective variety of dimension d, $x \in X$ a smooth point, and D a $\mathbb{Q}$-divisor on X.*

(1) *Let $D_1, \dots, D_q$ be Weil divisors on X and $V_i \subset H^0(X, \mathcal{O}_X(D_i))$ linear subspaces such that $s(V_i, x) \geq 0$ for each i. Then the image V of $V_1 \otimes \cdots \otimes V_q$ in $H^0(X, \mathcal{O}_X(D_1 + \cdots + D_q))$ satisfies $s(V, x) \geq s(V_1, x) + \cdots + s(V_q, x)$.*

(2) *For any positive integer I, we have*

$$\epsilon_{\mathrm{mov}}(D, x) = \limsup_{m \to +\infty,\ I \mid m} \frac{s(mD, x)}{m},$$

where m runs over the multiples of I such that mD is a Weil divisor.

(3) *For any positive integer m such that mD is a Weil divisor, any linear subspace $V \subset H^0(X, \mathcal{O}_X(mD))$, and any integer $s \geq 0$, the set of smooth points $y \in X$ such that $s(V, y) \geq s$ is open in X. In particular,*

$$\epsilon_{\mathrm{mov}}(D, y) = \epsilon_{\mathrm{mov}}(D)$$

for a very general point $y \in X$.

(4) *If D is $\mathbb{Q}$-Cartier and $\epsilon_{\mathrm{mov}}(D) > 0$, then D is big.*

(5) *Assume that D is a Cartier divisor, and let $W \subset X$ be a normal closed subvariety of positive dimension such that x is a smooth point of W. Then for any linear subspace $V \subset H^0(X, \mathcal{O}_X(D))$, the image $V|_W$ of V in $H^0(W, \mathcal{O}_X(D)|_W)$ satisfies $s(V|_W, x) \geq s(V, x)$.*

Proof. (1) Let $s_i := s(V_i, x)$ and $s := s_1 + \cdots + s_q$. Fix local parameters $z_1, \dots, z_d$ of X at x and a local generator of $\mathcal{O}_X(D_i)$ at x for each i, and use their product as a local generator of $\mathcal{O}_X(D_1 + \cdots + D_q)$ at x. Via the local generators of $\mathcal{O}_X(D_1), \dots, \mathcal{O}_X(D_q)$ and of $\mathcal{O}_X(D_1 + \cdots + D_q)$ fixed above, the maps in Definition 2.4 become maps to quotients of $\mathcal{O}_{X,x}$. Any monomial z^α with $|\alpha| \leq s$ can be written as $z^\alpha = z^{\alpha_1} \cdots z^{\alpha_q}$ with $|\alpha_i| \leq s_i$ for each i, where $\alpha_i = 0$ is allowed. Since V_i generates s_i-jets at x, there exists $v_i \in V_i$ such that $v_i - z^{\alpha_i} \in \mathfrak{m}_x^{s_i+1}$. Since

$$|\alpha| - |\alpha_i| + s_i + 1 \geq |\alpha| + 1$$

for each i, we have

$$v_1 \cdots v_q - z^\alpha \in \mathfrak{m}_x^{|\alpha|+1}.$$

Thus, by descending induction on j, the image of V in $\mathcal{O}_{X,x}/\mathfrak{m}_x^{s+1}$ contains $\mathfrak{m}_x^j/\mathfrak{m}_x^{s+1}$ for $j = s, s-1, \dots, 0$, because the monomials of degree j and $\mathfrak{m}_x^{j+1}$ generate $\mathfrak{m}_x^j$. For $j = 0$, we obtain $s(V, x) \geq s$.

(2) Let ϵ_I be the right-hand side. For any m such that mD is a Weil divisor and $s(mD, x) \geq 0$, (1) with $q = I$ and $V_1 = \cdots = V_I = H^0(X, \mathcal{O}_X(mD))$ implies

$$s(ImD, x) \geq I \cdot s(mD, x).$$

Since $s(mD, x) \geq -1$ for any m, and since there exist infinitely many integers m such that $I \mid m$ and mD is a Weil divisor, we have $\epsilon_I \geq 0$ and $\epsilon_{\mathrm{mov}}(D, x) \geq 0$. Moreover, $\epsilon_I \leq \epsilon_{\mathrm{mov}}(D, x)$, since ϵ_I is a limsup over a subsequence. If $\epsilon_{\mathrm{mov}}(D, x) = 0$, then $\epsilon_I = 0$. If $\epsilon_{\mathrm{mov}}(D, x) > 0$, then there exist integers $m_j \to +\infty$ such that $s(m_j D, x)/m_j \to \epsilon_{\mathrm{mov}}(D, x)$, and $s(m_j D, x) \geq 0$ for j large. Thus $s(Im_j D, x)/(Im_j) \geq s(m_j D, x)/m_j$ for j large, and $\epsilon_I \geq \epsilon_{\mathrm{mov}}(D, x)$.

(3) Let $v_1, \dots, v_N$ be a basis of V, and let $y \in X$ be a smooth point. Fix an open neighborhood $U \subset X_{\mathrm{reg}}$ of y, regular functions $z_1, \dots, z_d$ on U such that $dz_1, \dots, dz_d$ is a basis of Ω_U, and a generator of $\mathcal{O}_X(mD)|_U$. Via the generator of $\mathcal{O}_X(mD)|_U$ fixed above, each v_j is a regular function f_j on U. For any $y' \in U$, the functions $z_i - z_i(y')$ are local parameters at y', and by the Taylor expansion, V generates s-jets at y' if and only if the matrix

$$\left(\partial^\alpha f_j(y')\right)_{1 \leq j \leq N,\ |\alpha| \leq s}$$

has rank $\binom{d+s}{d}$. The rank of a matrix of regular functions on U is at least $\binom{d+s}{d}$ exactly where one of finitely many minors does not vanish, which is an open condition on U. Since X_{reg} is open in X, the first sentence of (3) follows.

For any m such that mD is a Weil divisor, if $H^0(X, \mathcal{O}_X(mD))$ generates s-jets at a smooth point, then $h^0(X, \mathcal{O}_X(mD)) \geq \binom{d+s}{d}$, so $s(mD, y)$ is bounded above for $y \in X_{\mathrm{reg}}$. Let s_m be

the maximum of $s(mD, y)$ over the smooth points y. By the first sentence of (3), the set U_m of smooth points y such that $s(mD, y) = s_m$ is a non-empty open subset of X, where $U_m = X_{\mathrm{reg}}$ if $s_m = -1$. For any y in the intersection of the countably many U_m, we have $s(mD, y) \geq s(mD, x')$ for any smooth point x' and any m, hence

$$\epsilon_{\mathrm{mov}}(D, y) \geq \epsilon_{\mathrm{mov}}(D, x').$$

Thus

$$\epsilon_{\mathrm{mov}}(D, y) = \epsilon_{\mathrm{mov}}(D).$$

(4) Let x' be a smooth point of X such that $\epsilon_{\mathrm{mov}}(D, x') > 0$, let $0 < c < \epsilon_{\mathrm{mov}}(D, x')$, and let I be a positive integer such that ID is Cartier. By (2), there exist infinitely many integers m such that $I \mid m$ and $s(mD, x') \geq cm$. For any such m, since $H^0(X, \mathcal{O}_X(mD))$ surjects onto

$$\mathcal{O}_X(mD) \otimes \mathcal{O}_X/\mathfrak{m}_{x'}^{\lceil cm \rceil + 1},$$

we have

$$h^0(X, \mathcal{O}_X(mD)) \geq \binom{d + \lceil cm \rceil}{d} \geq \frac{c^d m^d}{d!}.$$

Thus $\limsup_k h^0(X, \mathcal{O}_X(kID))/k^d > 0$, so the Cartier divisor ID is big, and D is big.

(5) We may assume that $s := s(V, x) \geq 0$. Since $\mathcal{O}_X(D)$ is invertible, the restriction map

$$\mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1} \to \mathcal{O}_X(D)|_W \otimes \mathcal{O}_W/\mathfrak{m}_{W,x}^{s+1}$$

is surjective, and it commutes with the maps from V and $V|_W$ in Definition 2.4. Thus $V|_W$ generates s-jets at x (cf. [Zhu18, Lemma 3.1]). □

Definition 2.7 ([Laz04, Definition 5.1.1], [Zhu18, Definition 1.1])**.** Let X be a projective variety, L a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and $x \in X$ a smooth point. Let $h : X' \to X$ be the blow-up of X at x and E the exceptional divisor of h. The *Seshadri constant* of L at x is

$$\epsilon(L, x) := \sup\{t \geq 0 \mid h^*L - tE \text{ is nef}\},$$

and the *Seshadri constant* of L is $\epsilon(L) := \sup_x \epsilon(L, x)$, where x runs over the smooth points of X.

Remark 2.8. Following [Zhu18, Definition 1.1], $\epsilon(L)$ is the supremum, not the infimum, of $\epsilon(L, x)$ over the smooth points x of X. Let $p_1 : X \times X_{\mathrm{reg}} \to X$ and $p_2 : X \times X_{\mathrm{reg}} \to X_{\mathrm{reg}}$ be the projections. By [Laz04, Example 5.1.11], applied to p_2, the diagonal section of p_2, and p_1^*L, we have $\epsilon(L, x) \leq \epsilon(L, x')$ for any smooth point x and a very general point x' of X. Thus $\epsilon(L)$ is the Seshadri constant of L at a very general point of X, and if X is normal and L is big, then $\epsilon(L) = \epsilon_{\mathrm{mov}}(L)$ by Lemma 2.9(4) and Lemma 2.6(3).

Lemma 2.9. *Let X be a normal projective variety of dimension d, L a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and $x \in X$ a smooth point. Then the following hold.*

(1) For any curve $C \subset X$ passing through x, we have $\epsilon(L, x) \leq (L \cdot C)/\operatorname{mult}_x C$.
(2) $(L^d) \geq \epsilon(L, x)^d$.
(3) If L is ample, then $\epsilon(L, x) = \epsilon_{\mathrm{mov}}(L, x)$.
(4) If L is big and x is a very general point of X, then $\epsilon(L, x) = \epsilon_{\mathrm{mov}}(L, x)$.

Proof. (1) follows from [Laz04, Proposition 5.1.5], applied to IL for a positive integer I with IL Cartier, by homogeneity [Laz04, Example 5.1.4].

(2) follows from [Laz04, Proposition 5.1.9], applied to IL for a positive integer I with IL Cartier and to the subvariety X of X, since $\operatorname{mult}_x X = 1$, by homogeneity [Laz04, Example 5.1.4].

(3) Let I be a positive integer such that IL is Cartier. By Lemma 2.6(2) and [Laz04, Theorem 5.1.17], we have

$$\epsilon_{\mathrm{mov}}(L, x) = \frac{1}{I} \limsup_{k \to +\infty} \frac{s(kIL, x)}{k} = \frac{1}{I} \epsilon(IL, x).$$

By Definition 2.7, $\epsilon(IL,x) = I\epsilon(L,x)$, and (3) follows.

(4) Let I be a positive integer such that IL is Cartier. Then $\epsilon(IL,x) = I\epsilon(L,x)$ by Definition 2.7, and

$$\epsilon_{\mathrm{mov}}(IL,x) = I\epsilon_{\mathrm{mov}}(L,x)$$

by Lemma 2.6(2). Possibly replacing L with IL, we may assume that L is Cartier.

Let $g : \widetilde{X} \to X$ be a resolution of singularities which is an isomorphism over X_{reg}, and let $\tilde{x} := g^{-1}(x)$. Since X is normal, $g_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$, so the projection formula [Har77, Exercise II.5.1(d)] implies

$$H^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(g^*(mL))) = H^0(X, \mathcal{O}_X(mL))$$

for any integer $m \geq 1$. Since g is an isomorphism over a neighborhood of x, it follows that $s(g^*(mL), \tilde{x}) = s(mL, x)$ for any $m \geq 1$, hence

$$\epsilon_{\mathrm{mov}}(g^*L, \tilde{x}) = \epsilon_{\mathrm{mov}}(L, x).$$

Moreover, taking the strict transform $g_*^{-1}C$ of a curve $C \subset X$ through x defines a bijection between the curves on X through x and the curves on $\widetilde{X}$ through $\tilde{x}$, with

$$(g^*L \cdot g_*^{-1}C) = (L \cdot C) \quad \text{and} \quad \operatorname{mult}_{\tilde{x}} g_*^{-1}C = \operatorname{mult}_x C.$$

Thus $\epsilon(g^*L, \tilde{x}) = \epsilon(L,x)$ by [Laz04, Proposition 5.1.5]. Finally, g^*L is nef and big, and if $\widetilde{Z}_1, \widetilde{Z}_2, \dots$ are proper closed subsets of $\widetilde{X}$, then $g(\widetilde{Z}_1), g(\widetilde{Z}_2), \dots$ are proper closed subsets of X, and $\tilde{x} \notin \bigcup_i \widetilde{Z}_i$ when $x \notin \bigcup_i g(\widetilde{Z}_i)$. Possibly replacing X, L, and x with $\widetilde{X}$, g^*L, and $\tilde{x}$, we may assume that X is smooth. In this case, (4) is [Dem92, Theorem 6.4] (cf. [Zhu18, equation (1)]). □

Definition 2.10 ([Zhu18, Definition 2.3])**.** Let X be a normal projective variety. A *terminal modification* of X is a projective birational morphism $h : X' \to X$ such that X' is $\mathbb{Q}$-factorial and terminal and $K_{X'}$ is nef over X.

Remark 2.11. Let $h : X' \to X$ be a terminal modification and m a positive integer. Since $K_{X'}$ is nef over X, the negativity lemma [KM98, Lemma 3.39] (cf. [Zhu18, Lemma 2.5]) shows that for any $D_X \in |-mK_X|$, there exists an h-exceptional divisor $E' \geq 0$ such that $h_*^{-1}D_X + E' \in |-mK_{X'}|$. Thus h_* induces an isomorphism

$$H^0(X', \mathcal{O}_{X'}(-mK_{X'})) \cong H^0(X, \mathcal{O}_X(-mK_X)).$$

Since h is an isomorphism over a non-empty open subset $U \subset X$, the isomorphism induced by h_* shows that $s(-mK_{X'}, x') = s(-mK_X, h(x'))$ for any $x' \in h^{-1}(U)$ such that $h(x')$ is a smooth point of X, and together with Lemma 2.6(3) this implies the equality in Lemma 2.12(1) (cf. [Zhu18, Lemma 3.4]). The condition that $K_{X'}$ is nef over X excludes, for example, the blow-up $h : X' \to X$ of X at a smooth point x when $d := \dim X \geq 2$. Indeed, let $E \cong \mathbb{P}^{d-1}$ be the exceptional divisor of h and $\ell \subset E$ a line. Then $K_{X'} = h^*K_X + (d-1)E$ and $(K_{X'} \cdot \ell) = -(d-1) < 0$, and the sections of $\mathcal{O}_{X'}(-mK_{X'})$ are the sections of $\mathcal{O}_X(-mK_X)$ vanishing to order at least $m(d-1)$ at x.

Lemma 2.12 ([Zhu18, Lemma 3.4, Corollary 3.3])**.**

(1) Let X be a normal projective variety. Then X has a terminal modification, and

$$\epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X)$$

for any terminal modification $X' \to X$.

(2) Let X be a $\mathbb{Q}$-factorial terminal projective variety and $X \dashrightarrow Y$ a composition of finitely many steps of a K_X-MMP. Then

$$\epsilon_{\mathrm{mov}}(-K_X) \leq \epsilon_{\mathrm{mov}}(-K_Y).$$

Proof. (1) The existence of a terminal modification of X follows from [BCHM10, Theorem 1.2(1)], applied to a resolution of X over X. See the paragraph after [Zhu18, Definition 2.4]. The equality is [Zhu18, Lemma 3.4].

(2) The inverse of a divisorial contraction or of a flip contracts no divisor, hence the inverse of $X \dashrightarrow Y$ contracts no divisor. Thus (2) follows from [Zhu18, Corollary 3.3]. □

Remark 2.13. Let Ω be an algebraically closed field of characteristic zero with $|\Omega| = |\mathbb{C}|$, for example an algebraic closure of the function field of a variety. Then Ω and $\mathbb{C}$ both have transcendence degree $|\mathbb{C}|$ over $\mathbb{Q}$, so Ω is isomorphic to $\mathbb{C}$ as a field. Thus the definitions and results of this section, and the results that we cite, hold for varieties over Ω.

2.3. **Boundedness.** We use the following notions of boundedness, see [HMX13, Definition 2.4.1] and [Bir21, Section 2.8].

Definition 2.14. Let $\mathfrak{X}$ be a set of projective varieties. We say that $\mathfrak{X}$ is *bounded* (resp. *birationally bounded*) if there exist a scheme T of finite type over $\mathbb{C}$ and a projective morphism $\mathcal{Z} \to T$ such that any $X \in \mathfrak{X}$ is isomorphic (resp. birational) to the fiber $\mathcal{Z}_t$ of $\mathcal{Z} \to T$ over some closed point $t \in T$.

Lemma 2.15.

(1) *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be sets of projective varieties such that $\mathfrak{Y}$ is birationally bounded and any member of $\mathfrak{X}$ is birational to a member of $\mathfrak{Y}$. Then $\mathfrak{X}$ is birationally bounded. Moreover, a bounded set of projective varieties is birationally bounded, and a finite union of birationally bounded sets of projective varieties is birationally bounded.*

(2) *For any positive integer m, the set of varieties $E \times \mathbb{P}^m$, where E runs over the smooth projective curves of genus one, is bounded.*

Proof. (1) The first sentence of (1) is [HMX13, Lemma 2.4.2(1)], and a set of projective varieties bounded by a family $\mathcal{Z} \to T$ is birationally bounded with the family $\mathcal{Z} \to T$. If $\mathfrak{X}_1, \dots, \mathfrak{X}_k$ are birationally bounded with families $\mathcal{Z}_i \to T_i$, then $\mathfrak{X}_1 \cup \cdots \cup \mathfrak{X}_k$ is birationally bounded with the family

$$\bigsqcup_{i=1}^{k} \mathcal{Z}_i \to \bigsqcup_{i=1}^{k} T_i,$$

and $\bigsqcup_{i=1}^{k} T_i$ is of finite type over $\mathbb{C}$.

(2) By [Har77, Proposition IV.4.6], any smooth projective curve E of genus one is isomorphic to the plane cubic curve

$$(y^2z = x(x-z)(x-\lambda z)) \subset \mathbb{P}^2$$

for some $\lambda \in \mathbb{C}\backslash\{0,1\}$. Let $T := \mathbb{A}^1\backslash\{0,1\}$ with coordinate λ, and let $\mathcal{E} \subset \mathbb{P}^2 \times T$ be the closed subscheme defined by $y^2z = x(x-z)(x-\lambda z)$. Then $\mathcal{E} \times \mathbb{P}^m \to T$ is a projective morphism, T is of finite type over $\mathbb{C}$, and the fiber over $\lambda \in T$ is $\mathcal{E}_\lambda \times \mathbb{P}^m$. Thus any $E \times \mathbb{P}^m$ is isomorphic to a fiber of $\mathcal{E} \times \mathbb{P}^m \to T$. □

Theorem 2.16 ([Bir21, Theorem 1.1])**.** *Let d be a positive integer. Then the set of terminal projective varieties X of dimension d such that $-K_X$ is nef and big is bounded.*

Theorem 2.16 is the case of [Bir21, Theorem 1.1] with zero boundary and 1-lc varieties (Definition 2.2): for a terminal variety X, the pair $(X, 0)$ is 1-lc, since any prime divisor over X has log discrepancy at least 1 with respect to $(X, 0)$ [KM98, Definition 2.34].

2.4. **Fibrations.**

Definition 2.17. Let $f : X \to Z$ be a contraction between normal projective varieties and (X, B) a pair. We say that (X, B) is *klt over a non-empty open subset of Z* if $(f^{-1}(U), B|_{f^{-1}(U)})$ is klt for some non-empty open subset $U \subset Z$.

The boundaries constructed in Theorem 2.19, Proposition 5.2, and Section 11 are klt near a general fiber, but they may fail to be lc over special points of the base. Lemma 2.18 only assumes that the pair is klt over a non-empty open subset of the base.

Lemma 2.18. *Let $f : X \to Z$ be a contraction from a normal projective variety X onto a normal projective variety Z of dimension $b \in \{1, 2\}$, $B \geq 0$ a $\mathbb{Q}$-divisor on X, and L a $\mathbb{Q}$-divisor on Z. Assume that Z is a smooth curve if $b = 1$, that Z is $\mathbb{Q}$-factorial if $b = 2$, that $K_X + B \sim_{\mathbb{Q}} f^*L$, and that (X, B) is klt over a non-empty open subset of Z. Then $((L - K_Z) \cdot H^{b-1}) \geq 0$ for any ample Cartier divisor H on Z. If $b = 1$, this means $\deg(L - K_Z) \geq 0$.*

Proof. By [Kol07, Theorem 8.5.1 and Proposition 8.4.9], we may write

$$L \sim_{\mathbb{Q}} K_Z + B_Z + M_Z,$$

where B_Z is the discriminant part and M_Z is the moduli part of $f : (X, B) \to Z$ [Kol07, Definitions 8.4.1–8.4.2]. Since Z is a smooth curve or a $\mathbb{Q}$-factorial surface, K_Z, B_Z, and M_Z are $\mathbb{Q}$-Cartier.

We show that $B_Z \geq 0$. The coefficient of B_Z at a prime divisor P on Z is $1 - c_P$, where c_P is the supremum of the real numbers c such that $(X, B + cf^*P)$ is lc over the generic point of P. Let T be a component of f^*P dominating P and $e := \operatorname{mult}_T f^*P \geq 1$ ($\mathcal{O}_{Z,P}$ is a discrete valuation ring [Har77, Theorem I.6.2A], so P is Cartier near the generic point of P and e is an integer). Since $B \geq 0$, we have $a(T, X, B) = 1 - \operatorname{mult}_T B \leq 1$. For any $c > a(T, X, B)/e$,

$$a(T, X, B + cf^*P) = a(T, X, B) - ce < 0,$$

so $c_P \leq a(T, X, B)/e \leq 1$. Thus $B_Z \geq 0$, and $(B_Z \cdot H^{b-1}) \geq 0$.

By [Kol07, Proposition 8.4.9], there exist a normal projective variety Z', a generically finite surjective morphism $g : Z' \to Z$, and a nef $\mathbb{Q}$-divisor M' on Z' such that $M_Z = (\deg g)^{-1} g_* M'$. By the projection formula,

$$(M_Z \cdot H^{b-1}) = (\deg g)^{-1} (M' \cdot (g^*H)^{b-1}) \geq 0.$$

Therefore

$$((L - K_Z) \cdot H^{b-1}) = (B_Z \cdot H^{b-1}) + (M_Z \cdot H^{b-1}) \geq 0.$$

□

Theorem 2.19 ([Zhu18, Corollary 5.2])**.** *Let $f : X \to Z$ be a contraction between normal projective varieties such that $0 < \dim Z < \dim X$, K_X is $\mathbb{Q}$-Cartier, and the general fiber of f is isomorphic to $\mathbb{P}^{a-1}$ for some integer $a \geq 2$. Let t be a rational number such that $0 < t \leq a$ and $\epsilon_{\mathrm{mov}}(-K_X) > a - t$. Then there exists a $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -t^{-1}K_X$ and (X, D) is klt over a non-empty open subset of Z.*

Remark 2.20. In [Zhu18], a fibertype morphism is a dominant morphism $f : X \to Z$ with connected fibers such that $0 < \dim Z < \dim X$. Thus the contraction f in Theorem 2.19 is a fibertype morphism. The invariant ϵ_m of [Zhu18] is ϵ_{mov}, and the numbers a and t of Theorem 2.19 are denoted by r and ϵ in [Zhu18, Lemma 5.1 and Corollary 5.2].

[Zhu18, Corollary 5.2] only assumes that $t > 0$. Theorem 2.19 has two more conditions, and both are used in the proof of [Zhu18, Lemma 5.1], from which [Zhu18, Corollary 5.2] is deduced.

(1) K_X is $\mathbb{Q}$-Cartier. The proof of [Zhu18, Lemma 5.1] uses the base ideal of $|-mK_X|$ for sufficiently divisible positive integers m.

(2) $t \leq a$. Let F be a general fiber of f, $x \in F$ a general point, and $W_m \subset H^0(F, \mathcal{O}_F(-mK_X|_F))$ the image of the restriction map

$$H^0(X, \mathcal{O}_X(-mK_X)) \to H^0(F, \mathcal{O}_F(-mK_X|_F)).$$

The proof of [Zhu18, Lemma 5.1] uses [Zhu18, Lemma 3.5] for W_m and x, and [Zhu18, Lemma 3.5] assumes that x is not a base point of W_m. Since $s(-mK_X, x) > m(a - t) \geq 0$ for the degrees m used there, W_m generates $\mathcal{O}_F(-mK_X|_F)$ at x.

Later we only use $t = 1/2$ and $t = 1/q$ for integers $q \geq 2$, so $t < 1 < a$.

The conclusion of [Zhu18, Corollary 5.2] is that (X, D) is klt along a general fiber F of f. Since $\mathrm{Nklt}(X, D)$ (Definition 2.3) is closed and f is proper, $f(\mathrm{Nklt}(X, D))$ is a closed subset of Z which does not contain $f(F)$. Thus (X, D) is klt over the non-empty open subset $Z \backslash f(\mathrm{Nklt}(X, D))$ in the sense of Definition 2.17.

2.5. **Brauer–Severi schemes.**

Definition 2.21. Let Z be a scheme of finite type over $\mathbb{C}$ and r a positive integer. A *Brauer–Severi scheme* of relative dimension r over Z is a smooth projective morphism $f : X \to Z$ such that any geometric fiber of f is isomorphic to $\mathbb{P}^r$.

Theorem 2.22 ([BF19, Theorem 1.1])**.** *Let C be an integral affine curve over $\mathbb{C}$ and $f : X \to C$ a Brauer–Severi scheme of relative dimension r. Then X is isomorphic to $\mathbb{P}^r \times C$ over C.*

Remark 2.23. [BF19, Theorem 1.1] holds for any semisimple group G such that $|\pi_1(G)|$ is invertible in the base field, and Theorem 2.22 is the case $G = \mathrm{PGL}_{r+1}$, where $|\pi_1(G)| = r + 1$. Indeed, by [Gro68, 8.1 and Corollary 8.3], f is isomorphic to $\mathbb{P}^r \times C \to C$ locally in the étale topology, and Brauer–Severi schemes of relative dimension r over C are classified up to isomorphism by $H^1_{\text{ét}}(C, \mathrm{PGL}_{r+1})$, so the triviality of PGL_{r+1}-torsors over C in [BF19, Theorem 1.1] implies $X \cong \mathbb{P}^r \times C$ over C.

3. Reduction to Mori fiber spaces

In this section, we prove Construction-Theorem 3.1, which reduces Theorem 1.2 to a terminal Mori fiber space $f : Y \to Z$. In case (2) of Construction-Theorem 3.1, X is birational to a terminal Fano variety, by [Zhu18, Theorem 3.7] when $\epsilon_{\mathrm{mov}}(-K_Y) > d - 1$. The Mori fiber spaces over curves and over surfaces in case (3) are studied in Sections 4–12.

Construction-Theorem 3.1. Let X be a normal projective variety of dimension $d \geq 2$ such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$. Then there exist a terminal modification $X' \to X$, a composition $X' \dashrightarrow Y$ of finitely many steps of a $K_{X'}$-MMP, and a Mori fiber space $f : Y \to Z$ such that Y is terminal and the following hold.

(1) The moving Seshadri constants satisfy
$$\epsilon_{\mathrm{mov}}(-K_Y) \geq \epsilon_{\mathrm{mov}}(-K_X).$$
(2) If $\epsilon_{\mathrm{mov}}(-K_Y) > d - 1$ or $\dim Z = 0$, then X is birational to a terminal projective variety X'' of dimension d such that $-K_{X''}$ is ample.
(3) If $\epsilon_{\mathrm{mov}}(-K_Y) = d - 1$ and $\dim Z > 0$, then the following hold.
   (i) $\dim Z \in \{1, 2\}$.
   (ii) If $\dim Z = 2$, then $d \geq 3$ and the geometric generic fiber of f is isomorphic to $\mathbb{P}^{d-2}$.

By (1), cases (2) and (3) cover all possibilities, and only case (3) is left to later sections, where for $\dim Z = 2$ we only know that the geometric generic fiber $Y_{\bar{\eta}}$ of f is isomorphic to $\mathbb{P}^{d-2}$, and not that the generic fiber Y_η is isomorphic to $\mathbb{P}^{d-2}$ over $\mathbb{C}(Z)$. The key point of the proof is the chain of inequalities
$$d - 1 \leq \epsilon_{\mathrm{mov}}(-K_Y) \leq \epsilon_{\mathrm{mov}}(-K_{Y_{\bar{\eta}}}) \leq \dim Y_{\bar{\eta}} + 1,$$
and
$$\epsilon_{\mathrm{mov}}(-K_{Y_{\bar{\eta}}}) > \dim Y_{\bar{\eta}}$$
only for projective space [Zhu18, Theorem 1.7]. We work on $Y_{\bar{\eta}}$ rather than on a general closed fiber of f so that the isomorphism $Y_{\bar{\eta}} \cong \mathbb{P}^{d-2}$ spreads out to a Brauer–Severi scheme over a dense open subset of Z (Definition-Lemma 9.4 in Section 9).

Proof of Construction-Theorem 3.1. The proof is long, so we divide it into several steps.

Step 1. In this step we construct X' and compare $\epsilon_{\mathrm{mov}}(-K_{X'})$ with $\epsilon_{\mathrm{mov}}(-K_X)$.

By Lemma 2.12(1), X has a terminal modification $h : X' \to X$ (Definition 2.10), and

$$\epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X) \geq d-1.$$

Step 2. In this step we construct $X' \dashrightarrow Y$ and $f : Y \to Z$, and we prove (1).

Since $d \geq 2$, **Step 1** implies that $\epsilon_{\mathrm{mov}}(-K_{X'}) \geq d-1 > 0$. Since X' is $\mathbb{Q}$-factorial, $-K_{X'}$ is $\mathbb{Q}$-Cartier, hence $-K_{X'}$ is big by Lemma 2.6(4). Let H be an ample Cartier divisor on X'. Then $(K_{X'} \cdot H^{d-1}) < 0$, while $(D \cdot H^{d-1}) \geq 0$ for any pseudo-effective $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X'. Thus $K_{X'}$ is not pseudo-effective. Since X' is terminal, the pair $(X', 0)$ is klt. By [BCHM10, Corollary 1.3.3], there exist a composition $X' \dashrightarrow Y$ of finitely many steps of a $K_{X'}$-MMP and a Mori fiber space $f : Y \to Z$. Since X' is $\mathbb{Q}$-factorial and terminal, any variety in the $K_{X'}$-MMP $X' \dashrightarrow Y$ is $\mathbb{Q}$-factorial and terminal by [KM98, Propositions 3.36 and 3.37, Corollaries 3.42 and 3.43]. In particular, Y is terminal, and Y is $\mathbb{Q}$-factorial as in [BCHM10, Definition 3.10.7]. By Lemma 2.12(2) and **Step 1**,

$$\epsilon_{\mathrm{mov}}(-K_Y) \geq \epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X),$$

which proves (1). We fix a positive integer I such that IK_Y is Cartier.

Step 3. In this step we assume that $b := \dim Z > 0$, and we prove that the geometric generic fiber of f is a terminal normal projective variety with ample anti-canonical divisor.

Let η be the generic point of Z, let $k := \overline{\mathbb{C}(Z)}$, and let $F := Y_{\bar{\eta}}$ be the geometric generic fiber of f. Then $\dim F = d - b$. We show that F is a normal projective variety over k, F is terminal, $\mathcal{O}_Y(mK_Y)|_F \cong \mathcal{O}_F(mK_F)$ for any integer m divisible by I, and $-K_F$ is ample.

Let $g : Y' \to Y$ be a log resolution with exceptional prime divisors E_j, and write $K_{Y'} = g^*K_Y + \sum_j (a(E_j, Y, 0) - 1)E_j$ with $a(E_j, Y, 0) > 1$. By [Har77, Corollary III.10.7], applied to $Y' \to Z$ and to the irreducible components of the intersections of the E_j, the variety $F' := Y' \times_Y F$ is smooth and $\sum_j E_j|_{F'}$ is a simple normal crossing divisor. By [Har77, Proposition III.9.3], the equalities $f_*\mathcal{O}_Y = \mathcal{O}_Z$ and $g_*\mathcal{O}_{Y'} = \mathcal{O}_Y$ are preserved by the base change to k. Thus F' is connected and smooth, hence integral, and the equality $\mathcal{O}_F = (g_F)_*\mathcal{O}_{F'}$ for the base change $g_F : F' \to F$ of g implies that F is integral and normal. By [Har77, Proposition II.8.10], relative differentials of f on the smooth locus of f commute with base change (on the smooth locus of f over Z_{reg}, $\omega_Y \cong \omega_{Y/Z} \otimes f^*\omega_Z$ [Har77, Proposition II.8.11], $f^*\omega_Z$ is trivial on Y_η, and the base change to k of the smooth locus of f is F_{reg} [Har77, Proposition II.8.10 and Example III.10.0.3]), so $\mathcal{O}_Y(mK_Y)|_F$ and $\mathcal{O}_F(mK_F)$ are isomorphic on F_{reg}, hence on F, as both are reflexive and F is normal. Similarly, by [Har77, Proposition II.8.10] applied to $Y' \to Z$, we have $\mathcal{O}_{Y'}(K_{Y'})|_{F'} \cong \mathcal{O}_{F'}(K_{F'})$. Restricting the formula for $K_{Y'}$ to F', the E_j that do not dominate Z disappear, the others restrict to divisors $E_j|_{F'}$ exceptional over F, and $K_{F'} = g_F^*K_F + \sum_j (a(E_j, Y, 0) - 1)E_j|_{F'}$ with $a(E_j, Y, 0) > 1$. Since $g_F : F' \to F$ is a log resolution of F whose exceptional prime divisors are the components of the $E_j|_{F'}$ for the E_j dominating Z, [KM98, Corollary 2.32(2)] with $\Delta = 0$ implies that the discrepancy of F is the minimum of 1 and the numbers $a(E_j, Y, 0) - 1$ for the E_j dominating Z, which is positive. Thus F is terminal. Finally, $-K_Y$ is f-ample [BCHM10, Definition 3.10.7], so $-K_F$ is ample.

Step 4. In this step we keep the assumption $b > 0$, and we prove that

$$\epsilon_{\mathrm{mov}}(-K_F) \geq \epsilon_{\mathrm{mov}}(-K_Y).$$

By Remark 2.13, we may use Definition 2.4 and Lemma 2.6 for F over k. For any positive integer m divisible by I, let $V_m := H^0(Y, \mathcal{O}_Y(-mK_Y))$, and let a_m be the maximum of $s(V_m, y)$ over the smooth points $y \in Y$. By Lemma 2.6(3), for a very general point $y \in Y$ we have $s(V_m, y) = a_m$ for any positive integer m divisible by I, and

$$\epsilon_{\mathrm{mov}}(-K_Y, y) = \epsilon_{\mathrm{mov}}(-K_Y).$$

Thus Lemma 2.6(2) implies that

$$\epsilon_{\mathrm{mov}}(-K_Y) = \limsup_{m\to+\infty,\ I|m} \frac{a_m}{m}.$$

We show below that for any positive integer m divisible by I, there exists a non-empty open subset $F_m \subset F$ such that $s(-mK_F, x) \geq a_m$ for any closed point $x \in F_m$. Since k is uncountable, a very general point $x \in F$ belongs to F_m for all such m, and Lemma 2.6(2) implies that

$$\epsilon_{\mathrm{mov}}(-K_F) \geq \epsilon_{\mathrm{mov}}(-K_F, x) = \limsup_{m\to+\infty,\ I|m} \frac{s(-mK_F, x)}{m} \geq \limsup_{m\to+\infty,\ I|m} \frac{a_m}{m} = \epsilon_{\mathrm{mov}}(-K_Y).$$

Fix a positive integer m divisible by I. If $a_m = -1$, we take $F_m = F_{\mathrm{reg}}$, which is a non-empty open subset of F since F is normal, as $s(-mK_F, x) \geq -1$ for any $x \in F_{\mathrm{reg}}$ by Definition 2.4. Assume that $a := a_m \geq 0$. For a smooth morphism $\pi : T \to S$ and a line bundle L on T, let

$$\mathcal{P}^a_{T/S}(L) := (p_1)_* \left(p_2^* L \otimes \mathcal{O}_{T\times_S T}/\mathcal{I}_\Delta^{a+1}\right)$$

be the sheaf of relative a-jets of L, where $p_1, p_2 : T \times_S T \to T$ are the projections and $\mathcal{I}_\Delta$ is the ideal sheaf of the diagonal. The sheaf $\mathcal{P}^a_{T/S}(L)$ is locally free, the formation of $\mathcal{P}^a_{T/S}(L)$ commutes with base change along any morphism $S' \to S$, and the fiber of $\mathcal{P}^a_{T/S}(L)$ at a closed point $t \in T$ is $L \otimes \mathcal{O}_{T_{\pi(t)}}/\mathfrak{m}_t^{a+1}$, where $T_{\pi(t)}$ is the fiber of π over $\pi(t)$ [Gro67, Propositions 16.4.5 and 17.12.4, and (16.7.2.1)]. Any linear subspace $V \subset H^0(T, L)$ defines a natural map $V \otimes \mathcal{O}_T \to \mathcal{P}^a_{T/S}(L)$. When S is the spectrum of an algebraically closed field, we write $\mathcal{P}^a_T(L) := \mathcal{P}^a_{T/S}(L)$, and by Nakayama's lemma, $V \otimes \mathcal{O}_T \to \mathcal{P}^a_T(L)$ is surjective at a closed point t if and only if $s(V, t) \geq a$.

Let Y° be the open subset of $f^{-1}(Z_{\mathrm{reg}})$ on which f is smooth. Then Y° is non-empty by [Har77, Lemma III.10.5], and $Y^\circ \subset Y_{\mathrm{reg}}$. Let $L := \mathcal{O}_Y(-mK_Y)|_{Y^\circ}$, which is invertible as mK_Y is Cartier. The closed immersion $Y^\circ \times_Z Y^\circ \subset Y^\circ \times Y^\circ$ induces a surjection

$$\mathcal{P}^a_{Y^\circ}(L) \to \mathcal{P}^a_{Y^\circ/Z}(L),$$

since p_1 is finite on the $(a+1)$-st infinitesimal neighborhood of the diagonal, and the surjection

$$\mathcal{P}^a_{Y^\circ}(L) \to \mathcal{P}^a_{Y^\circ/Z}(L)$$

commutes with the natural maps from $V_m \otimes \mathcal{O}_{Y^\circ}$. By Lemma 2.6(3), the set U of points $y \in Y^\circ$ such that $s(V_m, y) \geq a$ is a non-empty open subset of Y°. Thus

$$V_m \otimes \mathcal{O}_{Y^\circ} \to \mathcal{P}^a_{Y^\circ}(L)$$

is surjective on U, and so is

$$V_m \otimes \mathcal{O}_{Y^\circ} \to \mathcal{P}^a_{Y^\circ/Z}(L).$$

Since U contains the generic point of Y, the base change $F_m := U \times_Z \operatorname{Spec} k$ is a non-empty open subset of F, and F_m is smooth over k. Since pullback is right exact and the formation of $\mathcal{P}^a_{Y^\circ/Z}(L)$ commutes with base change, the map

$$V_m \otimes_{\mathbb{C}} \mathcal{O}_{F_m} \to \mathcal{P}^a_{F_m}(L|_{F_m})$$

is surjective, where $L|_{F_m} \cong \mathcal{O}_F(-mK_F)|_{F_m}$ by **Step 3**. Let $W_m \subset H^0(F, \mathcal{O}_F(-mK_F))$ be the image of the restriction map

$$V_m \otimes_{\mathbb{C}} k \to H^0(F, \mathcal{O}_F(-mK_F)).$$

The surjection onto $\mathcal{P}^a_{F_m}(L|_{F_m})$ factors through $W_m \otimes_k \mathcal{O}_{F_m}$, hence $s(W_m, x) \geq a$ for any closed point $x \in F_m$. Since $W_m \subset H^0(F, \mathcal{O}_F(-mK_F))$, we obtain $s(-mK_F, x) \geq s(W_m, x) \geq a$ for any closed point $x \in F_m$.

Step 5. In this step we keep the assumption $b > 0$, and we prove (3).

By Remark 2.13, we may use [Zhu18, Theorem 1.7] for the normal projective variety F over k: if $\epsilon_{\mathrm{mov}}(-K_F) > \dim F$, then $F \cong \mathbb{P}^{\dim F}$. For $r \geq 1$ and a hyperplane $H \subset \mathbb{P}^r$, we have $-K_{\mathbb{P}^r} \sim (r+1)H$ and $s(mH, x) = m$ for any $x \in \mathbb{P}^r$ and any integer $m \geq 1$: the monomials

of degree at most m in affine coordinates centered at x generate m-jets at x, and for a line $\ell \ni x$, Lemma 2.6(5) implies $s(mH, x) \leq s(\mathcal{O}_\ell(m), x) \leq m$, as $h^0(\ell, \mathcal{O}_\ell(m)) = m + 1$. Thus $\epsilon_{\mathrm{mov}}(-K_{\mathbb{P}^r}) = r + 1$. Hence $\epsilon_{\mathrm{mov}}(-K_F) \leq \dim F + 1 = d - b + 1$. By the hypothesis on X, (1), and **Step 4**, we have

$$d - 1 \leq \epsilon_{\mathrm{mov}}(-K_X) \leq \epsilon_{\mathrm{mov}}(-K_Y) \leq \epsilon_{\mathrm{mov}}(-K_F) \leq d - b + 1.$$

Thus $b \leq 2$, that is, $\dim Z \in \{1, 2\}$. Assume that $b = 2$. Since $\dim Z < \dim Y$ [BCHM10, Definition 3.10.7], we have $d \geq 3$. Moreover, $\epsilon_{\mathrm{mov}}(-K_F) \geq d - 1 > d - 2 = \dim F$, so the geometric generic fiber F of f is isomorphic to $\mathbb{P}^{d-2}$ by [Zhu18, Theorem 1.7]. This proves (3).

Step 6. In this step we prove (2).

Recall that X is birational to Y via $h : X' \to X$ and $X' \dashrightarrow Y$. Assume first that $\epsilon_{\mathrm{mov}}(-K_Y) > d - 1$. We use [Zhu18, Theorem 3.7] for the normal projective variety Y of dimension d, and not for X, since $\epsilon_{\mathrm{mov}}(-K_X)$ may be equal to $d - 1$. Thus Y is birational to a terminal projective variety X'' of dimension d such that $-K_{X''}$ is ample, and so is X. Assume next that $\dim Z = 0$. Since $-K_Y$ is f-ample, $-K_Y$ is ample, and Y is terminal by **Step 2**. Thus $X'' = Y$ satisfies (2). □

4. Rational points on generic fibers

In this section, we prove two results on the generic fiber of a contraction $f : X \to C$ onto a smooth projective curve. They are used in Section 5 and Section 11 to show that the generic fiber of f is rational over $\mathbb{C}(C)$ when $\dim X = d$ and $((-K_{X_{\bar{\eta}}})^{d-1}) > (d-1)^{d-1}$. In general, $X_{\bar{\eta}}$ is not necessarily isomorphic to $\mathbb{P}^{d-1}$, and $\mathrm{Br}(\mathbb{C}(C)) = 0$ only shows that a variety over $\mathbb{C}(C)$ which is isomorphic to $\mathbb{P}^{d-1}$ over the algebraic closure of $\mathbb{C}(C)$ is isomorphic to $\mathbb{P}^{d-1}$ over $\mathbb{C}(C)$ [Ser79, Chapter X, §6 and §7, Proposition 10 and Example c)].

The definitions of $s(V, x)$ and of generating s-jets in Definition 2.4 make sense for a k-rational smooth point x of a normal projective variety over any field k, and in this section we use them at $\mathbb{C}(C)$-rational points. When we use Definition 2.7, Lemma 2.9(3), and the results of [LZ18] over an algebraic closure of $\mathbb{C}(C)$, we refer to Remark 2.13.

The lower bound on Seshadri constants by generation of jets (Definition 2.4 and Lemma 2.9(4)) holds at very general geometric points of $X_{\bar{\eta}}$, which are in general not $\mathbb{C}(C)$-rational, while Proposition 4.2 assumes a rational point at which the classification in [LZ18, Lemma 13] can be used. Lemma 4.1 uses a parameter space of finite type for sections of f to turn countably many jet conditions into countably many non-empty open subsets of the parameter space, and thus finds such a rational point.

Lemma 4.1. *Let $f : X \to C$ be a contraction from a normal projective variety X onto a smooth projective curve C, let X_η be the generic fiber of f, and let $X_{\bar{\eta}}$ be the geometric generic fiber of f. Assume that a resolution of $X_{\bar{\eta}}$ is rationally connected. Let $m_1, m_2, \ldots$ be positive integers and $s_1, s_2, \ldots$ non-negative integers such that for any j, $-m_jK_X$ is Cartier and $H^0(X, \mathcal{O}_X(-m_jK_X))$ generates s_j-jets at any point of a non-empty open subset of X. Let $U \subset X$ be a non-empty open subset. Then there exists a $\mathbb{C}(C)$-rational point p of X_η such that $p \in U$, p is a smooth point of X_η, and for any j, the image of*

$$H^0(X, \mathcal{O}_X(-m_jK_X)) \otimes_{\mathbb{C}} \mathbb{C}(C) \to H^0(X_\eta, \mathcal{O}_{X_\eta}(-m_jK_{X_\eta}))$$

generates s_j-jets at p.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct a resolution $q : X' \to X$ and an integral scheme S of finite type over $\mathbb{C}$ parametrizing sections of $f \circ q$, such that the sections parametrized by S cover a dense subset of X'.

Let $q : X' \to X$ be a resolution of X such that X' is projective, let $f' := f \circ q$, and let $X^\circ \subset X$ be the non-empty open subset over which q is an isomorphism. Since X is normal,

$q_*\mathcal{O}_{X'} = \mathcal{O}_X$, and thus $f'_*\mathcal{O}_{X'} = f_*\mathcal{O}_X = \mathcal{O}_C$. In particular, the fibers of f' are connected [Har77, Corollary III.11.3]. By generic smoothness, there exists a non-empty open subset $C^\circ \subset C$ such that f' is smooth over C°. Thus for any closed point $c \in C^\circ$, the fiber X'_c is smooth and connected, hence irreducible.

Since $f_*\mathcal{O}_X = \mathcal{O}_C$ and $f'_*\mathcal{O}_{X'} = \mathcal{O}_C$, the generic fibers X_η and X'_η are normal and geometrically connected over $\mathbb{C}(C)$, hence geometrically integral in characteristic zero. In particular, $X_{\bar\eta}$ and $X'_{\bar\eta}$ are integral. Since X' is smooth, X'_η is regular, hence smooth over $\mathbb{C}(C)$. Since $X^\circ \cap X_\eta \neq \emptyset$, q induces a birational morphism $X'_{\bar\eta} \to X_{\bar\eta}$, and thus $X'_{\bar\eta}$ is a resolution of $X_{\bar\eta}$. By assumption, $X_{\bar\eta}$ has a resolution which is rationally connected in the sense of [Kol96, Definition IV.3.2]. Since any two resolutions of $X_{\bar\eta}$ are birational to each other, $X'_{\bar\eta}$ is rationally connected [Kol96, Proposition IV.3.3(3.3.3)].

Fix a closed point $c_0 \in C^\circ$ and a closed point $y \in X'_{c_0}$. We show that there exists a section $\sigma : C \to X'$ of f' such that $\sigma(c_0) = y$. Since f' is smooth at y, there exists a formal section of f' through y, that is, a point $\hat y$ of X'_η over the completion $\widehat{\mathbb{C}(C)}_{c_0}$ of $\mathbb{C}(C)$ at c_0 whose reduction at c_0 is y. Since X'_η is smooth and projective over $\mathbb{C}(C)$, $X'_{\bar\eta}$ is rationally connected, and f' is smooth over $C^\circ \ni c_0$, [HT06, Theorem 2] (weak approximation at places of good reduction) implies that there exists a $\mathbb{C}(C)$-rational point y_1 of X'_η which is congruent to $\hat y$ modulo the maximal ideal of $\widehat{\mathcal{O}}_{C,c_0}$. Since X' is projective, y_1 extends to a morphism $\sigma : C \to X'$ [Har77, Proposition I.6.8]. Since $f' \circ \sigma$ is the identity on a non-empty open subset of C, σ is a section of f', and $\sigma(c_0) = y$ by the choice of y_1.

By [Kol96, Theorems I.1.4 and I.1.10], the morphisms $C \to X'$ are parametrized, via graphs, by an open subscheme $\mathrm{Hom}(C, X')$ of the Hilbert scheme of $C \times X'$. The Hilbert scheme of $C \times X'$ is a disjoint union of countably many projective schemes. The sections of f' form the closed subscheme of $\mathrm{Hom}(C, X')$ which is the fiber over the point id_C of the morphism $\mathrm{Hom}(C, X') \to \mathrm{Hom}(C, C)$ given by composition with f'. The scheme of sections of f' has countably many irreducible components $S_1, S_2, \dots$, and we regard each S_i as an integral scheme of finite type over $\mathbb{C}$. For a closed point $t \in S_i$, let σ_t be the corresponding section, and let $g_i : S_i \times C \to X'$ be the morphism sending (t, c) to $\sigma_t(c)$. By the previous paragraph, the closed points of X'_{c_0} are contained in $\bigcup_i g_i(S_i \times \{c_0\})$. By Chevalley's theorem, any $g_i(S_i \times \{c_0\})$ is constructible. Since $\mathbb{C}$ is uncountable, the irreducible variety X'_{c_0} is not a countable union of nowhere dense constructible subsets. Thus there exists i such that $g_i(S_i \times \{c_0\})$ is dense in X'_{c_0}. Let $S := S_i$ and $g := g_i$.

We show that g is dominant. Let W be the closure of $g(S \times C)$ in X'. Then W is irreducible, $X'_{c_0} \subset W$, and $f'(W) = C$ since $f'(g(t, c)) = c$. If $W \neq X'$, then $\dim W \leq \dim X' - 1 = \dim X'_{c_0}$, so $W = X'_{c_0}$, which contradicts $f'(W) = C$. Thus $W = X'$.

Step 2. In this step we construct a section $\sigma : C \to X'$ and the point p.

For any j, let $U_j \subset X$ be a non-empty open subset such that $H^0(X, \mathcal{O}_X(-m_j K_X))$ generates s_j-jets at any point of U_j. By Definition 2.4, $U_j \subset X_{\mathrm{reg}}$. Let

$$O_j := q^{-1}(U_j \cap U \cap X^\circ),$$

which is a non-empty open subset of X'. Since g is dominant, $g^{-1}(O_j)$ is a non-empty open subset of $S \times C$. Since the projection $S \times C \to S$ is open, the image S_j of $g^{-1}(O_j)$ in S is a non-empty open subset of S. Since S is an integral scheme of finite type over the uncountable field $\mathbb{C}$, the set $\bigcap_j S_j$ contains a closed point t. Let $\sigma := \sigma_t$. For any j, $\sigma^{-1}(O_j)$ is a non-empty open subset of C, so it contains the generic point η of C, and $\sigma(\eta) \in O_j$.

The point $\sigma(\eta)$ is a $\mathbb{C}(C)$-rational point of X'_η. Let $p := q(\sigma(\eta))$. Since q is an isomorphism over X° and X'_η is smooth over $\mathbb{C}(C)$, p is a $\mathbb{C}(C)$-rational smooth point of X_η, and $p \in U \cap U_j$ for any j.

Step 3. In this step we show that for any j, the image of the restriction map in the lemma generates s_j-jets at p.

Fix j and let $\mathcal{L}_j := \mathcal{O}_X(-m_jK_X)$. Let $\mathcal{I}$ be the ideal sheaf of $\sigma(C^\circ)$ in $f'^{-1}(C^\circ)$, and let

$$\mathcal{J}_j := f'_*\big(q^*\mathcal{L}_j \otimes \mathcal{O}_{X'}/\mathcal{I}^{s_j+1}\big),$$

a coherent sheaf on C°. Since f' is smooth along $\sigma(C^\circ)$, $\mathcal{J}_j$ is locally free, and the fibers of $\mathcal{J}_j$ at a closed point $c \in C^\circ$ and at η are

$$q^*\mathcal{L}_j \otimes \mathcal{O}_{X'_c}/\mathfrak{m}^{s_j+1}_{X'_c,\sigma(c)} \quad \text{and} \quad q^*\mathcal{L}_j \otimes \mathcal{O}_{X'_\eta}/\mathfrak{m}^{s_j+1}_{X'_\eta,\sigma(\eta)},$$

respectively. We omit the verification, which is a computation in local coordinates of the smooth morphism f' along the section σ. Consider the natural map

$$\phi_j : H^0(X, \mathcal{L}_j) \otimes_{\mathbb{C}} \mathcal{O}_{C^\circ} \to \mathcal{J}_j.$$

Let $c \in C^\circ$ be a closed point such that $\sigma(c) \in O_j$. Such points exist since $\sigma^{-1}(O_j) \cap C^\circ$ is a non-empty open subset of C. Let $x := q(\sigma(c)) \in U_j \cap X^\circ$. Since q is an isomorphism near $\sigma(c)$, the fiber of ϕ_j at c is the composition

$$H^0(X, \mathcal{L}_j) \to \mathcal{L}_j \otimes \mathcal{O}_X/\mathfrak{m}_x^{s_j+1} \to \mathcal{L}_j \otimes \mathcal{O}_{X_c}/\mathfrak{m}^{s_j+1}_{X_c,x},$$

where the first map is surjective since $x \in U_j$, and the second map is surjective since $\mathcal{O}_{X_c}/\mathfrak{m}^{s_j+1}_{X_c,x}$ is a quotient of $\mathcal{O}_X/\mathfrak{m}_x^{s_j+1}$. Thus ϕ_j is surjective at c. By Nakayama's lemma, the support of the cokernel of ϕ_j is a closed subset of C° which does not contain c, so it does not contain η. Since q is an isomorphism near $\sigma(\eta)$ and $K_X|_{X_\eta} = K_{X_\eta}$, the fiber of ϕ_j at η is the surjective map

$$H^0(X, \mathcal{L}_j) \otimes_{\mathbb{C}} \mathbb{C}(C) \to \mathcal{O}_{X_\eta}(-m_jK_{X_\eta}) \otimes \mathcal{O}_{X_\eta}/\mathfrak{m}_p^{s_j+1}.$$

The map factors through the restriction map

$$H^0(X, \mathcal{L}_j) \otimes_{\mathbb{C}} \mathbb{C}(C) \to H^0(X_\eta, \mathcal{O}_{X_\eta}(-m_jK_{X_\eta})).$$

Thus the image of the restriction map generates s_j-jets at p (Definition 2.4). □

To prove Proposition 4.2, for a variety X of dimension d over $k = \mathbb{C}(C)$, we blow up a k-rational smooth point of X, whose exceptional divisor is $\mathbb{P}^{d-1}_k$. The exceptional divisor becomes the marked divisor E_0 in the classification of [LZ18, Lemma 13], and for each type in the classification, the k-rationality is deduced from the k-divisor $E_0 \cong \mathbb{P}^{d-1}_k$.

Proposition 4.2. *Let C be a smooth projective curve, $k := \mathbb{C}(C)$, and $\bar{k}$ an algebraic closure of k. Let X be a normal projective variety over k of dimension $d \geq 1$ such that $X_{\bar{k}}$ is a klt variety, $-K_X$ is ample, and $((-K_{X_{\bar{k}}})^d) > d^d$. Assume that there exists a k-rational point x in the smooth locus of X such that $\epsilon(-K_{X_{\bar{k}}}, x_{\bar{k}}) \geq d$, where $x_{\bar{k}}$ is the point of $X_{\bar{k}}$ over x. Then X is birational over k to $\mathbb{P}^d_k$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the proposition when $d = 1$ or $\epsilon(-K_{X_{\bar{k}}}, x_{\bar{k}}) > d$.

Canonical sheaves of normal varieties commute with flat base change. We omit the verification, which reduces to [Har77, Proposition II.8.10] on the smooth locus. Thus $-K_{X_{\bar{k}}} = (-K_X)_{\bar{k}}$ is ample, and $X_{\bar{k}}$ is a $\mathbb{Q}$-Fano variety. By [Ser79, Chapter X, §6, Exercise 1], a variety over k which is isomorphic to $\mathbb{P}^d$ over $\bar{k}$ and has a k-rational point is isomorphic to $\mathbb{P}^d_k$.

Suppose that $d = 1$. Then X is a regular curve over k, hence smooth over k as $\operatorname{char} k = 0$, and $X_{\bar{k}}$ is a smooth projective curve with $\deg K_{X_{\bar{k}}} < 0$. Thus $X_{\bar{k}} \cong \mathbb{P}^1$. Since X has the k-rational point x, $X \cong \mathbb{P}^1_k$.

Suppose that $\epsilon(-K_{X_{\bar{k}}}, x_{\bar{k}}) > d$. Since $x_{\bar{k}}$ is a smooth point of the $\mathbb{Q}$-Fano variety $X_{\bar{k}}$, $X_{\bar{k}} \cong \mathbb{P}^d$ by [LZ18, Theorem 2]. Since X has the k-rational point x, $X \cong \mathbb{P}^d_k$.

From now on, we assume that $d \geq 2$ and $\epsilon(-K_{X_{\bar{k}}}, x_{\bar{k}}) = d$.

Step 2. In this step we construct a birational morphism $g : \hat{X} \to Y$ over k, where $\hat{X}$ is the blow-up of X at x, and a prime divisor $E_0 \subset Y$ such that $E_0 \cong \mathbb{P}^{d-1}_k$, and we list the possible pairs $(Y_{\bar{k}}, (E_0)_{\bar{k}})$.

Let $\pi : \hat{X} \to X$ be the blow-up of X at x and E the exceptional divisor of π. Since x is a k-rational smooth point of X, $E \cong \mathbb{P}^{d-1}_k$, and $\pi_{\bar{k}}$ is the blow-up of $X_{\bar{k}}$ at $x_{\bar{k}}$ with exceptional divisor $E_{\bar{k}}$. Let

$$D := \pi^*(-K_X) - dE.$$

Since the nef cone is closed, $D_{\bar{k}}$ is nef by Definition 2.7. Since $(\pi^*_{\bar{k}}(-K_{X_{\bar{k}}})^{d-i} \cdot E^i_{\bar{k}}) = 0$ for $0 < i < d$ and $(E^d_{\bar{k}}) = (-1)^{d-1}$, we have

$$(D^d_{\bar{k}}) = ((-K_{X_{\bar{k}}})^d) - d^d > 0.$$

Thus $D_{\bar{k}}$ is big [Laz04, Theorem 2.2.16]. Since $x_{\bar{k}}$ is a smooth point of $X_{\bar{k}}$, we have [Har77, Exercise II.8.5(b)]

$$K_{\hat{X}_{\bar{k}}} = \pi^*_{\bar{k}} K_{X_{\bar{k}}} + (d-1)E_{\bar{k}},$$

so

$$D_{\bar{k}} - K_{\hat{X}_{\bar{k}}} = \frac{2d-1}{d} D_{\bar{k}} + \frac{1}{d}\pi^*_{\bar{k}}(-K_{X_{\bar{k}}}).$$

Since $D_{\bar{k}}$ is nef and big and $\pi^*_{\bar{k}}(-K_{X_{\bar{k}}})$ is nef, $D_{\bar{k}} - K_{\hat{X}_{\bar{k}}}$ is nef and big. Since $\hat{X}_{\bar{k}}$ is smooth near $E_{\bar{k}}$ and isomorphic to $X_{\bar{k}}$ outside $E_{\bar{k}}$, $\hat{X}_{\bar{k}}$ is klt. Thus the base point free theorem [KM98, Theorem 3.3], applied to a Cartier multiple of $D_{\bar{k}}$, implies that $D_{\bar{k}}$ is semi-ample. Let m be a positive integer such that mD is Cartier and $|mD_{\bar{k}}|$ is base point free. Since

$$H^0(\hat{X}_{\bar{k}}, lmD_{\bar{k}}) = H^0(\hat{X}, lmD) \otimes_k \bar{k}$$

for any positive integer l [Har77, Proposition III.9.3], $|mD|$ is base point free. Let

$$Y := \operatorname{Proj} \bigoplus_{l \geq 0} H^0(\hat{X}, lmD)$$

and let $g : \hat{X} \to Y$ be the induced morphism. Then Y is a normal projective variety over k, $D = g^*A$ for an ample $\mathbb{Q}$-Cartier divisor A on Y, and $g_{\bar{k}} : \hat{X}_{\bar{k}} \to Y_{\bar{k}}$ is the morphism induced by $mD_{\bar{k}}$, with

$$Y_{\bar{k}} = \operatorname{Proj} \bigoplus_{l \geq 0} H^0(\hat{X}_{\bar{k}}, lmD_{\bar{k}}).$$

We omit the verification, which follows from [Har77, Proposition III.9.3]. Since $D_{\bar{k}}$ is big, $g_{\bar{k}}$ is birational, and thus g is birational.

By [LZ18, Lemma 11], $g_{\bar{k}}|_{E_{\bar{k}}}$ is a closed immersion, $g(E)_{\bar{k}} \cong \mathbb{P}^{d-1}$ is a nef Cartier divisor contained in $(Y_{\bar{k}})_{\mathrm{reg}}$, $-(K_{Y_{\bar{k}}} + g(E)_{\bar{k}})$ is ample, and $Y_{\bar{k}}$ is a $\mathbb{Q}$-Fano variety. Let $E_0 := g(E)$. By [Gro65, Proposition 2.7.1], $g|_E : E \to Y$ is a closed immersion, so $E_0 \cong E \cong \mathbb{P}^{d-1}_k$. Moreover, E_0 is a Cartier divisor contained in Y_{reg} and $K_{Y_{\bar{k}}} = (K_Y)_{\bar{k}}$. We omit the verification: in characteristic zero smoothness can be checked over $\bar{k}$, invertibility of an ideal sheaf descends along the faithfully flat morphism $Y_{\bar{k}} \to Y$, and canonical sheaves commute with flat base change. Since a multiple of $-(K_Y + E_0)$ is Cartier and the base change of $-(K_Y + E_0)$ to $\bar{k}$ is ample, $-(K_Y + E_0)$ is ample by [Gro65, Corollary 2.7.2].

By [LZ18, Lemma 13], the pair $(Y_{\bar{k}}, (E_0)_{\bar{k}})$ is isomorphic to one of the following pairs.

(I) $(\mathbb{P}(1^d, a), \{w = 0\})$ for an integer $a \geq 1$, where $\mathbb{P}(1^d, a)$ is the weighted projective space with coordinates $z_1, \ldots, z_d$ of weight 1 and a coordinate w of weight a.

(II) the pair

$$(\mathbb{P}_{\mathbb{P}^{d-1}}(\mathcal{O} \oplus \mathcal{O}(-e)), E')$$

for an integer $e \geq 0$, where E' is a section of the projection

$$\mathbb{P}_{\mathbb{P}^{d-1}}(\mathcal{O} \oplus \mathcal{O}(-e)) \to \mathbb{P}^{d-1}.$$

(III) $d = 2$, and $(\mathbb{P}^2, E')$, where E' is a smooth conic.

Since X, $\hat{X}$, and Y are birational over k, it suffices to show that Y is birational over k to $\mathbb{P}^d_k$.

Step 3. In this step we show that Y is birational over k to $\mathbb{P}^d_k$ in case (I).

Suppose that $a = 1$. Then $Y_{\bar{k}} \cong \mathbb{P}^d$ and $\mathcal{O}_Y(E_0)_{\bar{k}} \cong \mathcal{O}_{\mathbb{P}^d}(1)$. By [Har77, Proposition III.9.3], $h^0(Y, \mathcal{O}_Y(E_0)) = d + 1$, $|E_0|$ is base point free, and the morphism $Y \to \mathbb{P}^d_k$ defined by $|E_0|$ becomes the isomorphism defined by $|\mathcal{O}_{\mathbb{P}^d}(1)|$ after base change to $\bar{k}$. Thus $Y \cong \mathbb{P}^d_k$ [Gro65, Proposition 2.7.1].

Suppose that $a \geq 2$. The space

$$H^0(Y_{\bar{k}}, \mathcal{O}_Y(E_0)_{\bar{k}}) = H^0(\mathbb{P}(1^d, a), \mathcal{O}(a))$$

is spanned by w and the monomials of degree a in $z_1, \ldots, z_d$. The monomials of degree a in $z_1, \ldots, z_d$ and w generate the a-th Veronese subring of $\bar{k}[z_1, \ldots, z_d, w]$, which is the polynomial ring in w over the a-th Veronese subring of $\bar{k}[z_1, \ldots, z_d]$. Thus, with $N + 1 := h^0(Y, \mathcal{O}_Y(E_0))$, the complete linear system $|\mathcal{O}_Y(E_0)_{\bar{k}}|$ embeds $Y_{\bar{k}}$ into $\mathbb{P}^N_{\bar{k}}$ as the projective cone over the a-th Veronese embedding of $(E_0)_{\bar{k}} = \{w = 0\}$, whose vertex is the image of the unique singular point $[0 : \cdots : 0 : 1]$ of $Y_{\bar{k}}$. By [Har77, Proposition III.9.3] and [Gro65, Proposition 2.7.1], $|E_0|$ defines a closed immersion $Y \to \mathbb{P}^N_k$, and we regard Y as a closed subvariety of $\mathbb{P}^N_k$. Since $(Y_{\mathrm{sing}})_{\bar{k}} = (Y_{\bar{k}})_{\mathrm{sing}}$ is one reduced point, Y_{sing} is a k-rational point v. Let $H_0 \subset \mathbb{P}^N_k$ be the hyperplane corresponding to a section of $\mathcal{O}_Y(E_0)$ with divisor E_0. Then $Y \cap H_0 = E_0$ and $v \notin H_0$. Choose homogeneous coordinates $u, y_1, \ldots, y_N$ on $\mathbb{P}^N_k$ such that $H_0 = \{u = 0\}$ and $v = [1 : 0 : \cdots : 0]$, and let $I \subset k[y_1, \ldots, y_N]$ be the homogeneous ideal of $E_0 \subset H_0$. The closed subscheme of $\mathbb{P}^N_k$ defined by $I \cdot k[u, y_1, \ldots, y_N]$ is the projective cone over E_0 with vertex v. After base change to $\bar{k}$, it becomes $Y_{\bar{k}}$, so it is equal to Y. We may assume that y_1 does not vanish identically on E_0. Then the morphism

$$Y \cap \{y_1 \neq 0\} \to (E_0 \cap \{y_1 \neq 0\}) \times \mathbb{A}^1_k$$

sending $[u : y_1 : \cdots : y_N]$ to $([y_1 : \cdots : y_N], u/y_1)$ is an isomorphism. Since $E_0 \cong \mathbb{P}^{d-1}_k$, Y is birational over k to $\mathbb{P}^d_k$.

Step 4. In this step we show that Y is birational over k to $\mathbb{P}^d_k$ in case (II).

Let $\rho : Y_{\bar{k}} \to \mathbb{P}^{d-1}$ be the projection, H the pullback of a hyperplane under ρ, and ξ a divisor with

$$\mathcal{O}_{Y_{\bar{k}}}(\xi) \cong \mathcal{O}_{\mathbb{P}(\mathcal{O} \oplus \mathcal{O}(-e))}(1).$$

Then $\mathrm{Pic}(Y_{\bar{k}}) = \mathbb{Z}\xi \oplus \mathbb{Z}H$ [Har77, Exercise II.7.9(a)]. By [Har77, Exercise III.8.4(b)],

$$\omega_{Y_{\bar{k}}/\mathbb{P}^{d-1}} \cong \mathcal{O}_{Y_{\bar{k}}}(-2\xi - eH),$$

and by the exact sequence of differentials of the smooth morphism ρ [Har77, Propositions II.8.11 and III.10.4],

$$\omega_{Y_{\bar{k}}} \cong \omega_{Y_{\bar{k}}/\mathbb{P}^{d-1}} \otimes \rho^* \omega_{\mathbb{P}^{d-1}}.$$

Thus

$$K_{Y_{\bar{k}}} \sim -2\xi - (d + e)H.$$

By [Har77, Proposition II.7.12], a section E' of ρ corresponds to a surjection $\mathcal{O} \oplus \mathcal{O}(-e) \to \mathcal{O}(b)$ for an integer b, and then $\mathcal{O}_{Y_{\bar{k}}}(\xi)|_{E'} \cong \mathcal{O}(b)$ via $\rho|_{E'}$. Let S be the section corresponding to the projection $\mathcal{O} \oplus \mathcal{O}(-e) \to \mathcal{O}(-e)$. Then

$$-K_{Y_{\bar{k}}}|_S \cong \mathcal{O}_{\mathbb{P}^{d-1}}(-2e + d + e) = \mathcal{O}_{\mathbb{P}^{d-1}}(d - e).$$

Since $-K_{Y_{\bar{k}}}$ is ample by **Step 2**, $e < d$.

Suppose that $d \geq 3$. Let $\mathcal{O} \oplus \mathcal{O}(-e) \to \mathcal{O}(b)$ be the surjection corresponding to $(E_0)_{\bar{k}}$. It is given by a section of $\mathcal{O}(b)$ and a section of $\mathcal{O}(b + e)$ without common zeros. If $b > 0$, then both sections are non-zero since a sheaf $\mathcal{O}(b')$ with $b' > 0$ has no nowhere vanishing section, and their zero loci are hypersurfaces in $\mathbb{P}^{d-1}$ which intersect since $d - 1 \geq 2$, a contradiction. If $b < 0$, then

the section of $\mathcal{O}(b)$ is zero, so the section of $\mathcal{O}(b+e)$ has no zeros, and $b=-e$. Thus $b\in\{0,-e\}$. The kernel of the surjection is isomorphic to $\mathcal{O}(-e-b)$. By the relative Euler sequence [Har77, Exercise III.8.4(b)], the restriction of the relative tangent sheaf of ρ to $(E_0)_{\bar{k}}$ is isomorphic to $\mathcal{O}(b)\otimes\mathcal{O}(-e-b)^{\vee}=\mathcal{O}(2b+e)$. Since the normal sheaf of a section of the smooth morphism ρ is the restriction of the relative tangent sheaf of ρ, we have

$$\mathcal{O}_{(E_0)_{\bar{k}}}((E_0)_{\bar{k}})\cong\mathcal{O}(2b+e).$$

Since E_0 is nef, $2b+e\geq 0$. If $b=-e$, then $2b+e=-e\geq 0$ forces $e=0$, so $b=0$. Thus $b=0$ in all cases. Write $(E_0)_{\bar{k}}\sim\xi+\beta H$ for an integer β. The coefficient of ξ is 1 since $((E_0)_{\bar{k}}\cdot l)=1$, $(\xi\cdot l)=1$, and $(H\cdot l)=0$ for a fiber l of ρ. Restricting to $(E_0)_{\bar{k}}$, we obtain $\mathcal{O}(e)\cong\mathcal{O}(b+\beta)=\mathcal{O}(\beta)$, so $\beta=e$. Thus

$$-K_{Y_{\bar{k}}}\sim 2(E_0)_{\bar{k}}+(d-e)H.$$

Suppose that $d=2$. Then $Y_{\bar{k}}$ is a ruled surface over $\mathbb{P}^1$ with invariant e, the section S defined above satisfies $(S^2)=-e$, and we let l be a fiber of ρ, so that $H\sim l$. By [Har77, Propositions V.2.3 and V.2.9] and [Har77, Corollary V.2.11], $\mathrm{Pic}(Y_{\bar{k}})=\mathbb{Z}S\oplus\mathbb{Z}l$, and

$$(S^2)=-e,\quad (S\cdot l)=1,\quad (l^2)=0,\quad K_{Y_{\bar{k}}}\sim -2S-(2+e)l.$$

Since $e<2$, $e\in\{0,1\}$. Since $((E_0)_{\bar{k}}\cdot l)=1$, we have $(E_0)_{\bar{k}}\sim S+\beta l$ for an integer β. Since E_0 is nef, $((E_0)_{\bar{k}}\cdot S)=\beta-e\geq 0$. Since $-(K_{Y_{\bar{k}}}+(E_0)_{\bar{k}})\sim S+(e+2-\beta)l$ is ample, $e+2-\beta>e$ by [Har77, Proposition V.2.20(b)], so $\beta<2$. Thus

$$(e,\beta)\in\{(0,0),(1,1),(0,1)\}.$$

If $\beta=e$, then $(E_0)_{\bar{k}}\sim S+el$, and $-K_{Y_{\bar{k}}}\sim 2(E_0)_{\bar{k}}+(2-e)H$.

Suppose that $d\geq 3$, or $d=2$ and $\beta=e$. Then $-K_{Y_{\bar{k}}}\sim 2(E_0)_{\bar{k}}+(d-e)H$ with $d-e>0$. Since $Y_{\bar{k}}$ is smooth, so is Y, and $L:=-K_Y-2E_0$ is a Cartier divisor such that $L_{\bar{k}}\sim(d-e)H$. The complete linear system $|(d-e)H|$ is base point free and defines the composition of ρ with the $(d-e)$-th Veronese embedding of $\mathbb{P}^{d-1}$. By [Har77, Proposition III.9.3], $|L|$ is base point free. Let $h:Y\to T$ be the morphism defined by $|L|$ onto the image T. Then $h_{\bar{k}}$ is the morphism defined by $|(d-e)H|$, so the fibers of $h_{\bar{k}}$ are the fibers of ρ, and $h_{\bar{k}}|_{(E_0)_{\bar{k}}}$ is an isomorphism onto $T_{\bar{k}}\cong\mathbb{P}^{d-1}$. By [Gro65, Proposition 2.7.1], $h|_{E_0}:E_0\to T$ is an isomorphism, so $T\cong\mathbb{P}^{d-1}_k$. Let η_T be the generic point of T. The generic fiber Y_{η_T} of h is a curve over $k(T)$ which is isomorphic to $\mathbb{P}^1$ over the algebraic closure of $k(T)$, and $E_0\cap Y_{\eta_T}$ is a $k(T)$-rational point of Y_{η_T} since $h|_{E_0}$ is an isomorphism. By [Ser79, Chapter X, §6, Exercise 1], $Y_{\eta_T}\cong\mathbb{P}^1_{k(T)}$. Thus Y is birational over k to $T\times\mathbb{P}^1_k$, hence to $\mathbb{P}^d_k$.

Suppose that $d=2$ and $(e,\beta)=(0,1)$. Then $Y_{\bar{k}}\cong\mathbb{P}^1\times\mathbb{P}^1$ and $(E_0)_{\bar{k}}$ is a divisor of bidegree $(1,1)$, so $\mathcal{O}_Y(E_0)_{\bar{k}}\cong\mathcal{O}(1,1)$, $h^0(Y,\mathcal{O}_Y(E_0))=4$, and the complete linear system $|\mathcal{O}(1,1)|$ embeds $Y_{\bar{k}}$ as a smooth quadric surface in $\mathbb{P}^3_{\bar{k}}$. By [Har77, Proposition III.9.3] and [Gro65, Proposition 2.7.1], $|E_0|$ defines a closed immersion of Y onto a smooth quadric surface in $\mathbb{P}^3_k$. Since $E_0\cong\mathbb{P}^1_k$, there exists a k-rational point $y_0\in E_0$. A general line through y_0 intersects Y at y_0 and at exactly one other point. Thus the projection from y_0 induces a birational map $Y\dashrightarrow\mathbb{P}^2_k$ over k.

Step 5. In this step we show that Y is birational over k to $\mathbb{P}^d_k$ in case (III), and conclude the proof.

In case (III), $d=2$ and $Y_{\bar{k}}\cong\mathbb{P}^2$ is smooth, so Y is smooth and $-K_Y-E_0$ is a Cartier divisor such that

$$\mathcal{O}_Y(-K_Y-E_0)_{\bar{k}}\cong\mathcal{O}_{\mathbb{P}^2}(3-2)=\mathcal{O}_{\mathbb{P}^2}(1).$$

Replacing $\mathcal{O}_Y(E_0)$ with $\mathcal{O}_Y(-K_Y-E_0)$, the argument of **Step 3** for $a=1$ implies that $Y\cong\mathbb{P}^2_k$.

Thus in any of the cases (I), (II), and (III), Y is birational over k to $\mathbb{P}^d_k$. Since X is birational over k to Y, X is birational over k to $\mathbb{P}^d_k$. □

5. Mori fiber spaces over curves

In this section, we prove Theorem 5.1, that is, the case of Theorem 1.2 in which the base of the Mori fiber space is a curve. Unlike the case when $\epsilon_{\mathrm{mov}}(-K_X) > d-1$, a general fiber of the Mori fiber space need not be a projective space, and the base can be an elliptic curve.

Theorem 5.1. *Let $d \geq 2$ be an integer, and let $\mathfrak{X}$ be the set of terminal projective varieties X of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d-1$ and there exists a Mori fiber space $f : X \to C$ onto a smooth projective curve C. Then $\mathfrak{X}$ is birationally bounded. Moreover, let $X \in \mathfrak{X}$, let $f : X \to C$ be such a Mori fiber space, and let F be a general fiber of f. If $((-K_F)^{d-1}) > (d-1)^{d-1}$, then X is birational over C to $C \times \mathbb{P}^{d-1}$.*

To bound the genus of C, we need, for any rational number $\lambda \in (0,1)$, a $\mathbb{Q}$-divisor $D_\lambda \geq 0$ with $D_\lambda \sim_{\mathbb{Q}} -\lambda K_X$ such that (X, D_λ) is klt over a non-empty open subset of C. When $F \cong \mathbb{P}^{d-1}$, [Zhu18, Corollary 5.2] deduces such a divisor from the bound of [Zhu18, Lemma 5.1] on multiplicities of the base ideal of $|-mK_X|$ along F, but when F is not a projective space, we bound from below the log canonical threshold of the base ideal of the image of $H^0(X, \mathcal{O}_X(-mK_X))$ in $H^0(F, \mathcal{O}_F(-mK_F))$, using the connectedness of non-klt loci [Zhu18, Lemma 2.12]. The lower bound uses a $\mathbb{Q}$-divisor $G \geq 0$ with $G \sim_{\mathbb{Q}} -K_F$ and $\mathrm{mult}_x G > d-1$ at some point x, which exists if $((-K_F)^{d-1}) > (d-1)^{d-1}$. When $((-K_F)^{d-1}) = (d-1)^{d-1}$, we do not construct G. Instead, $-K_X$ is nef and big by [Zhu18, Lemma 5.4], and a klt $\mathbb{Q}$-divisor $B \sim_{\mathbb{Q}} -K_X$ bounds the genus of C.

Proposition 5.2. *Let $f : X \to C$ be a Mori fiber space onto a smooth projective curve C such that X is terminal of dimension $d \geq 2$ and $\epsilon_{\mathrm{mov}}(-K_X) \geq d-1$. Let F be a general fiber of f. Then the genus of C is at most 1 and $((-K_F)^{d-1}) \geq (d-1)^{d-1}$. Moreover,*

(1) if $((-K_F)^{d-1}) = (d-1)^{d-1}$, then $-K_X$ is nef and big, and
(2) if $((-K_F)^{d-1}) > (d-1)^{d-1}$, then for any rational number $\lambda \in (0,1)$, there exists a $\mathbb{Q}$-divisor $D_\lambda \geq 0$ on X such that $D_\lambda \sim_{\mathbb{Q}} -\lambda K_X$ and (X, D_λ) is klt over a non-empty open subset of C.

Proof of Proposition 5.2. The proof is long, so we divide it into several steps.

Step 1. In this step, we choose F and a point $p \in F$, introduce the restricted linear systems W_m and the integers m_j and k_j, and prove that

$$(d-1)^{d-1} \leq \epsilon_{\mathrm{mov}}(-K_X)^{d-1} \leq ((-K_F)^{d-1}).$$

Fix a positive integer R such that $-RK_X$ is an f-very ample Cartier divisor. Since X is integral and f is a dominant morphism onto a smooth curve, f is flat [Har77, Proposition III.9.7]. Hence for closed points $c \in C$, the Hilbert polynomial of the fiber $X_c := f^{-1}(c)$ with respect to $\mathcal{O}_X(-RK_X)|_{X_c}$ does not depend on c [Har77, Theorem III.9.9]. We show that there exists a non-empty open subset $C^\circ \subset C$ such that for any closed point $c \in C^\circ$, X_c is a normal terminal variety and $K_{X_c} = K_X|_{X_c}$. Since X is terminal, X has rational singularities [KM98, Theorem 5.22], so X is Cohen–Macaulay [KM98, Theorem 5.10], and $\mathrm{codim}(X_{\mathrm{sing}}, X) \geq 3$ [KM98, Corollary 5.18]. Let $c \in C$ be a general closed point. By generic smoothness [Har77, Corollary III.10.7] applied to $f|_{X_{\mathrm{reg}}}$, $X_c \cap X_{\mathrm{reg}}$ is smooth. The Cartier divisor X_c of X is Cohen–Macaulay, and $\mathrm{codim}(X_c \cap X_{\mathrm{sing}}, X_c) \geq 2$ since $\mathrm{codim}(X_{\mathrm{sing}}, X) \geq 3$. Thus X_c is normal by Serre's criterion, and X_c is connected since f is a contraction. Since $X_c = f^*c$ is a Cartier divisor and $\mathcal{O}_{X_c}(X_c) \cong \mathcal{O}_{X_c}$, adjunction implies $K_{X_c} = K_X|_{X_c}$. Let A_0 be a very ample divisor on C. A general member of the free linear system $f^*|A_0|$ is the disjoint union of $\deg A_0$ general fibers of f, so X_c is terminal by [KM98, Lemma 5.17(1)]. For $c \in C^\circ$, the number $((-K_{X_c})^{d-1})$ is $(d-1)!R^{1-d}$ times the leading coefficient of the Hilbert polynomial of X_c, so it does not depend on $c \in C^\circ$. Since the conclusions of Proposition 5.2 depend on F only through $((-K_F)^{d-1})$, it suffices to prove them for $F = X_c$, where $c \in C^\circ$ avoids the countable subset $\Sigma \subset C$ constructed below.

By Lemma 2.6(3), there exists a smooth point $y_0 \in X$ such that

$$\epsilon_{\mathrm{mov}}(-K_X, y_0) = \epsilon_{\mathrm{mov}}(-K_X).$$

By Lemma 2.6(2) with $I = R$, there exist positive integers $m_1 < m_2 < \cdots$ divisible by R such that

$$\lim_{j\to+\infty} \frac{s(-m_jK_X, y_0)}{m_j} = \epsilon_{\mathrm{mov}}(-K_X) \geq d-1.$$

Dropping finitely many j, we may assume that $s(-m_jK_X, y_0) \geq 0$ for any j. For any j, let U_j be the set of smooth points $y \in X$ such that $s(-m_jK_X, y) \geq s(-m_jK_X, y_0)$. Then U_j is a non-empty open subset of X by Lemma 2.6(3). Since $X\backslash U_j$ is a proper closed subset of X, it contains only finitely many fibers of f. Thus there exists a countable subset $\Sigma \subset C$ such that $X_c \cap U_j \neq \emptyset$ for any j and any closed point $c \in C\backslash\Sigma$.

Let $F := X_c$ for a closed point $c \in C^\circ\backslash\Sigma$. Then F is a normal terminal variety, $K_F = K_X|_F$, and $-K_F$ is ample since $-K_X$ is f-ample. Since $F \cap U_j$ is a non-empty open subset of F for any j, a very general point $p \in F$ satisfies $p \in F_{\mathrm{reg}}$ and $p \in U_j$ for any j. We fix such a point p.

For any positive integer m divisible by R, let W_m be the image of the restriction map

$$H^0(X, \mathcal{O}_X(-mK_X)) \to H^0(F, \mathcal{O}_F(-mK_F)),$$

and let $\mathfrak{b}_m \subset \mathcal{O}_F$ be the base ideal of W_m. We write $H \in W_m$ if H is the divisor of a non-zero element of W_m. By Lemma 2.6(5) and $p \in U_j$,

$$s(W_{m_j}, p) \geq s(-m_jK_X, p) \geq s(-m_jK_X, y_0) \geq 0.$$

In particular, p is not a base point of W_{m_j}, that is, $\mathfrak{b}_{m_j}\mathcal{O}_{F,p} = \mathcal{O}_{F,p}$. Let

$$k_j := \min\{s(W_{m_j}, p), (d-1)m_j\}.$$

Since

$$\min\{s(-m_jK_X, y_0), (d-1)m_j\} \leq k_j \leq (d-1)m_j$$

and $\epsilon_{\mathrm{mov}}(-K_X) \geq d-1$, we have

$$\lim_{j\to+\infty} \frac{k_j}{m_j} = d-1.$$

Dropping finitely many j, we may assume that $k_j \geq 1$ for any j.

Since W_{m_j} generates $s(-m_jK_X, y_0)$-jets at the smooth point p of F and $\dim F = d-1$,

$$h^0(F, \mathcal{O}_F(-m_jK_F)) \geq \dim W_{m_j} \geq \dim_{\mathbb{C}} \mathcal{O}_{F,p}/\mathfrak{m}_p^{s(-m_jK_X,y_0)+1} \geq \frac{s(-m_jK_X, y_0)^{d-1}}{(d-1)!}.$$

Since $-RK_F$ is an ample Cartier divisor, [Laz04, Corollary 1.4.41] implies

$$h^0(F, \mathcal{O}_F(-m_jK_F)) = \frac{((-K_F)^{d-1})}{(d-1)!}m_j^{d-1} + O(m_j^{d-2}).$$

Dividing by $m_j^{d-1}/(d-1)!$ and letting $j \to +\infty$, we obtain

$$((-K_F)^{d-1}) \geq \epsilon_{\mathrm{mov}}(-K_X)^{d-1} \geq (d-1)^{d-1}.$$

Step 2. In this step, we assume that $((-K_F)^{d-1}) = (d-1)^{d-1}$, and we prove (1) and that the genus $g(C)$ of C is at most 1.

By **Step 1** and the hypothesis $\epsilon_{\mathrm{mov}}(-K_X) \geq d-1$, we have $\epsilon_{\mathrm{mov}}(-K_X) = d-1$. Since $-K_F$ is ample,

$$\mathrm{vol}(-K_F) = ((-K_F)^{d-1}) = \epsilon_{\mathrm{mov}}(-K_X)^{\dim F}.$$

Since f is a Mori fiber space onto a curve and $\mathrm{vol}(-K_F) = \epsilon_{\mathrm{mov}}(-K_X)^{\dim F}$, [Zhu18, Lemma 5.4] implies that $-K_X$ is nef and big. This proves (1).

Since X is klt and $-RK_X - K_X = -(R+1)K_X$ is nef and big, the base point free theorem [KM98, Theorem 3.3] implies that $|-qRK_X|$ is base point free for some integer $q \geq 2$. Let D be a general member of $|-qRK_X|$ and $B := \frac{1}{qR}D$. By [KM98, Lemma 5.17(2)], (X, D) is lc. Since

X is klt and $\frac{1}{qR} < 1$, (X, B) is klt. Since $K_X + B \sim_{\mathbb{Q}} 0$, Lemma 2.18 applied to B and the zero divisor on C implies $\deg(-K_C) \geq 0$, that is, $2g(C) - 2 \leq 0$. Thus $g(C) \leq 1$.

Step 3. In this step, we assume that $((-K_F)^{d-1}) > (d-1)^{d-1}$, and we prove that for any rational number $\lambda \in (0, 1)$,

$$m_j \operatorname{lct}(F; \mathfrak{b}_{m_j}) > \lambda$$

for any sufficiently large j.

Fix a rational number $\lambda \in (0, 1)$. We first construct G. Fix a real number u such that $d - 1 < u < ((-K_F)^{d-1})^{1/(d-1)}$. For positive integers l,

$$\dim_{\mathbb{C}} \mathcal{O}_{F,p}/\mathfrak{m}_p^{\lfloor ul \rfloor} = \binom{\lfloor ul \rfloor + d - 2}{d-1} = \frac{u^{d-1}}{(d-1)!} l^{d-1} + O(l^{d-2}).$$

Since $u^{d-1} < ((-K_F)^{d-1})$, [Laz04, Corollary 1.4.41] implies that

$$h^0(F, \mathcal{O}_F(-lK_F)) > \dim_{\mathbb{C}} \mathcal{O}_{F,p}/\mathfrak{m}_p^{\lfloor ul \rfloor}$$

for any sufficiently large integer l divisible by R. For such l, there exists a non-zero element $\tau \in H^0(F, \mathcal{O}_F(-lK_F))$ whose image in

$$\mathcal{O}_F(-lK_F) \otimes \mathcal{O}_F/\mathfrak{m}_p^{\lfloor ul \rfloor}$$

is zero. Let G be $\frac{1}{l}$ times the divisor of τ. Then $G \geq 0$, $G \sim_{\mathbb{Q}} -K_F$, and $\operatorname{mult}_p G \geq \lfloor ul \rfloor / l$. Since $u > d - 1$, we have $\operatorname{mult}_p G > d - 1$ for l sufficiently large, and we fix such l and G. Let $U \subset F_{\mathrm{reg}}$ be an open neighborhood of p such that $\operatorname{mult}_y G \leq \operatorname{mult}_p G$ for any $y \in U$, by upper semicontinuity of multiplicity. Then $\operatorname{mult}_y(G/(\operatorname{mult}_p G + 1)) < 1$ for any $y \in U$, so $(U, G/(\operatorname{mult}_p G + 1))$ is klt by [Zhu18, Lemma 2.13].

Next, we fix rational numbers μ and δ such that $\lambda < \mu < 1$, $0 < \delta < 1$, and

$$\mu(1 - \delta) > \lambda, \quad \delta \operatorname{mult}_p G < 1, \quad \mu\big((1 - \delta) + \delta(\operatorname{mult}_p G + 1)\big) < 1.$$

Such μ and δ exist: fix a rational number $\mu \in (\lambda, 1)$. Then the three inequalities hold strictly for $\delta = 0$, hence for any sufficiently small rational number $\delta > 0$.

Let E_p be the exceptional divisor of the blow-up of F at p if $d \geq 3$, and let E_p be the prime divisor p on F if $d = 2$. For any $\mathbb{Q}$-divisor $D \geq 0$ on U, we have $a(E_p, F, D) = d - 1 - \operatorname{mult}_p D$. If $d \geq 3$, the equality is [KM98, Lemma 2.29], where the discrepancy of [KM98, Definition 2.25] is our log discrepancy minus one, and if $d = 2$, the equality follows from Definition 2.1.

For any j, fix local coordinates $z_1, \ldots, z_{d-1}$ of F at p and a local generator of $\mathcal{O}_F(-m_j K_F)$ at p. Since $k_j \leq s(W_{m_j}, p)$, there exists $H_j \in W_{m_j}$ whose local equation at p is congruent to $z_1^{k_j} + \cdots + z_{d-1}^{k_j}$ modulo $\mathfrak{m}_p^{k_j+1}$. Thus $\operatorname{mult}_p H_j = k_j$. We show that there exists an open neighborhood $\Omega_j \subset U$ of p such that $\operatorname{mult}_y H_j \leq 1$ for any $y \in \Omega_j \backslash \{p\}$ and $(\Omega_j, H_j/m_j)$ is lc. If $d = 2$, then F is a curve, and we take Ω_j such that $\operatorname{Supp} H_j \cap \Omega_j \subset \{p\}$. In this case, $(\Omega_j, H_j/m_j)$ is lc since $\operatorname{mult}_p(H_j/m_j) = k_j/m_j \leq 1$ by **Step 1**. If $d \geq 3$, then the projectivized tangent cone of H_j at p is the smooth Fermat hypersurface $(z_1^{k_j} + \cdots + z_{d-1}^{k_j} = 0) \subset \mathbb{P}^{d-2}$. Hence the strict transform of H_j on the blow-up of F at p is smooth along E_p and intersects E_p transversally. Thus there exists an open neighborhood $\Omega_j \subset U$ of p such that H_j is smooth on $\Omega_j \backslash \{p\}$ and the blow-up of Ω_j at p is a log resolution of $(\Omega_j, H_j|_{\Omega_j})$. The coefficient of the strict transform of H_j is $1/m_j \leq 1$, and $a(E_p, F, H_j/m_j) = d - 1 - k_j/m_j \geq 0$ since $k_j \leq (d-1)m_j$. Thus $(\Omega_j, H_j/m_j)$ is lc.

Let

$$\Delta_j := (1 - \delta)\frac{H_j}{m_j} + \delta G.$$

Then $\Delta_j \geq 0$ and $\Delta_j \sim_{\mathbb{Q}} -K_F$. Let E be a prime divisor over F whose center on F intersects Ω_j. Since $(\Omega_j, H_j/m_j)$ is lc and $(U, G/(\operatorname{mult}_p G + 1))$ is klt,

$$\frac{1}{m_j} \operatorname{mult}_E H_j \leq a(E, F, 0) \quad \text{and} \quad \operatorname{mult}_E G < (\operatorname{mult}_p G + 1) a(E, F, 0).$$

Hence

$$\operatorname{mult}_E(\mu\Delta_j) < \mu\big((1-\delta)+\delta(\operatorname{mult}_p G+1)\big)a(E,F,0) < a(E,F,0),$$

so $(\Omega_j, \mu\Delta_j)$ is klt. Let $c_j := \operatorname{lct}(F \ni p; \Delta_j)$. Then $c_j > \mu$ (for any open subset $V \subset F$, klt and lc of $(V, t\Delta_j|_V)$ are tested on one log resolution of $(V, \Delta_j|_V)$, so $(V, t\Delta_j|_V)$ is klt if and only if $t < \operatorname{lct}(V; \Delta_j|_V)$, see [KM98, Corollary 2.32(2)], and for $V = \Omega_j$ we obtain $\mu < \operatorname{lct}(\Omega_j; \Delta_j|_{\Omega_j}) \le c_j$). Since $a(E_p, F, t\Delta_j) = d - 1 - t\operatorname{mult}_p \Delta_j$ for any $t \ge 0$, we have $c_j \le (d-1)/\operatorname{mult}_p \Delta_j$, where

$$\operatorname{mult}_p \Delta_j = (1-\delta)\frac{k_j}{m_j} + \delta \operatorname{mult}_p G \to (1-\delta)(d-1) + \delta \operatorname{mult}_p G > d-1 \quad (j \to +\infty).$$

Since $\delta \operatorname{mult}_p G < 1$, there exists j_0 such that for any $j \ge j_0$,

$$\operatorname{mult}_p \Delta_j > d-1 \quad \text{and} \quad \frac{1-\delta}{m_j} + \delta \operatorname{mult}_p G < 1.$$

From now on, let $j \ge j_0$. Then $\mu < c_j < 1$.

We show that p is an isolated point of $\operatorname{Nklt}(F, c_j\Delta_j)$. For any open neighborhood $U'' \subset \Omega_j$ of p, the pair $(U'', c_j\Delta_j|_{U''})$ is not klt, since otherwise $\operatorname{lct}(U''; \Delta_j|_{U''}) > c_j$. Since $\operatorname{Nklt}(F, c_j\Delta_j)$ is closed, $p \in \operatorname{Nklt}(F, c_j\Delta_j)$. For any $y \in \Omega_j \backslash \{p\}$, we have $\operatorname{mult}_y H_j \le 1$ and $\operatorname{mult}_y G \le \operatorname{mult}_p G$, so

$$\operatorname{mult}_y(c_j\Delta_j) \le \operatorname{mult}_y \Delta_j \le \frac{1-\delta}{m_j} + \delta \operatorname{mult}_p G < 1.$$

By [Zhu18, Lemma 2.13], $(F, c_j\Delta_j)$ is klt near y. Thus $\operatorname{Nklt}(F, c_j\Delta_j) \cap \Omega_j = \{p\}$.

Suppose that $m_j \operatorname{lct}(F; \mathfrak{b}_{m_j}) \le \lambda$. Since $c_j(1-\delta) > \mu(1-\delta) > \lambda$, we have

$$\operatorname{lct}(F; \mathfrak{b}_{m_j}) < \frac{c_j(1-\delta)}{m_j}.$$

By Definition 2.3, there exists a prime divisor E over F such that $\operatorname{mult}_E \mathfrak{b}_{m_j} > 0$ and

$$a(E,F,0) < \frac{c_j(1-\delta)}{m_j} \operatorname{mult}_E \mathfrak{b}_{m_j}.$$

Since $H_j \in W_{m_j}$, we have $\operatorname{mult}_E H_j \ge \operatorname{mult}_E \mathfrak{b}_{m_j}$. Since $G \ge 0$,

$$c_j \operatorname{mult}_E \Delta_j \ge \frac{c_j(1-\delta)}{m_j} \operatorname{mult}_E H_j \ge \frac{c_j(1-\delta)}{m_j} \operatorname{mult}_E \mathfrak{b}_{m_j} > a(E,F,0),$$

that is, $a(E, F, c_j\Delta_j) < 0$. Thus the center V_E of E on F is contained in $\operatorname{Nklt}(F, c_j\Delta_j)$. Since $\operatorname{mult}_E \mathfrak{b}_{m_j} > 0$, V_E is contained in the zero locus of $\mathfrak{b}_{m_j}$, so $p \notin V_E$ by **Step 1**. Hence $\operatorname{Nklt}(F, c_j\Delta_j)$ is the disjoint union of the closed subsets $\{p\}$ and $\operatorname{Nklt}(F, c_j\Delta_j)\backslash\Omega_j$, where $V_E \subset \operatorname{Nklt}(F, c_j\Delta_j)\backslash\Omega_j$ and $V_E \ne \emptyset$. Thus $\operatorname{Nklt}(F, c_j\Delta_j)$ is not connected. On the other hand, since $c_j < 1$,

$$-(K_F + c_j\Delta_j) \sim_{\mathbb{Q}} (1-c_j)(-K_F)$$

is ample, so $\operatorname{Nklt}(F, c_j\Delta_j)$ is connected by [Zhu18, Lemma 2.12], a contradiction. Thus $m_j \operatorname{lct}(F; \mathfrak{b}_{m_j}) > \lambda$ for any $j \ge j_0$.

Step 4. In this step, we assume that $((-K_F)^{d-1}) > (d-1)^{d-1}$, and we prove (2).

Fix a rational number $\lambda \in (0,1)$. By **Step 3**, there exists j such that $m := m_j$ satisfies $\lambda/m < \operatorname{lct}(F; \mathfrak{b}_m)$. Let $\varphi : F' \to F$ be a log resolution of $(F, \mathfrak{b}_m)$, and let $\Phi \ge 0$ be the divisor on F' such that $\mathfrak{b}_m \mathcal{O}_{F'} = \mathcal{O}_{F'}(-\Phi)$. Then $\varphi^* H = \Phi + M_H$ for any $H \in W_m$, where M_H runs over the members of a free linear system on F'. Let E be a prime divisor on F' which is φ-exceptional or a component of Φ. If $\operatorname{mult}_E \mathfrak{b}_m > 0$, then

$$\frac{\lambda}{m} \operatorname{mult}_E \mathfrak{b}_m < \operatorname{lct}(F; \mathfrak{b}_m) \operatorname{mult}_E \mathfrak{b}_m \le a(E,F,0),$$

and if $\operatorname{mult}_E \mathfrak{b}_m = 0$, then $a(E, F, 0) > 0$ since F is klt. Thus for a general $H \in W_m$,

$$a\big(E, F, \tfrac{\lambda}{m}H\big) = a(E, F, 0) - \frac{\lambda}{m}\operatorname{mult}_E \mathfrak{b}_m > 0.$$

Since the linear system $\{M_H\}$ on F' is free, by Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2] applied to F' and to each stratum of the snc divisor $\operatorname{Exc}(\varphi) \cup \operatorname{Supp} \Phi$, for a general $H \in W_m$ the divisor M_H is smooth, has no common component with $\operatorname{Exc}(\varphi) \cup \operatorname{Supp} \Phi$, and $\operatorname{Exc}(\varphi) \cup \operatorname{Supp} \Phi \cup \operatorname{Supp} M_H$ is snc. Thus φ is also a log resolution of $(F, \frac{\lambda}{m}H)$. All coefficients of the $\mathbb{Q}$-divisor B' with $K_{F'} + B' = \varphi^*(K_F + \frac{\lambda}{m}H)$ are less than 1 (the components of M_H have coefficient $\lambda/m < 1$, the φ-exceptional prime divisors E and the components E of Φ have coefficient $1 - a(E, F, \frac{\lambda}{m}H) < 1$, and all other coefficients are 0), so $(F, \frac{\lambda}{m}H)$ is klt by [KM98, Lemma 2.30 and Corollary 2.31(3)]. Thus $(F, \frac{\lambda}{m}H_F)$ is klt for a general $H_F \in W_m$.

Since W_m is the image of $H^0(X, \mathcal{O}_X(-mK_X))$, a general element $\sigma \in H^0(X, \mathcal{O}_X(-mK_X))$ has non-zero restriction to F, and the divisor H_F of $\sigma|_F$ satisfies that $(F, \frac{\lambda}{m}H_F)$ is klt. Let H_X be the divisor of σ. Then $F \not\subset \operatorname{Supp} H_X$ and $H_X|_F = H_F$. Let $D_\lambda := \frac{\lambda}{m}H_X$. Then $D_\lambda \geq 0$ and $D_\lambda \sim_{\mathbb{Q}} -\lambda K_X$. Since $F = f^*c$ is a normal Cartier divisor and $K_F = K_X|_F$,

$$(K_X + F + D_\lambda)|_F \sim_{\mathbb{Q}} K_F + \frac{\lambda}{m}H_F.$$

By inversion of adjunction [KM98, Theorem 5.50(1)], $(X, F + D_\lambda)$ is plt near F, so (X, D_λ) is klt near F. Since $\operatorname{Nklt}(X, D_\lambda)$ is closed and disjoint from F, and f is proper, $f(\operatorname{Nklt}(X, D_\lambda))$ is a closed subset of C which does not contain c. Thus (X, D_λ) is klt over the non-empty open subset $C \backslash f(\operatorname{Nklt}(X, D_\lambda))$ of C. This proves (2).

Step 5. In this step, we assume that $((-K_F)^{d-1}) > (d-1)^{d-1}$, and we prove that $g(C) \leq 1$.

Let M_C be an ample divisor on C. Recall from **Step 1** that $-RK_X$ is an f-very ample Cartier divisor. Thus $-RK_X + f^*(NM_C)$ is ample for some positive integer N [KM98, Proposition 1.45]. Let $A_C := \frac{N}{R}M_C$. Then

$$L := -K_X + f^*A_C$$

is ample. We fix A_C. Let $\lambda \in (0, 1)$ be a rational number, let D_λ be as constructed in **Step 4**, and let $U_\lambda \subset C$ be a non-empty open subset such that $(f^{-1}(U_\lambda), D_\lambda)$ is klt. Let q be a positive integer such that qL is a very ample Cartier divisor, let A be a general member of $|qL|$, and let $A_\lambda := \frac{1}{q}A$. By [KM98, Lemma 5.17(2)], $(f^{-1}(U_\lambda), D_\lambda + A)$ is lc. Since $(1-\lambda)/q < 1$, the pair $(f^{-1}(U_\lambda), D_\lambda + (1-\lambda)A_\lambda)$ is klt. Let $B_\lambda := D_\lambda + (1-\lambda)A_\lambda$. Then $B_\lambda \geq 0$, (X, B_λ) is klt over a non-empty open subset of C, and

$$K_X + B_\lambda \sim_{\mathbb{Q}} K_X - \lambda K_X + (1-\lambda)(-K_X + f^*A_C) = (1-\lambda)f^*A_C.$$

By Lemma 2.18 applied to B_λ and the $\mathbb{Q}$-divisor $(1-\lambda)A_C$ on C, we have $\deg((1-\lambda)A_C - K_C) \geq 0$, that is,

$$2g(C) - 2 \leq (1-\lambda)\deg A_C.$$

Since A_C does not depend on λ, letting $\lambda \to 1$ implies $2g(C) - 2 \leq 0$. Thus $g(C) \leq 1$.

Finally, by **Step 1**, $((-K_F)^{d-1}) \geq (d-1)^{d-1}$. Thus **Steps 2 and 5** imply $g(C) \leq 1$, which completes the proof. □

Proof of Theorem 5.1. Let $\mathfrak{Y}_1$ be the set of terminal projective varieties X' of dimension d such that $-K_{X'}$ is nef and big, let $\mathfrak{Y}_2 := \{\mathbb{P}^d\}$, and let $\mathfrak{Y}_3$ be the set of varieties $E \times \mathbb{P}^{d-1}$, where E runs over the smooth projective curves of genus one. By Theorem 2.16, $\mathfrak{Y}_1$ is bounded. By Lemma 2.15(2) with $m = d-1$, $\mathfrak{Y}_3$ is bounded. $\mathfrak{Y}_2$ is bounded since it consists of one variety. Thus

$$\mathfrak{Y} := \mathfrak{Y}_1 \cup \mathfrak{Y}_2 \cup \mathfrak{Y}_3$$

is birationally bounded by Lemma 2.15(1). By Lemma 2.15(1) again, the first part of Theorem 5.1 follows once we show that any $X \in \mathfrak{X}$ is birational to a member of $\mathfrak{Y}$. We also prove the second part, namely that X is birational over C to $C \times \mathbb{P}^{d-1}$ when $((-K_F)^{d-1}) > (d-1)^{d-1}$.

Let $X \in \mathfrak{X}$, let $f : X \to C$ be a Mori fiber space onto a smooth projective curve C, and let F be a general fiber of f. By Proposition 5.2, $((-K_F)^{d-1}) \geq (d-1)^{d-1}$ and $g(C) \leq 1$, where $g(C)$ is the genus of C. If $((-K_F)^{d-1}) = (d-1)^{d-1}$, then $-K_X$ is nef and big by Proposition 5.2(1), so $X \in \mathfrak{Y}_1$.

From now on, we assume that $((-K_F)^{d-1}) > (d-1)^{d-1}$. Let η be the generic point of C, X_η the generic fiber of f, and $X_{\bar{\eta}}$ the geometric generic fiber of f. Since f is a contraction and the characteristic is zero, $X_{\bar{\eta}}$ is a normal projective variety of dimension $d-1$ over $\overline{\mathbb{C}(C)}$. Let R be as in **Step 1** of the proof of Proposition 5.2. By **Step 1** of that proof, the Hilbert polynomial with respect to $\mathcal{O}_X(-RK_X)$ of the fiber of f over a closed point of C does not depend on the point, and a general fiber F of f is normal with $K_F = K_X|_F$. Since f is flat, this is also the Hilbert polynomial of the generic fiber X_η [Har77, Theorem III.9.9], and the Hilbert polynomial does not change under the base change to $\overline{\mathbb{C}(C)}$ [Har77, Proposition III.9.3]. By [Har77, Proposition II.8.10], relative differentials of f on the smooth locus of f commute with base change (on the smooth locus of f, $\omega_X \cong \omega_{X/C} \otimes f^*\omega_C$ [Har77, Proposition II.8.11], $f^*\omega_C$ is trivial on X_η, and the base change to $\overline{\mathbb{C}(C)}$ of the smooth locus of f is the smooth locus of $X_{\bar{\eta}}$ [Har77, Proposition II.8.10 and Example III.10.0.3]), so $\mathcal{O}_X(RK_X)|_{X_{\bar{\eta}}}$ and $\mathcal{O}_{X_{\bar{\eta}}}(RK_{X_{\bar{\eta}}})$ are isomorphic on the smooth locus of $X_{\bar{\eta}}$, hence on $X_{\bar{\eta}}$, as both are reflexive and $X_{\bar{\eta}}$ is normal. Hence

$$((-K_{X_{\bar{\eta}}})^{d-1}) = ((-K_F)^{d-1}) > (d-1)^{d-1}.$$

Next, we show that $X_{\bar{\eta}}$ is a terminal Fano variety. Let $h : X' \to X$ be a log resolution, and write $K_{X'} = h^*K_X + \sum_i a_i E_i$, where the E_i are the h-exceptional prime divisors. Since X is terminal, $a_i > 0$ for any i. The base change $X'_{\bar{\eta}} \to X_{\bar{\eta}}$ of h is a log resolution (by [Har77, Corollary III.10.7] applied to $X' \to C$ and to the strata of $\sum_i E_i$, and by [Har77, Proposition II.8.10]), the exceptional prime divisors of $X'_{\bar{\eta}} \to X_{\bar{\eta}}$ are the components of the restrictions of the E_i to $X'_{\bar{\eta}}$, and the discrepancy of such a component is $a_i > 0$ (since $K_{X'}|_{X'_{\bar{\eta}}} = K_{X'_{\bar{\eta}}}$ by the argument of the previous paragraph applied to $X' \to C$). Thus [KM98, Corollary 2.32(2)] with $\Delta = 0$ implies

$$\operatorname{discrep}(X_{\bar{\eta}}) \geq \min\{\min_i a_i, 1\} > 0,$$

so $X_{\bar{\eta}}$ is terminal. Since $-K_X$ is f-ample, $-K_{X_{\bar{\eta}}}$ is ample. Since $X_{\bar{\eta}}$ is klt, hence dlt, and $-K_{X_{\bar{\eta}}}$ is ample, [HM07, Corollary 1.3(1) and Corollary 1.5(2)], applied with S a point and $\Delta = 0$, implies that $X_{\bar{\eta}}$ is rationally connected [Kol96, Definition IV.3.2] (see Remark 2.13). Since rational connectedness is invariant under birational maps of proper varieties [Kol96, Proposition IV.3.3], a resolution of $X_{\bar{\eta}}$ is rationally connected.

Next, we construct a $\mathbb{C}(C)$-rational point of X_η with a large Seshadri constant. Let y_0, the integers $m_1 < m_2 < \cdots$, and the open subsets $U_j \subset X$ be as in **Step 1** of the proof of Proposition 5.2, and let $s_j := s(-m_jK_X, y_0)$. Then $s_j \geq 0$,

$$\lim_{j\to+\infty} \frac{s_j}{m_j} = \epsilon_{\mathrm{mov}}(-K_X) \geq d-1,$$

and by Lemma 2.6(3), the space $H^0(X, \mathcal{O}_X(-m_jK_X))$ generates s_j-jets at any point of U_j. By Lemma 4.1 with $U = X$, there exists a $\mathbb{C}(C)$-rational point p of X_η such that p is a smooth point of X_η and for any j, the image V_j of

$$H^0(X, \mathcal{O}_X(-m_jK_X)) \otimes_{\mathbb{C}} \mathbb{C}(C) \to H^0(X_\eta, \mathcal{O}_{X_\eta}(-m_jK_{X_\eta}))$$

generates s_j-jets at p. Let $p_{\bar{\eta}}$ be the point of $X_{\bar{\eta}}$ over p. The jet map at $p_{\bar{\eta}}$ of

$$V_j \otimes_{\mathbb{C}(C)} \overline{\mathbb{C}(C)} \subset H^0(X_{\bar{\eta}}, \mathcal{O}_{X_{\bar{\eta}}}(-m_jK_{X_{\bar{\eta}}}))$$

is the base change of the jet map of V_j at p, so $s(-m_jK_{X_{\bar{\eta}}}, p_{\bar{\eta}}) \geq s_j$. Hence

$$\epsilon_{\mathrm{mov}}(-K_{X_{\bar{\eta}}}, p_{\bar{\eta}}) \geq \lim_{j\to+\infty} \frac{s_j}{m_j} \geq d-1.$$

Since $-K_{X_{\bar{\eta}}}$ is ample, Lemma 2.9(3) (see Remark 2.13) implies $\epsilon(-K_{X_{\bar{\eta}}}, p_{\bar{\eta}}) \geq d-1$.

By Proposition 4.2 applied to X_η, with $d-1$ in place of d, X_η is birational over $\mathbb{C}(C)$ to $\mathbb{P}^{d-1}_{\mathbb{C}(C)}$. Since the function field of X is the function field of X_η, X is birational over C to $C \times \mathbb{P}^{d-1}$. This proves the second part of Theorem 5.1.

Finally, recall that $g(C) \leq 1$. If $g(C) = 0$, then $C \cong \mathbb{P}^1$, so X is birational to $\mathbb{P}^1 \times \mathbb{P}^{d-1}$, hence to $\mathbb{P}^d \in \mathfrak{Y}_2$. If $g(C) = 1$, then X is birational to $C \times \mathbb{P}^{d-1} \in \mathfrak{Y}_3$. Thus any $X \in \mathfrak{X}$ is birational to a member of $\mathfrak{Y}$, and $\mathfrak{X}$ is birationally bounded. □

6. Surfaces with large moving Seshadri constants

In this section, we prove Proposition 6.1. It is used only in Section 12, where we combine it with Theorem 9.2 to show that if the base surface Z of a Mori fiber space as in Construction-Theorem 3.1(3) with $\dim Z = 2$ (see Set-up 9.1 in Section 9) admits a contraction onto a smooth projective curve C whose general fiber is isomorphic to $\mathbb{P}^1$, then the genus of C is at most one.

Proposition 6.1. *Let X be a normal projective surface such that $\epsilon_{\mathrm{mov}}(-K_X) \geq 1$. Then X is birational to a projective surface Y with canonical singularities such that $-K_Y$ is ample and $\epsilon(-K_Y) \geq 1$, or to $E \times \mathbb{P}^1$ for some smooth projective curve E of genus one.*

Proof. By Lemma 2.12(1), X has a terminal modification $h : X' \to X$ (Definition 2.10), and

$$\epsilon_{\mathrm{mov}}(-K_{X'}) = \epsilon_{\mathrm{mov}}(-K_X) \geq 1.$$

Since X' is a terminal surface, X' is smooth by [KM98, Theorem 4.5(1)] (with zero boundary). Since h is birational, possibly replacing X with X', we may assume that X is smooth. In particular, $-K_X$ is Cartier, and $-K_X$ is big by Lemma 2.6(4).

Since $-K_X$ is big, it is pseudo-effective. By [Laz04, Theorem 2.3.19], we may write the Zariski decomposition

$$-K_X = P + N,$$

where P is a nef $\mathbb{Q}$-divisor and $N \geq 0$ is a $\mathbb{Q}$-divisor. By [Laz04, Proposition 2.3.21], for any integer $m \geq 1$, the natural map

$$H^0(X, \mathcal{O}_X(-mK_X - \lceil mN \rceil)) \to H^0(X, \mathcal{O}_X(-mK_X))$$

is bijective.

Let q be the smallest positive integer such that qP is a divisor. Then $qN = -qK_X - qP$ is also a divisor. By [Laz04, Corollary 2.3.22],

$$(P^2) = \mathrm{vol}(-K_X) > 0.$$

Thus qP is a nef divisor such that $((qP)^2) = q^2(P^2) > 0$, so qP is big by [Laz04, Theorem 2.2.16]. Hence P is big.

Let $y \in X \setminus \operatorname{Supp} N$ be a very general point. By Lemma 2.6(3),

$$\epsilon_{\mathrm{mov}}(-K_X, y) = \epsilon_{\mathrm{mov}}(-K_X).$$

Let m be a positive integer such that $q \mid m$. Then mN is a divisor, so $-mK_X - \lceil mN \rceil = mP$, and the bijection above implies that

$$H^0(X, \mathcal{O}_X(mP)) = H^0(X, \mathcal{O}_X(-mK_X)).$$

Since $mN \geq 0$ and $y \notin \operatorname{Supp} N$, the inclusion $\mathcal{O}_X(mP) \subset \mathcal{O}_X(-mK_X)$ is an isomorphism near y. Therefore, $s(mP, y) = s(-mK_X, y)$. By the choice of q, the positive integers m such that mP is a divisor are exactly the multiples of q. By Definition 2.4 and Lemma 2.6(2) (with $I = q$), we have

$$\epsilon_{\mathrm{mov}}(P, y) = \limsup_{m \to +\infty,\ q \mid m} \frac{s(mP, y)}{m} = \limsup_{m \to +\infty,\ q \mid m} \frac{s(-mK_X, y)}{m} = \epsilon_{\mathrm{mov}}(-K_X, y).$$

Since P is nef and big and y is a very general point of X, Lemma 2.9(4) implies that $\epsilon(P, y) = \epsilon_{\mathrm{mov}}(P, y)$. Thus

$$\epsilon(P) \geq \epsilon(P, y) = \epsilon_{\mathrm{mov}}(-K_X) \geq 1.$$

Since $N \geq 0$ and $K_X + N = -P$ is $\mathbb{Q}$-Cartier, (X, N) is a pair (Definition 2.1). Moreover, $-(K_X + N) = P$ is nef and $\epsilon(P) \geq 1 = \dim X - 1$. By [Zhu18, Theorem 1.6], which does not assume that the coefficients of the boundary are at most one, X is birational to a projective surface Y with canonical singularities such that $-K_Y$ is ample and $\epsilon(-K_Y) \geq 1$, or to $E \times \mathbb{P}^1$ for some smooth projective curve E of genus one. □

7. Nefness, volume, and birationality

In this section, we prove four lemmas used in Sections 10–12.

Remark 7.1. In [Zhu18, Lemma 5.4], X is the total space of a Mori fiber space onto a curve, so $\rho(X) = 2$. In [Zhu18, proof of Lemma 5.4], the nefness of $-K_X$ is obtained by a length argument. If $-K_X$ is not nef, then for a $\mathbb{Q}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{Q}} -qK_X$, and (X, D) is klt along a general fiber, there exists a $(K_X + D)$-negative extremal ray generated by a rational curve Γ such that

$$0 < -((K_X + D) \cdot \Gamma) \leq 2 \dim X,$$

while $-((K_X + D) \cdot \Gamma) = (q - 1)(K_X \cdot \Gamma)$ tends to $+\infty$ as $q \to +\infty$. In Lemma 7.2 below, $f : X \to C$ is a contraction onto a smooth projective curve, L is ample over C, $K_X + L \sim_{\mathbb{Q}} f^*A$, and $D_q \sim_{\mathbb{Q}} qL$, so

$$K_X + D_q \sim_{\mathbb{Q}} (q - 1)L + f^*A.$$

If $\deg A > 0$, then $(f^*A \cdot \Gamma) > 0$ for any horizontal curve Γ, and $(f^*A \cdot \Gamma)$ may cancel the negative number $(q - 1)(L \cdot \Gamma)$, so the length argument does not lead to a contradiction. If C has positive genus, then any rational curve on X is vertical, and $((K_X + D_q) \cdot \Gamma) = (q - 1)(L \cdot \Gamma) > 0$ for any vertical curve Γ. Thus, in the proof of Lemma 7.2, if $C \cong \mathbb{P}^1$, we first replace C by a double cover $g : C' \to C$ such that C' has genus one, so that any rational curve on $X \times_C C'$ is vertical.

Lemma 7.2. *Let $f : X \to C$ be a contraction from a normal projective variety X onto a smooth projective curve C, and L a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Assume that*

(1) L is ample over C, and $K_X + L \sim_{\mathbb{Q}} f^*A$ for some $\mathbb{Q}$-divisor A on C,
(2) f is smooth over a non-empty open subset of C, and
(3) for any integer $q \geq 2$, there exists a $\mathbb{Q}$-divisor $D_q \geq 0$ on X such that $D_q \sim_{\mathbb{Q}} qL$ and (X, D_q) is klt over a non-empty open subset of C.

Then L is nef.

Proof. By (1), K_X is $\mathbb{Q}$-Cartier, so (X, D_q) is a pair for any integer $q \geq 2$.

We first replace C by a curve of positive genus. If the genus of C is positive, let $C' := C$, let $g : C' \to C$ be the identity, and let $P := 0$ and $R(g) := 0$. Otherwise, $C \cong \mathbb{P}^1$, and by (2) there exists a reduced divisor P of degree 4 on C such that f is smooth over a neighborhood of $\operatorname{Supp} P$. Since $P \sim 2M$ for a divisor M of degree 2 on C, there exists a finite double cover $g : C' \to C$ from a smooth projective curve C' which is branched exactly over $\operatorname{Supp} P$ [Laz04, Proposition 4.1.6]. The curve C' is connected, since otherwise g would be étale. Let $R(g)$ be the ramification divisor of g. Then $\deg R(g) = 4$, and by the Hurwitz formula [Har77, Corollary IV.2.4],

$$\deg K_{C'} = 2 \deg K_C + \deg R(g) = -4 + 4 = 0,$$

so C' has genus one.

Let $X' := X \times_C C'$, and let $h : X' \to X$ and $f' : X' \to C'$ be the projections. Since g is étale over $C \setminus \operatorname{Supp} P$ and f is smooth over a neighborhood of $\operatorname{Supp} P$, X' is a normal projective variety (X' is irreducible: f is flat with geometrically integral generic fiber, so $X' \to C'$ is flat with integral generic fiber). Again since g is étale over $C \setminus \operatorname{Supp} P$ and f is smooth over a neighborhood of $\operatorname{Supp} P$, the relative canonical divisor of f pulls back to the relative canonical divisor of f', and $K_{C'} \sim g^*K_C + R(g)$ [Har77, Proposition IV.2.3], so

$$K_{X'} \sim h^*K_X + f'^*R(g).$$

Let $L' := h^*L$, and let $D'_q := h^*D_q$ for any integer $q \geq 2$. Since $X' = X \times_C C'$ and L is ample over C, L' is ample over C'. Moreover, $D'_q \geq 0$, $D'_q \sim_{\mathbb{Q}} qL'$, and

$$K_{X'} + L' \sim_{\mathbb{Q}} f'^*(g^*A + R(g)).$$

Since $K_{X'} \sim h^*K_X + f'^*R(g)$ is $\mathbb{Q}$-Cartier, (X', D'_q) is a pair. Since h is étale over $g^{-1}(C \setminus \operatorname{Supp} P)$ and (X, D_q) is klt over a non-empty open subset of C, (X', D'_q) is klt over a non-empty open subset of C'. In particular, $\operatorname{Nlc}(X', D'_q)$ is contained in finitely many fibers of f'.

For any integer $q \geq 2$, let

$$Q_q := K_{X'} + D'_q \sim_{\mathbb{Q}} (q-1)L' + f'^*(g^*A + R(g)).$$

For any curve $\Gamma \subset X'$ such that $f'(\Gamma)$ is a point, we have

$$(Q_q \cdot \Gamma) = (q-1)(L' \cdot \Gamma) > 0.$$

We show that Q_q is nef. Suppose not. Any curve contained in $\operatorname{Nlc}(X', D'_q)$ is contracted by f', so Q_q is non-negative on $\overline{\operatorname{NE}}(X')_{\operatorname{Nlc}(X', D'_q)}$. By the cone theorem [Fuj11, Theorem 1.1], applied to (X', D'_q) and the morphism from X' to a point, $\overline{\operatorname{NE}}(X')$ is the sum of $\overline{\operatorname{NE}}(X')_{Q_q \geq 0}$, $\overline{\operatorname{NE}}(X')_{\operatorname{Nlc}(X', D'_q)}$, and Q_q-negative extremal rays R_j such that $R_j \cap \overline{\operatorname{NE}}(X')_{\operatorname{Nlc}(X', D'_q)} = \{0\}$. Since Q_q is non-negative on the first two cones and Q_q is not nef, such a ray R_j exists, and by [Fuj11, Theorem 1.1(5)], R_j is spanned by the class of a rational curve Γ. Since C' has positive genus, any morphism from $\mathbb{P}^1$ to C' is constant by the Hurwitz formula [Har77, Corollary IV.2.4], so $f'(\Gamma)$ is a point. Thus $(Q_q \cdot \Gamma) > 0$, a contradiction. Therefore Q_q is nef for any integer $q \geq 2$.

Let $\Gamma' \subset X'$ be any curve. For any integer $q \geq 2$, we have

$$(L' \cdot \Gamma') = \frac{(Q_q \cdot \Gamma') - (f'^*(g^*A + R(g)) \cdot \Gamma')}{q-1} \geq -\frac{(f'^*(g^*A + R(g)) \cdot \Gamma')}{q-1}.$$

Letting $q \to +\infty$, we obtain $(L' \cdot \Gamma') \geq 0$. Thus L' is nef. Since h is finite and surjective, L is nef [Laz04, Example 1.4.4(ii)]. □

For a contraction $f : X \to Z$ and an ample Cartier divisor H on Z, the next lemma bounds the rational numbers $a > 0$ such that $-K_X - af^*H$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. The proof mixes D' with a $\mathbb{Q}$-divisor $D \sim_{\mathbb{Q}} -K_X$ such that $(X, 2D)$ is klt over a non-empty open subset of Z, and then uses Lemma 2.18.

Lemma 7.3. *Let $f : X \to Z$ be a contraction from a normal projective variety X onto a normal projective variety Z of dimension $b \in \{1, 2\}$, such that K_X is $\mathbb{Q}$-Cartier, Z is a smooth curve if $b = 1$, and Z is $\mathbb{Q}$-factorial if $b = 2$. Let $X_{\bar{\eta}}$ be the geometric generic fiber of f, H an ample Cartier divisor on Z, and $\alpha \in (0, 1]$ a rational number. Assume that*

(1) there exists a $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$ and $(X, 2D)$ is klt over a non-empty open subset of Z, and
(2) $(X_{\bar{\eta}}, \alpha\Delta)$ is lc for any $\mathbb{Q}$-divisor $\Delta \geq 0$ on $X_{\bar{\eta}}$ such that $\Delta \sim_{\mathbb{Q}} -K_{X_{\bar{\eta}}}$.

Then $a\alpha(H^b) \leq 4((-K_Z) \cdot H^{b-1})$ for any rational number $a > 0$ such that $-K_X - af^*H$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. If $b = 1$, this means $a\alpha \deg H \leq 4 \deg(-K_Z)$.

Proof. Let $a > 0$ be a rational number and $D' \geq 0$ a $\mathbb{Q}$-divisor on X such that $D' \sim_{\mathbb{Q}} -K_X - af^*H$. Let $t := \alpha/4$ and

$$B_a := (1-t)D + tD'.$$

Then $B_a \geq 0$ and

$$K_X + B_a \sim_{\mathbb{Q}} (1-t)(K_X + D) + t(K_X + D') \sim_{\mathbb{Q}} f^*(-taH).$$

We show that (X, B_a) is klt over a non-empty open subset of Z. Denote by $D_{\bar{\eta}}$, $D'_{\bar{\eta}}$, and $B_{a,\bar{\eta}}$ the restrictions of D, D', and B_a to $X_{\bar{\eta}}$. Since f^*H restricts to the trivial divisor on $X_{\bar{\eta}}$, we have $D'_{\bar{\eta}} \sim_{\mathbb{Q}} -K_{X_{\bar{\eta}}}$ (canonical divisors of X, and of log resolutions of open subsets of X, restrict to canonical divisors of their geometric generic fibers over Z [Har77, Propositions II.8.10 and

II.8.11, Example III.10.0.3]). For a prime divisor E over $X_{\bar\eta}$ and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor G on $X_{\bar\eta}$, we denote by $\operatorname{mult}_E G$ the coefficient of E in the pullback of G to a higher model of $X_{\bar\eta}$ on which E is a divisor. Thus mult_E is the divisorial valuation of E, and if $G \geq 0$, then

$$a(E, X_{\bar\eta}, G) = a(E, X_{\bar\eta}, 0) - \operatorname{mult}_E G$$

by Definition 2.1. Let $U \subset Z$ be a non-empty open subset such that $(f^{-1}(U), 2D|_{f^{-1}(U)})$ is klt. Let $\mu : Y \to f^{-1}(U)$ be a log resolution of $(f^{-1}(U), 2D|_{f^{-1}(U)})$, and write $K_Y + B_Y := \mu^*(K_{f^{-1}(U)} + 2D|_{f^{-1}(U)})$. The base change $\mu_{\bar\eta} : Y_{\bar\eta} \to X_{\bar\eta}$ of μ is a log resolution of $(X_{\bar\eta}, 2D_{\bar\eta})$, the restriction $B_{Y,\bar\eta}$ of B_Y to $Y_{\bar\eta}$ satisfies

$$K_{Y_{\bar\eta}} + B_{Y,\bar\eta} = \mu_{\bar\eta}^*(K_{X_{\bar\eta}} + 2D_{\bar\eta}),$$

and any coefficient of $B_{Y,\bar\eta}$ is a coefficient of B_Y (smoothness and simple normal crossings are preserved by the base change to the geometric generic fiber in characteristic zero, and each component P of $B_{Y,\bar\eta}$ maps into a component P_0 of B_Y dominating Z, and the coefficient of P in $B_{Y,\bar\eta}$ equals the coefficient of P_0 in B_Y). Thus $(X_{\bar\eta}, 2D_{\bar\eta})$ is klt [KM98, Lemma 2.30 and Corollary 2.31(3)]. In particular, $a(E, X_{\bar\eta}, 0) > 0$ and

$$\operatorname{mult}_E D_{\bar\eta} < \frac{a(E, X_{\bar\eta}, 0)}{2}$$

for any prime divisor E over $X_{\bar\eta}$. By (2), $(X_{\bar\eta}, \alpha D'_{\bar\eta})$ is lc, so

$$\operatorname{mult}_E D'_{\bar\eta} \leq \frac{a(E, X_{\bar\eta}, 0)}{\alpha}$$

for any prime divisor E over $X_{\bar\eta}$. Since $0 < t \leq 1/4$, we obtain

$$\operatorname{mult}_E B_{a,\bar\eta} < \left(\frac{1-t}{2} + \frac{t}{\alpha}\right) a(E, X_{\bar\eta}, 0) = \left(\frac{3}{4} - \frac{\alpha}{8}\right) a(E, X_{\bar\eta}, 0) < a(E, X_{\bar\eta}, 0).$$

Thus $(X_{\bar\eta}, B_{a,\bar\eta})$ is klt.

Let $h : X' \to X$ be a log resolution of (X, B_a), and write $K_{X'} + B'_a := h^*(K_X + B_a)$. The base change of h to $X_{\bar\eta}$ is a log resolution of $(X_{\bar\eta}, B_{a,\bar\eta})$ (as for $\mu_{\bar\eta}$ in the previous paragraph), so any component of B'_a with coefficient ≥ 1 does not dominate Z [KM98, Lemma 2.30 and Corollary 2.31(3)]. Removing from Z the images of the components of B'_a with coefficient ≥ 1, we see that (X, B_a) is klt over a non-empty open subset of Z.

By Lemma 2.18 applied to B_a and the $\mathbb{Q}$-divisor $-taH$ on Z, we have

$$((-taH - K_Z) \cdot H^{b-1}) \geq 0,$$

that is, $ta(H^b) \leq ((-K_Z) \cdot H^{b-1})$. Since $t = \alpha/4$, we obtain $a\alpha(H^b) \leq 4((-K_Z) \cdot H^{b-1})$. □

The condition $H^0(T, \mathcal{O}_T(T)) \neq 0$ in the next lemma cannot be dropped. For example, if X is the blow-up of $\mathbb{P}^d$ at a point, T is the exceptional divisor, and L is the pullback of a hyperplane, then $\mathcal{O}_T(T) \cong \mathcal{O}_{\mathbb{P}^{d-1}}(-1)$, $\operatorname{vol}(L|_T) = 0$, and $\operatorname{vol}(L) = 1$.

Lemma 7.4. *Let X be a normal projective variety of dimension $d \geq 2$, L a big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and $T \subset X$ a prime divisor such that T is Cartier and normal and $H^0(T, \mathcal{O}_T(T)) \neq 0$. Let τ be the supremum of the rational numbers $a \geq 0$ such that $L - aT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D \geq 0$. Then $\operatorname{vol}(L) \leq d\tau \operatorname{vol}(L|_T)$.*

Proof. Let H be an ample Cartier divisor on X. If $a \geq 0$ is a rational number and $D \geq 0$ is a $\mathbb{Q}$-divisor such that $L - aT \sim_{\mathbb{Q}} D$, then

$$0 \leq (D \cdot H^{d-1}) = (L \cdot H^{d-1}) - a(T \cdot H^{d-1}).$$

Since T is a prime divisor and H is ample, $(T \cdot H^{d-1}) > 0$, so $a \leq (L \cdot H^{d-1})/(T \cdot H^{d-1})$. Since L is big, L is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D \geq 0$. Thus $0 \leq \tau < +\infty$.

Let I be a positive integer such that IL is Cartier, and let m be a positive integer divisible by I. If $H^0(X, mL) \neq 0$, let b_m be the largest integer $k \geq 0$ such that $H^0(X, mL - kT) \neq 0$. By

the previous paragraph applied to $L-(k/m)T$, b_m exists and $b_m/m \le \tau$. For any integer $k \ge 0$, the exact sequence

$$0 \to \mathcal{O}_X(mL-(k+1)T) \to \mathcal{O}_X(mL-kT) \to \mathcal{O}_T((mL-kT)|_T) \to 0$$

[Har77, Proposition II.6.18] implies

$$h^0(X, mL-kT) - h^0(X, mL-(k+1)T) \le h^0(T, (mL-kT)|_T). \tag{7.1}$$

Let $\sigma \in H^0(T, \mathcal{O}_T(T))$ be a non-zero section. Since T is integral, multiplication by σ^k is an injection from $H^0(T, (mL-kT)|_T)$ into $H^0(T, mL|_T)$. Summing (7.1) over $0 \le k \le b_m$ and using $h^0(X, mL-(b_m+1)T) = 0$, we obtain

$$h^0(X, mL) \le (b_m+1)h^0(T, mL|_T) \le (m\tau+1)h^0(T, mL|_T).$$

If $H^0(X, mL) = 0$, then the last inequality also holds. Hence

$$\frac{h^0(X, mL)}{m^d/d!} \le d\left(\tau + \frac{1}{m}\right)\frac{h^0(T, mL|_T)}{m^{d-1}/(d-1)!}$$

for any positive integer m divisible by I. By [Laz04, Definition 2.2.31, Lemma 2.2.38, and Remark 2.2.39], we have

$$\mathrm{vol}(L) = \limsup_{m\to+\infty,\ I|m} \frac{h^0(X, mL)}{m^d/d!} \quad \text{and} \quad \mathrm{vol}(L|_T) = \limsup_{m\to+\infty,\ I|m} \frac{h^0(T, mL|_T)}{m^{d-1}/(d-1)!}.$$

Taking lim sup over the positive integers m divisible by I, we obtain $\mathrm{vol}(L) \le d\tau\,\mathrm{vol}(L|_T)$. □

Remark 7.5. [Zhu18, Corollary 4.2] proves that for any $\epsilon > 0$, the set of normal projective varieties X of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) > d-1+\epsilon$ is birationally bounded. In [Zhu18, proof of Corollary 4.2], for a very general point $x \in X$, a $\mathbb{Q}$-divisor $\Delta \ge 0$ such that $\Delta \sim_{\mathbb{Q}} -aK_X$ for some $0 < a < 2$ is constructed, such that (X, Δ) is lc, $\{x\}$ is an lc center of (X, Δ), and x is an isolated point of $\mathrm{Nklt}(X, \Delta)$. Then [HMX13, Lemma 2.3.4] implies that $|-3K_X|$ defines a birational map, and [HMX13, Lemma 2.4.2] implies birational boundedness. Lemma 7.6 only assumes $\epsilon_{\mathrm{mov}}(-K_X) > d/2$ and $\mathrm{vol}(-K_X) \le v$, which, for $d \ge 3$, do not imply the assumption of [Zhu18, Corollary 4.2]. Thus, for Lemma 7.6, we construct the boundary directly from jets of $|-mK_X|$ at two very general points x and y, for a suitable positive integer m such that mK_X is Cartier. The conclusion then follows from [HMX13, Lemma 2.3.4(1) and Lemma 2.4.2(2)] as in [Zhu18, proof of Corollary 4.2]. Lemma 7.6 is used only in Theorem 12.1 and Theorem 11.1, where $d \ge 3$ and $\epsilon_{\mathrm{mov}}(-K_X) = d-1 > d/2$.

Lemma 7.6. *Let d be a positive integer and v a positive real number. Let $\mathfrak{X}$ be the set of normal projective varieties X of dimension d such that K_X is $\mathbb{Q}$-Cartier, $\epsilon_{\mathrm{mov}}(-K_X) > d/2$, and $\mathrm{vol}(-K_X) \le v$. Then $|-3K_X|$ defines a birational map for any $X \in \mathfrak{X}$, and $\mathfrak{X}$ is birationally bounded.*

Proof. Let $X \in \mathfrak{X}$. By Lemma 2.6(4), $-K_X$ is big. Let I be a positive integer such that IK_X is Cartier. By Lemma 2.6(3), there exists a smooth point $x_0 \in X$ such that

$$\epsilon_{\mathrm{mov}}(-K_X, x_0) = \epsilon_{\mathrm{mov}}(-K_X) > d/2.$$

By Lemma 2.6(2), there exists an integer $m \ge 2$ divisible by I such that $k := s(-mK_X, x_0)$ satisfies $k/m > d/2$. Then $-mK_X$ is Cartier and $k > dm/2 \ge d$. By Lemma 2.6(3), the set U of smooth points $y \in X$ such that $s(-mK_X, y) \ge k$ is a non-empty open subset of X.

Claim 7.7. *Let $x \in U$ and let $G \in |-mK_X|$ be a divisor such that $\mathrm{mult}_x G = k$ and the projectivized tangent cone of G at x (the hypersurface of $\mathbb{P}^{d-1}$ defined by the degree-k leading form of a local equation of G at x) is smooth. Then there exists an open neighborhood V of x such that $(V, \frac{d}{k}G|_V)$ is lc and $\mathrm{Nklt}(V, \frac{d}{k}G|_V) = \{x\}$. In particular, $\{x\}$ is an lc center of $(X, \frac{d}{k}G)$.*

Proof. Let $h : X' \to X$ be the blow-up of X at x, E the exceptional divisor, and G' the strict transform of G. By [KM98, Lemma 2.29], where the discrepancy of [KM98, Definition 2.25] is our log discrepancy minus one, we have

$$a\left(E, X, \tfrac{d}{k}G\right) = d - \frac{d}{k}\operatorname{mult}_x G = d - \frac{d}{k}k = 0.$$

Since $G' \cap E \subset E \cong \mathbb{P}^{d-1}$ is the projectivized tangent cone of G at x, G' is smooth near E and $G' + E$ is a simple normal crossing divisor near E. Thus there exists an open neighborhood $V \subset X_{\mathrm{reg}}$ of x such that h is a log resolution of $(V, \frac{d}{k}G|_V)$ and

$$K_{X'} + \frac{d}{k}G' + E = h^*\left(K_X + \frac{d}{k}G\right)$$

over V. Since $d/k < 1$, $(V, \frac{d}{k}G|_V)$ is lc and $\operatorname{Nklt}(V, \frac{d}{k}G|_V) = h(E) = \{x\}$. In particular, $\{x\}$ is an lc center of $(X, \frac{d}{k}G)$. □

Proof of Lemma 7.6 continued. Next we show that for (x, y) in a non-empty open subset of $U \times U$, there exist $H_x, H_y \in |-mK_X|$ such that $\operatorname{mult}_x H_x = \operatorname{mult}_y H_y = k$, the projectivized tangent cones of H_x at x and of H_y at y are smooth, and $x \notin H_y$. Fix local coordinates $z_1, \dots, z_d$ at a point $y \in U$ and a local generator of $\mathcal{O}_X(-mK_X)$ at y. Since $s(-mK_X, y) \geq k$, for any homogeneous form ϕ of degree k in $z_1, \dots, z_d$, there exists a section of $\mathcal{O}_X(-mK_X)$ whose local expression is congruent to ϕ modulo $\mathfrak{m}_y^{k+1}$. Moreover, the forms ϕ defining smooth hypersurfaces of $\mathbb{P}^{d-1}$, such as $z_1^k + \cdots + z_d^k$, form a non-empty open subset of the space of forms of degree k. Let $V_y \subset H^0(X, \mathcal{O}_X(-mK_X))$ be the subspace of sections vanishing to order at least k at y. Then the map from V_y to the space of forms of degree k is surjective. In particular, $V_y \neq 0$, and $\dim V_y = h^0(X, -mK_X) - \binom{d+k-1}{d}$ does not depend on $y \in U$. Thus the subspaces V_y form a vector subbundle of the trivial bundle with fiber $H^0(X, \mathcal{O}_X(-mK_X))$ over U, and the set W of pairs $(x, y) \in U \times U$ such that any section in V_y vanishes at x is closed in $U \times U$. For any $y \in U$, the set of points $x \in U$ such that $(x, y) \in W$ is the intersection of U with the base locus of V_y, which is a proper closed subset of U since $V_y \neq 0$. Thus $(U \times U)\backslash W$ is a non-empty open subset of $U \times U$.

Let $(x, y) \in (U \times U)\backslash W$. By the previous paragraph, with y replaced by x, there exists $H_x \in |-mK_X|$ such that $\operatorname{mult}_x H_x = k$ and the projectivized tangent cone of H_x at x is smooth. The sections in V_y which do not vanish at x and the sections in V_y whose degree-k leading form at y defines a smooth hypersurface of $\mathbb{P}^{d-1}$ form two non-empty open subsets of V_y. Thus there exists $H_y \in |-mK_X|$ such that $x \notin H_y$, $\operatorname{mult}_y H_y = k$, and the projectivized tangent cone of H_y at y is smooth. Let

$$B_x := \frac{d}{k}H_x, \quad B_y := \frac{d}{k}H_y, \quad \text{and} \quad \Delta := B_x + B_y.$$

Then $\Delta \geq 0$ and

$$\Delta \sim_{\mathbb{Q}} -\frac{2dm}{k}K_X = \frac{dm}{2k}(-4K_X), \quad 0 < \frac{dm}{2k} < 1.$$

By Claim 7.7 applied to $G = H_x$, (X, B_x) is lc near x and x is an isolated point of $\operatorname{Nklt}(X, B_x)$. Since $x \notin \operatorname{Supp} B_y$, (X, Δ) coincides with (X, B_x) near x. Thus (X, Δ) is lc near x, $\{x\}$ is an lc center of (X, Δ), and x is an isolated point of $\operatorname{Nklt}(X, \Delta)$. Since $B_x \geq 0$ and, by Claim 7.7 applied at y to $G = H_y$, $\{y\}$ is an lc center of (X, B_y), $y \in \operatorname{Nklt}(X, \Delta)$. Since $(U \times U)\backslash W$ is a non-empty open subset of $X \times X$, any pair (x, y) of very general points of X satisfies $(x, y) \in (U \times U)\backslash W$. Since $-K_X$ is big and $\Delta \sim_{\mathbb{Q}} c(-4K_X)$ with $c := \frac{dm}{2k} \in (0, 1)$, $-4K_X$ is potentially birational [HMX13, Definition 2.3.3]. By [HMX13, Lemma 2.3.4(1)], $|K_X + \lceil -4K_X \rceil|$ defines a birational map. Since $-4K_X$ is a Weil divisor, $|K_X + \lceil -4K_X \rceil| = |-3K_X|$.

Finally, $\operatorname{vol}(-3K_X) = 3^d \operatorname{vol}(-K_X) \leq 3^d v$ for any $X \in \mathfrak{X}$ [Laz04, Proposition 2.2.35(i) and Remark 2.2.39]. Since $|-3K_X|$ defines a birational map for any $X \in \mathfrak{X}$, $\mathfrak{X}$ is birationally bounded by [HMX13, Lemma 2.4.2(2)]. □

8. Relative Wronskians

In this section, we construct relative Wronskians (Construction-Theorem 8.1). Section 9 uses them to turn jets of a Mori fiber space along the fiber directions into jets of the anti-canonical divisor of the base along sections.

Restricting a single section of a multiple of $-K_X$ to a section of f leaves a part along the fiber directions without uniform sign. In the determinant of vertical derivatives of several sections, for suitable orders of derivatives and of jets, a multiple of the canonical class of the vertical foliation cancels the part along the fiber directions, and the determinant becomes a section of a negative multiple of the sum of the pullback of the canonical class of the base and the ramification divisor. In Section 9, Lemma 9.5 uses orders for which the cancellation holds in the limit of large degrees.

Construction-Theorem 8.1. Let $f : X \to Z$ and r be as follows.

(i) $f : X \to Z$ is a contraction between normal projective varieties, and $r := \dim X - \dim Z \geq 1$.
(ii) $f^{-1}(Z_{\mathrm{sing}})$ has codimension at least 2 in X.

Let $\mathcal{F}$ be the *vertical foliation* of f, that is, the saturation of the kernel of

$$df : T_{X_{\mathrm{reg}} \cap f^{-1}(Z_{\mathrm{reg}})} \to f^* T_{Z_{\mathrm{reg}}}.$$

Let $X^\circ \subset X_{\mathrm{reg}} \cap f^{-1}(Z_{\mathrm{reg}})$ be the largest open subset on which $\mathcal{F}$ and $T_{X_{\mathrm{reg}} \cap f^{-1}(Z_{\mathrm{reg}})}/\mathcal{F}$ are locally free. Let $K_{\mathcal{F}}$ be the Weil divisor class on X such that

$$\mathcal{O}_X(-K_{\mathcal{F}})|_{X^\circ} \cong \det(\mathcal{F}|_{X^\circ}).$$

Let the *ramification divisor* $R(f)$ of f be the closure in X of the divisor of zeros of the natural map

$$\det(T_{X^\circ}/\mathcal{F}|_{X^\circ}) \to (f|_{X^\circ})^* \det T_{Z_{\mathrm{reg}}}.$$

Let f^*K_Z be the Weil divisor class on X whose restriction to X° is the pullback of $K_{Z_{\mathrm{reg}}}$ under $f|_{X^\circ} : X^\circ \to Z_{\mathrm{reg}}$. Then the following hold.

(1) $R(f) \geq 0$, $f(\operatorname{Supp} R(f)) \neq Z$, and $K_{\mathcal{F}} \sim K_X - f^*K_Z - R(f)$.
(2) Let D, V, and k be as follows.
   (a) D is a Weil divisor on X.
   (b) $V \subset H^0(X, \mathcal{O}_X(D))$ is a linear subspace.
   (c) $k \geq 0$ is an integer, $M := \binom{k+r}{r}$, and $c := \binom{k+r}{r+1}$.

   Then there exists a linear map

$$w_k : \bigwedge^M V \to H^0(X, \mathcal{O}_X(MD + cK_{\mathcal{F}})),$$

   which we call the *Wronskian map*, with the following property. Let $x \in X^\circ$ be a point at which f is smooth, let $s \geq k$ be an integer such that $s(V, x) \geq s$, and let σ be a *formal section* of f through x, that is, a section of the base change of f to $\operatorname{Spec} \widehat{\mathcal{O}}_{Z,f(x)}$ which maps the closed point to x. Then the restriction of $w_k(\bigwedge^M V)$ to σ generates $(Ms - rc)$-jets at $f(x)$, that is, the natural map

$$w_k\big(\bigwedge^M V\big) \to \sigma^* \mathcal{O}_X(MD + cK_{\mathcal{F}}) \otimes \widehat{\mathcal{O}}_{Z,f(x)}/\widehat{\mathfrak{m}}^{Ms-rc+1}$$

   is surjective, where $\widehat{\mathfrak{m}}$ is the maximal ideal of $\widehat{\mathcal{O}}_{Z,f(x)}$.

Condition (ii) guarantees that the codimension one points of X are over the smooth locus of Z. Thus $K_{\mathcal{F}}$, $R(f)$, and f^*K_Z are defined on X° and extend to X by normality, and the base may be singular. In Section 9, the base is a surface which is only known to be klt, and condition (ii) holds by Lemma 9.3.

Proof of Construction-Theorem 8.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that $\mathcal{F}$ is involutive of rank r, that $X \backslash X^\circ$ has codimension at least 2 in X, and that $K_{\mathcal{F}}$, $R(f)$, and f^*K_Z are well defined.

Let $d := \dim X$ and $U := X_{\mathrm{reg}} \cap f^{-1}(Z_{\mathrm{reg}})$. Since X is normal, X_{sing} has codimension at least 2 in X. Together with (ii), $X \backslash U$ has codimension at least 2 in X. The quotient $T_U / \operatorname{Ker}(df)$ embeds into the locally free sheaf $(f|_U)^* T_{Z_{\mathrm{reg}}}$, so it is torsion free. Thus $\operatorname{Ker}(df)$ is saturated in T_U, that is, $\mathcal{F} = \operatorname{Ker}(df)$, and $Q := T_U / \mathcal{F}$ is torsion free. A local vector field θ on U is a section of $\mathcal{F}$ if and only if $\theta(f^* g) = 0$ for any local regular function g on Z_{reg}. For two such vector fields θ and θ', we have

$$[\theta, \theta'](f^* g) = \theta(\theta'(f^* g)) - \theta'(\theta(f^* g)) = 0.$$

Hence $\mathcal{F}$ is involutive.

Let ξ be a codimension one point of X. Then $\xi \in U$, and $\mathcal{O}_{X,\xi}$ is a discrete valuation ring because X is normal. Since finitely generated torsion free modules over a discrete valuation ring are free, $\mathcal{F}$ and Q are locally free near ξ, so $\xi \in X^\circ$. Thus the closed subset $X \backslash X^\circ$ contains no codimension one point of X, that is, $X \backslash X^\circ$ has codimension at least 2 in X. By [Har77, Proposition II.6.5(b)], the restriction map $\mathrm{Cl}(X) \to \mathrm{Cl}(X^\circ)$ is an isomorphism. In particular, the Weil divisor classes $K_{\mathcal{F}}$ and $f^* K_Z$ exist and are unique, and two Weil divisors on X are linearly equivalent if and only if their restrictions to X° are linearly equivalent.

By generic smoothness [Har77, Corollary III.10.7], applied to $f|_U : U \to Z_{\mathrm{reg}}$, there exists a non-empty open subset $Z_1 \subset Z_{\mathrm{reg}}$ such that f is smooth on $U \cap f^{-1}(Z_1)$. Since f is surjective and U is dense in X, $U \cap f^{-1}(Z_1) \neq \emptyset$. On $U \cap f^{-1}(Z_1)$, df is surjective, so $\mathcal{F}$ is locally free of rank r and the natural map $Q \to (f|_U)^* T_{Z_{\mathrm{reg}}}$ is an isomorphism. Thus $\mathcal{F}$ has rank r, $U \cap f^{-1}(Z_1) \subset X^\circ$, and the natural map

$$\det(Q|_{X^\circ}) \to (f|_{X^\circ})^* \det T_{Z_{\mathrm{reg}}}$$

is a map of line bundles on X° which is an isomorphism on $U \cap f^{-1}(Z_1)$. In particular, $R(f)$ is well defined.

Step 2. In this step, we prove (1).

By definition, $R(f)|_{X^\circ}$ is the divisor of zeros of the map

$$\det(Q|_{X^\circ}) \to (f|_{X^\circ})^* \det T_{Z_{\mathrm{reg}}},$$

which is ≥ 0. Thus $R(f) \geq 0$ and

$$\det(Q|_{X^\circ}) \cong (f|_{X^\circ})^* \det T_{Z_{\mathrm{reg}}} \otimes \mathcal{O}_{X^\circ}(-R(f)|_{X^\circ}).$$

Since $X^\circ \cap f^{-1}(Z_1) \subset U \cap f^{-1}(Z_1)$ and the map is an isomorphism on $U \cap f^{-1}(Z_1)$, we have $\operatorname{Supp} R(f) \cap X^\circ \subset f^{-1}(Z \backslash Z_1)$. Since any component of $R(f)$ intersects X° and $f^{-1}(Z \backslash Z_1)$ is closed, $\operatorname{Supp} R(f) \subset f^{-1}(Z \backslash Z_1)$. Hence $f(\operatorname{Supp} R(f)) \subset Z \backslash Z_1 \neq Z$.

Taking determinants in the exact sequence of locally free sheaves

$$0 \to \mathcal{F}|_{X^\circ} \to T_{X^\circ} \to Q|_{X^\circ} \to 0,$$

we obtain

$$\mathcal{O}_{X^\circ}(-K_X|_{X^\circ}) \cong \det T_{X^\circ} \cong \det(\mathcal{F}|_{X^\circ}) \otimes \det(Q|_{X^\circ}) \cong \mathcal{O}_{X^\circ}\big((-K_{\mathcal{F}} - f^* K_Z - R(f))|_{X^\circ}\big),$$

where the last isomorphism follows from the definitions of $K_{\mathcal{F}}$ and $f^* K_Z$, and from $\det T_{Z_{\mathrm{reg}}} \cong \mathcal{O}_{Z_{\mathrm{reg}}}(-K_{Z_{\mathrm{reg}}})$. By **Step 1**, $K_{\mathcal{F}} \sim K_X - f^* K_Z - R(f)$ on X (cf. [Dru17, Example 2.4], where the formula for $K_{\mathcal{F}}$ appears for equidimensional morphisms). This proves (1).

Step 3. In this step, we construct w_k.

For $\alpha = (\alpha_1, \dots, \alpha_r) \in \mathbb{N}^r$, we write $|\alpha| := \alpha_1 + \cdots + \alpha_r$ and $\alpha! := \alpha_1! \cdots \alpha_r!$. For vector fields $v_1, \dots, v_r$ on an open subset of X, we let $v^\alpha := v_1^{\alpha_1} \circ \cdots \circ v_r^{\alpha_r}$ be the composition, which is a differential operator on regular functions (v^0 is the identity). Let

$$\mathcal{A} := \{\alpha \in \mathbb{N}^r \mid |\alpha| \leq k\}.$$

Then $\#\mathcal{A} = \binom{k+r}{r} = M$, and we fix a listing $\mathcal{A} = \{\alpha^{(1)}, \dots, \alpha^{(M)}\}$ such that $|\alpha^{(i)}| \leq |\alpha^{(i')}|$ whenever $i \leq i'$.

Let $U_1 \subset X^\circ$ be an open subset on which there exist a local generator ℓ of $\mathcal{O}_X(D)|_{U_1}$ and a local basis $v_1, \dots, v_r$ of $\mathcal{F}|_{U_1}$. Such open subsets cover X° because X° is smooth and $\mathcal{F}|_{X^\circ}$ is

locally free of rank r. For $\varphi \in V$, write $\varphi|_{U_1} = h_\varphi \ell$ with $h_\varphi \in \mathcal{O}_X(U_1)$, and for $\varphi_1, \dots, \varphi_M \in V$, set

$$W_{\ell,v}(\varphi_1, \dots, \varphi_M) := \det\left(v^{\alpha^{(i)}}(h_{\varphi_j})\right)_{1\le i,j\le M} \in \mathcal{O}_X(U_1).$$

Then $W_{\ell,v}$ is multilinear and alternating in $\varphi_1, \dots, \varphi_M$. Let $(v_1 \wedge \cdots \wedge v_r)^{-c}$ be the local generator of $(\det \mathcal{F}|_{U_1})^{\otimes(-c)}$ dual to $(v_1 \wedge \cdots \wedge v_r)^{\otimes c}$. Let

$$\omega_{\ell,v}(\varphi_1, \dots, \varphi_M) := W_{\ell,v}(\varphi_1, \dots, \varphi_M)\ell^M \otimes (v_1 \wedge \cdots \wedge v_r)^{-c}.$$

Since

$$\mathcal{O}_X(MD + cK_{\mathcal{F}})|_{X^\circ} \cong \mathcal{O}_X(D)|_{X^\circ}^{\otimes M} \otimes (\det \mathcal{F}|_{X^\circ})^{\otimes(-c)},$$

$\omega_{\ell,v}(\varphi_1, \dots, \varphi_M)$ is a section of $\mathcal{O}_X(MD + cK_{\mathcal{F}})$ over U_1. We show that $\omega_{\ell,v}$ does not depend on ℓ and v.

First, let $\lambda \in \mathcal{O}_X(U_1)^*$ and $\ell' := \lambda\ell$, so that $\varphi|_{U_1} = \lambda^{-1} h_\varphi \ell'$. For $\gamma, \alpha \in \mathbb{N}^r$, we write $\gamma \le \alpha$ if $\gamma_i \le \alpha_i$ for each i. By the Leibniz rule, for any $h \in \mathcal{O}_X(U_1)$ and $\alpha \in \mathcal{A}$,

$$v^\alpha(\lambda^{-1}h) = \lambda^{-1}v^\alpha(h) + \sum_{\gamma\le\alpha,\ \gamma\ne\alpha} b_{\alpha\gamma} v^\gamma(h),$$

where the functions $b_{\alpha\gamma} \in \mathcal{O}_X(U_1)$ depend only on λ, $v_1, \dots, v_r$, α, and γ. This holds because v^α is a composition of $|\alpha|$ derivations, and when the factors of v^α are distributed between λ^{-1} and h, the factors applied to h keep their order and form v^γ for some $\gamma \le \alpha$. Since $\gamma \le \alpha$ and $\gamma \ne \alpha$ imply $|\gamma| < |\alpha|$, the matrix $\left(v^{\alpha^{(i)}}(\lambda^{-1}h_{\varphi_j})\right)_{i,j}$ is the product of a lower triangular matrix with diagonal entries λ^{-1} and the matrix $\left(v^{\alpha^{(i)}}(h_{\varphi_j})\right)_{i,j}$. Thus

$$W_{\ell',v} = \lambda^{-M} W_{\ell,v}. \tag{8.1}$$

Second, let $A = (a_{ii'})$ be an invertible $r \times r$ matrix with entries in $\mathcal{O}_X(U_1)$, and let $v_i' := \sum_{i'=1}^r a_{ii'} v_{i'}$ for $1 \le i \le r$. For an integer $j \ge 0$, let $\mathrm{Sym}^j(A)$ be the matrix with rows and columns indexed by $\{\alpha \in \mathbb{N}^r \mid |\alpha| = j\}$ such that

$$\prod_{i=1}^r \Big(\sum_{i'=1}^r a_{ii'} y_{i'}\Big)^{\alpha_i} = \sum_{|\beta|=j} \mathrm{Sym}^j(A)_{\alpha\beta} y^\beta$$

in the polynomial ring $\mathcal{O}_X(U_1)[y_1, \dots, y_r]$. We show that for any $\alpha \in \mathcal{A}$,

$$(v')^\alpha = \sum_{|\beta|=|\alpha|} \mathrm{Sym}^{|\alpha|}(A)_{\alpha\beta} v^\beta + \sum_{|\gamma|<|\alpha|} b'_{\alpha\gamma} v^\gamma$$

for some $b'_{\alpha\gamma} \in \mathcal{O}_X(U_1)$. Let $n := |\alpha|$. Replacing each factor v_i' of the composition $(v')^\alpha$ with $\sum_{i'} a_{ii'} v_{i'}$ and moving the coefficients $a_{ii'}$ to the left by the Leibniz rule, we write $(v')^\alpha$ as the sum of the operators

$$a_{i_1 i_1'} \cdots a_{i_n i_n'} v_{i_1'} \circ \cdots \circ v_{i_n'},$$

where $(i_1, \dots, i_n)$ is the sequence of indices of the factors of $(v')^\alpha$ and $i_1', \dots, i_n'$ run over $\{1, \dots, r\}$, and of a combination with regular coefficients of compositions of at most $n-1$ of the vector fields $v_1, \dots, v_r$. Since $\mathcal{F}$ is involutive (**Step 1**), any bracket $[v_{i'}, v_{i''}]$ is a combination of $v_1, \dots, v_r$ with regular coefficients. Thus interchanging two adjacent factors of a composition of n of the $v_{i'}$ changes the composition by a combination with regular coefficients of compositions of $n-1$ of the $v_{i'}$. By induction on n, $v_{i_1'} \circ \cdots \circ v_{i_n'}$ is v^β plus a combination with regular coefficients of operators v^γ with $|\gamma| < n$, where $\beta_{i'}$ is the number of indices l such that $i_l' = i'$. Collecting the terms, the coefficient of v^β for $|\beta| = n$ is the coefficient of y^β in $\prod_{l=1}^n \left(\sum_{i'} a_{i_l i'} y_{i'}\right)$, which is $\mathrm{Sym}^n(A)_{\alpha\beta}$.

Since the listing of $\mathcal{A}$ is nondecreasing in $|\alpha|$, the matrix $\left((v')^{\alpha^{(i)}}(h_{\varphi_j})\right)_{i,j}$ is the product of a block lower triangular matrix, whose diagonal blocks are $\operatorname{Sym}^j(A)$ for $j = 0, \dots, k$, and the matrix $\left(v^{\alpha^{(i)}}(h_{\varphi_j})\right)_{i,j}$. We have

$$\det \operatorname{Sym}^j(A) = (\det A)^{\binom{j+r-1}{r}} \qquad \text{and} \qquad \sum_{j=0}^{k} \binom{j+r-1}{r} = \binom{k+r}{r+1} = c.$$

For the first identity, both sides are polynomials with integer coefficients in the entries of A. By the definition of Sym^j by substitution,

$$\operatorname{Sym}^j(BE) = \operatorname{Sym}^j(B) \operatorname{Sym}^j(E)$$

for $r \times r$ matrices B and E, and Sym^j of the identity matrix is the identity matrix. Thus

$$\det \operatorname{Sym}^j(PEP^{-1}) = \det \operatorname{Sym}^j(E)$$

for any $r \times r$ matrix E and any invertible $r \times r$ matrix P. If E is diagonal with diagonal entries $a_1, \dots, a_r$, then $\operatorname{Sym}^j(E)$ is diagonal with diagonal entries $a_1^{\alpha_1} \cdots a_r^{\alpha_r}$ for $|\alpha| = j$, so

$$\det \operatorname{Sym}^j(E) = \prod_{i=1}^{r} a_i^{e_i} \qquad \text{with} \qquad e_i := \sum_{|\alpha|=j} \alpha_i.$$

We have

$$\#\{\alpha \in \mathbb{N}^r \mid |\alpha| = j\} = \binom{j+r-1}{r-1} \qquad \text{and} \qquad j\binom{j+r-1}{r-1} = r\binom{j+r-1}{r}.$$

Since e_i does not depend on i by symmetry,

$$e_i = \frac{1}{r} \sum_{|\alpha|=j} |\alpha| = \frac{j}{r}\binom{j+r-1}{r-1} = \binom{j+r-1}{r}.$$

Hence $\det \operatorname{Sym}^j(A) = (\det A)^{\binom{j+r-1}{r}}$ holds for the complex $r \times r$ matrices A with distinct eigenvalues, which are diagonalizable and form a non-empty Zariski open, hence dense, subset of the space of complex $r \times r$ matrices. Thus both sides are equal as polynomials, and we may substitute a matrix A with entries in $\mathcal{O}_X(U_1)$. The second identity follows from Pascal's rule $\binom{k+r}{r+1} + \binom{k+r}{r} = \binom{k+r+1}{r+1}$ by induction on k. Thus

$$W_{\ell,v'} = (\det A)^c W_{\ell,v}. \tag{8.2}$$

Since $\ell'^M = \lambda^M \ell^M$ and $v_1' \wedge \cdots \wedge v_r' = (\det A) v_1 \wedge \cdots \wedge v_r$, (8.1) and (8.2) imply that $\omega_{\ell',v} = \omega_{\ell,v} = \omega_{\ell,v'}$. Hence for $\varphi_1, \dots, \varphi_M \in V$, the sections $\omega_{\ell,v}(\varphi_1, \dots, \varphi_M)$ on the open subsets U_1 glue to a section $\omega(\varphi_1, \dots, \varphi_M)$ of $\mathcal{O}_X(MD + cK_{\mathcal{F}})$ over X°. Since $\mathcal{O}_X(MD + cK_{\mathcal{F}})$ is reflexive and $X \backslash X^\circ$ has codimension at least 2 in X (**Step 1**), $\omega(\varphi_1, \dots, \varphi_M)$ extends uniquely to a section

$$w_k(\varphi_1 \wedge \cdots \wedge \varphi_M) \in H^0(X, \mathcal{O}_X(MD + cK_{\mathcal{F}}))$$

by [Har80, Proposition 1.6]. Since $W_{\ell,v}$ is multilinear and alternating, w_k is a well defined linear map on $\bigwedge^M V$.

Step 4. In this step, we prove the jet property in (2).

Let x, s, and σ be given as in (2). Since $x \in X^\circ$, Z is smooth at $f(x)$ and $\mathcal{O}_X(D)$ is invertible near x. Let $z_1, \dots, z_{d-r}$ be a regular system of parameters of $\mathcal{O}_{Z,f(x)}$, where $d = \dim X$ as in **Step 1**. We also write z_i for $f^* z_i$. Since f is smooth at x, there exist $u_1, \dots, u_r \in \mathcal{O}_{X,x}$ such that $z_1, \dots, z_{d-r}, u_1, \dots, u_r$ is a regular system of parameters of $\mathcal{O}_{X,x}$. Let $U_1 \subset X^\circ$ be an open neighborhood of x on which z_i and $u_{i'}$ are regular, $dz_1, \dots, dz_{d-r}, du_1, \dots, du_r$ form a basis of Ω_{U_1}, and $\mathcal{O}_X(D)|_{U_1}$ has a local generator ℓ. Let $\partial_1, \dots, \partial_r$ be the vector fields on U_1 such that $\partial_{i'}(z_i) = 0$ and $\partial_{i'}(u_{i''}) = \delta_{i'i''}$ for all i, i', i''.

We show that the vector fields $\partial_1, \dots, \partial_r$ form a local basis of $\mathcal{F}|_{U_1}$. Shrinking U_1, for any local regular function g near $f(x)$ the differential dg is a combination of $dz_1, \dots, dz_{d-r}$, so $d(f^*g)$ is a combination of $dz_1, \dots, dz_{d-r}$ and $\partial_{i'}(f^*g) = 0$. Thus $\partial_{i'}$ is a section of $\mathrm{Ker}(df) = \mathcal{F}$. Moreover, the vector fields on U_1 dual to $dz_1, \dots, dz_{d-r}$ map under df to the pullbacks of $\partial/\partial z_1, \dots, \partial/\partial z_{d-r}$, which form a basis of $f^*T_{Z_{\mathrm{reg}}}$ near x. Hence $\mathrm{Ker}(df)$ is spanned by $\partial_1, \dots, \partial_r$ near x.

We have

$$\widehat{\mathcal{O}}_{Z,f(x)} = \mathbb{C}[[z_1, \dots, z_{d-r}]] \qquad \text{and} \qquad \widehat{\mathcal{O}}_{X,x} = \mathbb{C}[[z_1, \dots, z_{d-r}, u_1, \dots, u_r]]$$

by [Har77, Theorems I.5.4A and I.5.5A], and $\partial_{i'}$ acts on $\widehat{\mathcal{O}}_{X,x}$ as the partial derivative with respect to $u_{i'}$, since a derivation is continuous for the $\mathfrak{m}_x$-adic topology, so it is determined by its values on z and u. The formal section σ corresponds to a homomorphism of $\widehat{\mathcal{O}}_{Z,f(x)}$-algebras

$$\sigma^* : \widehat{\mathcal{O}}_{X,x} \to \widehat{\mathcal{O}}_{Z,f(x)}.$$

Let $\sigma_{i'} := \sigma^*(u_{i'}) \in \widehat{\mathfrak{m}}$ and $u'_{i'} := u_{i'} - \sigma_{i'}$. Then

$$\widehat{\mathcal{O}}_{X,x} = \mathbb{C}[[z_1, \dots, z_{d-r}, u'_1, \dots, u'_r]], \qquad (z, u') = (z, u) = \mathfrak{m}_x \widehat{\mathcal{O}}_{X,x},$$

σ^* is the substitution $u' = 0$, and $\partial_{i'}$ is the partial derivative with respect to $u'_{i'}$, since $\partial_{i'}(z_i) = 0$ and $\partial_{i'}(u'_{i''}) = \delta_{i'i''}$. For $\varphi \in V$, write $\varphi = h_\varphi \ell$ near x and

$$h_\varphi = \sum_{\gamma \in \mathbb{N}^r} a_{\gamma,\varphi} u'^{\gamma} \quad \text{with } a_{\gamma,\varphi} \in \mathbb{C}[[z_1, \dots, z_{d-r}]].$$

Then

$$\sigma^*\big(\partial^\alpha(h_\varphi)\big) = \alpha! a_{\alpha,\varphi},$$

where

$$\partial^\alpha := \partial_1^{\alpha_1} \circ \cdots \circ \partial_r^{\alpha_r}.$$

The $\widehat{\mathcal{O}}_{Z,f(x)}$-module $\sigma^*\mathcal{O}_X(MD + cK_{\mathcal{F}})$ is free of rank one with generator

$$e := \sigma^*\big(\ell^M \otimes (\partial_1 \wedge \cdots \wedge \partial_r)^{-c}\big).$$

By **Step 3**, with respect to e, the restriction of $w_k(\varphi_1 \wedge \cdots \wedge \varphi_M)$ to σ corresponds to

$$\sigma^* W_{\ell,\partial}(\varphi_1, \dots, \varphi_M) = \Big(\prod_{\alpha \in \mathcal{A}} \alpha!\Big) \det\big(a_{\alpha^{(i)},\varphi_j}\big)_{1 \le i,j \le M}.$$

Since $w_k(\bigwedge^M V)$ is spanned by the elements $w_k(\varphi_1 \wedge \cdots \wedge \varphi_M)$, it suffices to show that the power series $\det\big(a_{\alpha^{(i)},\varphi_j}\big)_{i,j}$, for $\varphi_1, \dots, \varphi_M \in V$, span

$$\widehat{\mathcal{O}}_{Z,f(x)}/\widehat{\mathfrak{m}}^{Ms-rc+1}.$$

For $\alpha \in \mathcal{A}$, let $t_\alpha := s - |\alpha|$. Since $|\alpha| \le k \le s$, we have $t_\alpha \ge 0$. Since $s(V,x) \ge s$, the map

$$V \to \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1}$$

is surjective. Thus for any polynomials $p_\alpha \in \mathbb{C}[z_1, \dots, z_{d-r}]$ of degree at most t_α, $\alpha \in \mathcal{A}$, there exists $\varphi \in V$ such that

$$h_\varphi - \sum_{\alpha \in \mathcal{A}} p_\alpha u'^{\alpha} \in (z, u')^{s+1},$$

because the polynomial $\sum_\alpha p_\alpha u'^{\alpha}$ has degree at most s. The coefficient of u'^{α} of an element of $(z, u')^{s+1}$ is contained in $\widehat{\mathfrak{m}}^{s+1-|\alpha|}$, so

$$a_{\alpha,\varphi} - p_\alpha \in \widehat{\mathfrak{m}}^{t_\alpha+1} \quad \text{for any } \alpha \in \mathcal{A}.$$

By the counting identities for $\alpha \in \mathbb{N}^r$ with $|\alpha| = j$ and the second identity in **Step 3**,

$$\sum_{\alpha \in \mathcal{A}} t_\alpha = Ms - \sum_{j=0}^{k} j \binom{j+r-1}{r-1} = Ms - r \sum_{j=0}^{k} \binom{j+r-1}{r} = Ms - rc.$$

Let z^β be a monomial of degree $q := |\beta| \leq Ms - rc$. Distributing the q factors of z^β among the elements of $\mathcal{A}$ so that α receives at most t_α of them, we write $\beta = \sum_{\alpha\in\mathcal{A}} \beta_\alpha$ with $\beta_\alpha \in \mathbb{N}^{d-r}$ and $q_\alpha := |\beta_\alpha| \leq t_\alpha$. By the previous paragraph, there exist $\varphi_1, \ldots, \varphi_M \in V$ such that for all $1 \leq i, j \leq M$,

$$n_{ij} := a_{\alpha^{(i)},\varphi_j} \in \delta_{ij} z^{\beta_{\alpha^{(i)}}} + \widehat{\mathfrak{m}}^{t_{\alpha^{(i)}}+1}.$$

Since $t_{\alpha^{(i)}} + 1 > q_{\alpha^{(i)}}$, we have

$$n_{ii} \in z^{\beta_{\alpha^{(i)}}} + \widehat{\mathfrak{m}}^{q_{\alpha^{(i)}}+1}$$

and $n_{ij} \in \widehat{\mathfrak{m}}^{q_{\alpha^{(i)}}+1}$ for $i \neq j$. In the expansion

$$\det(n_{ij}) = \sum_\pi \operatorname{sgn}(\pi) \prod_{i=1}^M n_{i\pi(i)}$$

over the permutations π of $\{1, \ldots, M\}$, the term of the identity is contained in $z^\beta + \widehat{\mathfrak{m}}^{q+1}$, and any other term has a factor $n_{i\pi(i)}$ with $i \neq \pi(i)$, so it is contained in $\widehat{\mathfrak{m}}^{q+1}$ because $\sum_i q_{\alpha^{(i)}} = q$. Thus

$$\det(n_{ij}) \in z^\beta + \widehat{\mathfrak{m}}^{q+1}.$$

Hence for any monomial z^β of degree at most $Ms - rc$, there exists an element ψ_β in the span of the power series $\det\left(a_{\alpha^{(i)},\varphi_j}\right)_{i,j}$ such that

$$\psi_\beta - z^\beta \in \widehat{\mathfrak{m}}^{|\beta|+1}.$$

Order the monomials of degree at most $Ms - rc$ by total degree. In the basis of

$$\widehat{\mathcal{O}}_{Z,f(x)}/\widehat{\mathfrak{m}}^{Ms-rc+1}$$

given by the monomials of degree at most $Ms - rc$, the matrix of the images of the elements ψ_β is block triangular with identity diagonal blocks, so it is invertible. Thus the elements ψ_β span

$$\widehat{\mathcal{O}}_{Z,f(x)}/\widehat{\mathfrak{m}}^{Ms-rc+1},$$

and the natural map in (2) is surjective. □

9. The base of a Mori fiber space over a surface

In this section, we prove Theorem 9.2, which says that the base surface Z of a Mori fiber space as in Set-up 9.1 satisfies $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$. Theorem 9.2 is used in two ways: it implies that $-K_Z$ is big, and, by Proposition 6.1, it implies in Section 12 that if Z admits a contraction onto a smooth projective curve C whose general fiber is isomorphic to $\mathbb{P}^1$, then the genus of C is at most one.

Set-up 9.1. Let d and $f : X \to Z$ be as follows.

(1) $d \geq 3$ is an integer.
(2) $f : X \to Z$ is a Mori fiber space onto a normal projective surface Z such that X is terminal of dimension d.
(3) The geometric generic fiber of f is isomorphic to $\mathbb{P}^{d-2}$.
(4) $\epsilon_{\mathrm{mov}}(-K_X) = d - 1$.

In Sections 9–12, we work under Set-up 9.1.

Theorem 9.2. *Notation and conditions as in Set-up 9.1. Then $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$. In particular, $-K_Z$ is big.*

Lemma 9.3. *Notation and conditions as in Set-up 9.1. Then Z is $\mathbb{Q}$-factorial and klt, and no prime divisor on X is mapped by f to a point. In particular, $f^{-1}(Z_{\mathrm{sing}})$ has codimension at least 2 in X.*

Proof. Since X is terminal, X is klt. Thus Z is $\mathbb{Q}$-factorial and klt by [Fuj99, Corollary 4.6].

Suppose that there exists a prime divisor E on X such that $f(E)$ is a point z_0. Since f is a Mori fiber space, X is $\mathbb{Q}$-factorial, so there exists a positive integer l such that lE is Cartier. Let γ be a curve contained in a fiber of f over a point of $Z\backslash\{z_0\}$. Then $(E\cdot\gamma)=0$ and $(-K_X\cdot\gamma)>0$. Since $\rho(X/Z)=1$, any curve contracted by f is numerically proportional to γ, hence $E\equiv_Z 0$. Thus lE is f-nef and $lE-K_X$ is f-ample. Since X is klt, the non-lc locus of $(X,0)$ is empty, and [Fuj11, Theorem 13.1] implies that $\mathcal{O}_X(klE)$ is generated over Z for any sufficiently large integer k.

On the other hand, we show that $f_*\mathcal{O}_X(klE)=\mathcal{O}_Z$. Let $V\subset Z$ be an open subset and φ a section of $\mathcal{O}_X(klE)$ over $f^{-1}(V)$, that is, a rational function on X such that $\mathrm{div}(\varphi)+klE\geq 0$ on $f^{-1}(V)$. Since $f^{-1}(V\backslash\{z_0\})$ does not intersect E, φ is regular on $f^{-1}(V\backslash\{z_0\})$. Since $f_*\mathcal{O}_X=\mathcal{O}_Z$, we have $\varphi=f^*\psi$ for a regular function ψ on $V\backslash\{z_0\}$. Since Z is normal and z_0 is a point of the surface Z, ψ extends to a regular function on V, so φ is regular on $f^{-1}(V)$. Thus the image of the natural map $f^*f_*\mathcal{O}_X(klE)\to\mathcal{O}_X(klE)$ is the subsheaf $\mathcal{O}_X\subset\mathcal{O}_X(klE)$, which differs from $\mathcal{O}_X(klE)$ at any point of E. This contradicts the generation of $\mathcal{O}_X(klE)$ over Z. Hence no prime divisor on X is mapped by f to a point.

Since Z is a normal surface, Z_{sing} is a finite set. For any $z\in Z_{\mathrm{sing}}$, $f^{-1}(z)$ contains no prime divisor of X, so $\dim f^{-1}(z)\leq d-2$. Thus $f^{-1}(Z_{\mathrm{sing}})$ has codimension at least 2 in X. □

Definition-Lemma 9.4. Notation and conditions as in Set-up 9.1. Let $Z^\circ\subset Z$ be the union of the open subsets $U\subset Z$ such that $f^{-1}(U)\to U$ is a Brauer–Severi scheme of relative dimension $d-2$. Then Z° is non-empty, and $f^{-1}(Z^\circ)\to Z^\circ$ is a Brauer–Severi scheme of relative dimension $d-2$. In particular, the general fiber of f is isomorphic to $\mathbb{P}^{d-2}$.

Proof. The morphism $f^{-1}(Z^\circ)\to Z^\circ$ is the restriction of the projective morphism f, so it is projective. Since smoothness and the condition on geometric fibers are local on the base, $f^{-1}(Z^\circ)\to Z^\circ$ is a Brauer–Severi scheme of relative dimension $d-2$.

The isomorphism $X_{\bar\eta}\cong\mathbb{P}^{d-2}$ is defined over a finite extension of $\mathbb{C}(Z)$. Spreading it out, we obtain a non-empty open subset $U\subset Z$ and a finite surjective morphism $U'\to U$ from an integral scheme U' whose function field is the extension above such that $X\times_Z U'\cong\mathbb{P}^{d-2}\times U'$ over U' (by a standard limit argument, see [Gro66, Théorème 8.8.2]). Thus any geometric fiber of f over a point of U, closed or not, is isomorphic to $\mathbb{P}^{d-2}$. By generic flatness, possibly shrinking U, we may assume that f is flat over U. Since the geometric fibers of f over U are smooth, f is smooth over U [Har77, Theorem III.10.2]. Thus $U\subset Z^\circ$, and Z° is non-empty. In particular, the fiber of f over any closed point of Z° is isomorphic to $\mathbb{P}^{d-2}$. □

The Wronskians of Construction-Theorem 8.1 applied to $|-mK_X|$ have divisors in the class $f^*(-cK_Z)-cR(f)$, with c as in Construction-Theorem 8.1(2), that is, pullbacks of divisors on Z up to the fixed vertical divisor $cR(f)\geq 0$, only when the jet order is exactly $d-1$ times the degree m. Otherwise there remains a positive multiple of $-K_X+f^*K_Z$, which does not descend to Z. Therefore we normalize the classes on a fixed smooth projective surface T with a morphism $h:T\to X$ satisfying the conditions of Lemma 9.5 and pass to the limit, using the continuity in [ELM+09, Theorem 6.2].

Lemma 9.5. *Notation and conditions as in Set-up 9.1. Let $x\in X$ be a very general point. Let T be a smooth projective surface, $h:T\to X$ a morphism, and $t\in T$ a point such that $h(t)=x$, the restriction of h to an open neighborhood of t is a closed immersion into an open subset of X, and $f\circ h$ is étale at t. Then $\epsilon_{\mathrm{mov}}((f\circ h)^*(-K_Z),t)\geq 1$.*

Proof. By Lemma 9.3, Z is $\mathbb{Q}$-factorial and $f^{-1}(Z_{\mathrm{sing}})$ has codimension at least 2 in X. Thus the conditions of Construction-Theorem 8.1 hold for f, with $r=d-2\geq 1$. Let $\mathcal{F}$, X°, $K_{\mathcal{F}}$, and $R(f)$ be as in Construction-Theorem 8.1.

We first compare the class f^*K_Z of Construction-Theorem 8.1 with the usual pullback. Let l be a positive integer such that lK_Z is Cartier. The usual pullback $f^*(lK_Z)$ and the class $l\cdot f^*K_Z$

of Construction-Theorem 8.1 coincide on X°. The complement $X\backslash X^\circ$ has codimension at least 2 in X, because X is normal, $f^{-1}(Z_{\mathrm{sing}})$ has codimension at least 2 in X, and at any codimension one point of $X_{\mathrm{reg}} \cap f^{-1}(Z_{\mathrm{reg}})$ the local ring is a discrete valuation ring, so the saturated sheaf $\mathcal{F}$ and the torsion-free sheaf $T_X/\mathcal{F}$ are free there. Thus $f^*(lK_Z)$ and $l \cdot f^*K_Z$ coincide in $\mathrm{Cl}(X)$ by [Har77, Proposition II.6.5(b)]. By Construction-Theorem 8.1(1), we obtain

$$lK_{\mathcal{F}} \sim lK_X - f^*(lK_Z) - lR(f),$$

where f^* is the usual pullback, $R(f) \geq 0$, and $f(\mathrm{Supp}\, R(f))$ is a proper closed subset of Z.

Since x is very general, we may assume that $x \in X^\circ$, that $f(x) \in Z^\circ \backslash f(\mathrm{Supp}\, R(f))$, and, by Lemma 2.6(3) and Set-up 9.1(4), that $\epsilon_{\mathrm{mov}}(-K_X, x) = d-1$. By Definition-Lemma 9.4, f is smooth at x. Let $b := f(x)$. There exists an increasing sequence of positive integers m_i such that $s_i := s(-m_iK_X, x)$ satisfies $s_i/m_i \to d-1$ as $i \to +\infty$. Since $d-1 > 0$, we may assume that $s_i \geq 1$ for any i. Let

$$M_i := \binom{s_i+d-2}{d-2}, \quad c_i := \binom{s_i+d-2}{d-1}, \quad a_i := m_iM_i, \quad \lambda_i := \frac{c_i}{a_i}.$$

Since $\binom{s+r}{r+1} = \frac{s}{r+1}\binom{s+r}{r}$ for any integers $s \geq 1$ and $r \geq 0$, applied with $s = s_i$ and $r = d-2$, we have $c_i = s_iM_i/(d-1)$. Thus $\lambda_i = s_i/((d-1)m_i) > 0$ and $\lambda_i \to 1$.

Let $V^{(i)} := H^0(X, \mathcal{O}_X(-m_iK_X))$, let D_i be a Weil divisor in the class $-a_iK_X + c_iK_{\mathcal{F}}$, and let $V_i \subset H^0(X, \mathcal{O}_X(D_i))$ be the image of the Wronskian map

$$w_{s_i} : \bigwedge^{M_i} V^{(i)} \to H^0(X, \mathcal{O}_X(D_i))$$

of Construction-Theorem 8.1(2), applied with $D = -m_iK_X$ and $k = s = s_i$. Since $M_is_i - (d-2)c_i = c_i$, for any formal section σ of f through x, the restriction of V_i to σ generates c_i-jets at b.

Since X is $\mathbb{Q}$-factorial, there exists a positive integer q such that qK_X and $qK_{\mathcal{F}}$ are Cartier. Thus qD_i is Cartier. We pass to qD_i because D_i is not Cartier in general, so sections of $\mathcal{O}_X(D_i)$ cannot be pulled back to T. Let $V_i^q \subset H^0(X, \mathcal{O}_X(qD_i))$ be the image of $V_i^{\otimes q}$. Since $x \in X^\circ$ is a smooth point of X, $\mathcal{O}_X(D_i)$ is invertible near x. Fix a local generator ξ of $\mathcal{O}_X(D_i)$ at x. Then ξ^q is a local generator of $\mathcal{O}_X(qD_i)$ at x, and after dividing by ξ or ξ^q, restriction to a formal section σ of f through x is the ring homomorphism

$$\sigma^* : \mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{Z,b}.$$

Let $\widehat{\mathfrak{m}}$ be the maximal ideal of $\widehat{\mathcal{O}}_{Z,b}$. Any monomial of degree at most qc_i in a regular system of parameters of $\widehat{\mathcal{O}}_{Z,b}$ is a product of q monomials of degree at most c_i, so the descending induction in the proof of Lemma 2.6(1), with

$$\mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathfrak{m}_x^{s+1}$$

replaced by σ^* followed by the quotient map to $\widehat{\mathcal{O}}_{Z,b}/\widehat{\mathfrak{m}}^{qc_i+1}$, shows that the restriction of V_i^q to σ generates qc_i-jets at b.

Since $f \circ h$ is étale at t, it induces an isomorphism

$$\widehat{\mathcal{O}}_{Z,b} \to \widehat{\mathcal{O}}_{T,t}.$$

Since h is a closed immersion near t, the composite of

$$\mathrm{Spec}\, \widehat{\mathcal{O}}_{Z,b} \cong \mathrm{Spec}\, \widehat{\mathcal{O}}_{T,t} \to T \to X$$

is a formal section σ_T of f through x, and for any $u \in H^0(X, \mathcal{O}_X(qD_i))$ the image of h^*u in $\widehat{\mathcal{O}}_{T,t} \otimes h^*\mathcal{O}_X(qD_i)$ is the restriction of u to σ_T. Therefore $s(h^*(qD_i), t) \geq qc_i$. Let

$$L_i := \frac{1}{qa_i} h^*(qD_i).$$

By Lemma 2.6(1) on T, $s(nh^*(qD_i), t) \geq nqc_i$ for any positive integer n. Thus $\epsilon_{\mathrm{mov}}(L_i, t) \geq \lambda_i > 0$, and L_i is big by Lemma 2.6(4).

Since $x \notin \operatorname{Supp} R(f)$ and $R(f)$ is $\mathbb{Q}$-Cartier, $h^*R(f)$ is a $\mathbb{Q}$-divisor such that $h^*R(f) \geq 0$ and $t \notin \operatorname{Supp} h^*R(f)$. Let

$$L_\infty := (f \circ h)^*(-K_Z) - h^*R(f).$$

By the linear equivalence above, in $N^1(T)_\mathbb{R}$ we have

$$L_i \equiv h^*(-K_X) + \lambda_i h^* K_\mathcal{F} \to h^*(-K_X + K_\mathcal{F}) \equiv L_\infty \quad (i \to +\infty).$$

We use two facts on the smooth projective surface T at the fixed point t. By [ELM+09, Proposition 6.6], for a big $\mathbb{Q}$-divisor L on T, $\epsilon_{\mathrm{mov}}(L, t)$ is equal to the moving Seshadri constant of L at t in the sense of [ELM+09, Definition 6.1]. By [ELM+09, Theorem 6.2], the moving Seshadri constant at t in the sense of [ELM+09, Definition 6.1] is a continuous function on $N^1(T)_\mathbb{R}$ which is zero outside the big cone. Since $\epsilon_{\mathrm{mov}}(L_i, t) \geq \lambda_i$ and $\lambda_i \to 1$, the moving Seshadri constant of L_∞ at t in the sense of [ELM+09, Definition 6.1] is at least 1. Thus L_∞ is big and $\epsilon_{\mathrm{mov}}(L_\infty, t) \geq 1$.

Finally, let I be a positive integer such that IL_∞ and $Ih^*R(f)$ are divisors. For any positive integer m divisible by I, multiplication by the canonical section of $mh^*R(f)$, which does not vanish at t, maps $H^0(T, \mathcal{O}_T(mL_\infty))$ into $H^0(T, \mathcal{O}_T(m(f \circ h)^*(-K_Z)))$ and multiplies germs at t by a unit. Thus $s(m(f \circ h)^*(-K_Z), t) \geq s(mL_\infty, t)$. By Lemma 2.6(2),

$$\epsilon_{\mathrm{mov}}((f \circ h)^*(-K_Z), t) \geq \epsilon_{\mathrm{mov}}(L_\infty, t) \geq 1.$$

□

Lemma 9.5 bounds a moving Seshadri constant on T only, and since f is not assumed to have a rational section, the curves in T through t over a curve C on Z through $f(x)$ need not map birationally onto C, in which case the bound on T loses a factor equal to the degree over C. Therefore, in the proof of Theorem 9.2, for each such C we use Theorem 2.22 to find a section of f through x over an affine open subset of C, and we choose T so that $h(T)$ contains the image of the section. Since $-K_Z$ need not be nef, we work with the positive part of the Zariski decomposition of the pullback of $-K_Z$ to the minimal resolution of Z, and we show that the pullback of the decomposition to T is a Zariski decomposition.

Proof of Theorem 9.2. The proof is long, so we divide it into several steps.

Step 1. In this step we construct, through a very general point $x \in X$, triples (T, h, t) satisfying the conditions of Lemma 9.5, and prove that $-K_Z$ is big.

Let $x \in X$ be a very general point. We may assume that x is not contained in the countable union of proper closed subsets of X excluded in Lemma 9.5, so that the conclusion of Lemma 9.5 holds for x and any T, h, t as there, and by Definition-Lemma 9.4 we may assume that $f(x) \in Z^\circ \backslash Z_{\mathrm{sing}}$. Let $b := f(x)$. Thus f is smooth at x and Z is smooth at b.

Let $\Gamma \subset X$ be a closed subset containing x, with the reduced structure, and let $\mathcal{I}_\Gamma$ and $\mathcal{I}_x$ be the ideal sheaves of Γ and x in $\mathcal{O}_X$. Assume that there exist $v_1, \ldots, v_{d-2} \in \mathcal{I}_{\Gamma,x}$ whose vertical differentials, that is, images in $\Omega_{X/Z} \otimes \kappa(x)$, form a basis. Let A be a very ample divisor on X. By Serre vanishing,

$$H^1(X, (\mathcal{I}_\Gamma \cap \mathcal{I}_x^2) \otimes \mathcal{O}_X(nA)) = 0$$

for any sufficiently large integer n, so the natural map

$$H^0(X, \mathcal{I}_\Gamma \otimes \mathcal{O}_X(nA)) \to \left(\mathcal{I}_{\Gamma,x}/(\mathcal{I}_{\Gamma,x} \cap \mathcal{I}_{x,x}^2)\right) \otimes \mathcal{O}_X(nA)$$

is surjective. Thus there exist

$$u_1, \ldots, u_{d-2} \in H^0(X, \mathcal{I}_\Gamma \otimes \mathcal{O}_X(nA))$$

whose germs at x, divided by a local generator of $\mathcal{O}_X(nA)$, are congruent to $v_1, \ldots, v_{d-2}$ modulo $\mathcal{I}_{x,x}^2$. Let $W \subset X$ be the common zero locus of $u_1, \ldots, u_{d-2}$. Then $\Gamma \subset W$. The images of the local equations of W in $\mathcal{I}_{x,x}/\mathcal{I}_{x,x}^2$ are linearly independent, so W is smooth at x and $\dim_x W = 2$. Since the vertical differentials of the local equations form a basis, the tangent space of W at x intersects the tangent space of $f^{-1}(b)$ at x only in 0. Thus $d(f|_W)$ maps the tangent space

of W at x isomorphically onto that of Z at b, and $f|_W$ is étale at x. Let T_0 be the irreducible component of W through x. Then T_0 dominates Z, and since $T_0 = W$ near x, any irreducible component of Γ through x is contained in T_0. Let $T \to T_0$ be a resolution of singularities which is an isomorphism over a neighborhood of x (by Hironaka's resolution of singularities, which is an isomorphism over the smooth locus of T_0), let $h : T \to X$ be the composite with the inclusion $T_0 \subset X$, and let $t := h^{-1}(x)$. Then T is a smooth projective surface, the restriction of h to an open neighborhood of t is a closed immersion into an open subset of X, and $f \circ h$ is étale at t. We call (T, h, t) a *surface multisection* of f through x constructed from $(\Gamma; v_1, \dots, v_{d-2})$.

Since f is smooth at x, there exist a regular system of parameters z_1, z_2 of $\mathcal{O}_{Z,b}$ and $v_1, \dots, v_{d-2} \in \mathcal{I}_{x,x}$ such that $f^*z_1, f^*z_2, v_1, \dots, v_{d-2}$ is a regular system of parameters of $\mathcal{O}_{X,x}$. The vertical differentials of $v_1, \dots, v_{d-2}$ form a basis. Let (T, h, t) be a surface multisection constructed from $(\{x\}; v_1, \dots, v_{d-2})$. Then T, h, and t satisfy the conditions of Lemma 9.5. By Lemma 9.5, $\epsilon_{\mathrm{mov}}((f \circ h)^*(-K_Z), t) \geq 1$, so $(f \circ h)^*(-K_Z)$ is big by Lemma 2.6(4).

Let e be the degree of the generically finite surjective morphism $f \circ h : T \to Z$, let l be a positive integer such that lK_Z is Cartier, and let $H \geq 0$ be a very ample divisor on Z. By Kodaira's lemma [Laz04, Proposition 2.2.6], there exist a positive integer m and a divisor $G \geq 0$ on T such that

$$G \sim (f \circ h)^*(-mlK_Z - H).$$

Pushforward by $f \circ h$ preserves linear equivalence, by taking norms of rational functions under the finite extension $\mathbb{C}(Z) \subset \mathbb{C}(T)$, and $(f \circ h)_*(f \circ h)^*D = eD$ [Ful98, Proposition 2.5(c)] for any Cartier divisor D on Z. Thus $-emlK_Z \sim eH + (f \circ h)_*G$, where $(f \circ h)_*G \geq 0$. Hence $-K_Z$ is big.

Step 2. In this step we take the Zariski decomposition of the pullback of $-K_Z$ to the minimal resolution of Z.

Let $g : Z' \to Z$ be the minimal resolution. Since $-K_Z$ is big by **Step 1**, $-g^*K_Z$ is big. Let l be a positive integer such that lK_Z is Cartier. Taking the Zariski decomposition of $-lg^*K_Z$ [Laz04, Theorem 2.3.19] and dividing by l, we obtain

$$-g^*K_Z = P + N,$$

where P is a nef $\mathbb{Q}$-divisor, $N \geq 0$ is a $\mathbb{Q}$-divisor, the intersection matrix of the components of N is negative definite if $N \neq 0$, and $(P \cdot S) = 0$ for any component S of N. By [Laz04, Corollary 2.3.22], $(P^2) = \mathrm{vol}(-g^*K_Z) > 0$, so P is big [Laz04, Theorem 2.2.16].

Step 3. In this step we show that the Zariski decomposition of **Step 2** is preserved under pullback by generically finite morphisms and compare moving Seshadri constants.

Let T be a smooth projective surface and $h' : T \to Z'$ a generically finite surjective morphism. We show that

$$h'^*(-g^*K_Z) = h'^*P + h'^*N$$

is the Zariski decomposition, and that

$$\epsilon_{\mathrm{mov}}(h'^*(-g^*K_Z), t) = \epsilon_{\mathrm{mov}}(h'^*P, t) \quad \text{for any } t \in T \setminus \operatorname{Supp} h'^*N. \tag{9.1}$$

The $\mathbb{Q}$-divisor h'^*P is nef and $((h'^*P)^2) = \deg(h') \cdot (P^2) > 0$. Let S be a component of h'^*N. Then $h'(S)$ is a point or a component of N, so by the projection formula $(h'^*P \cdot S) = (P \cdot h'_*S) = 0$. By the Hodge index theorem [Har77, Theorem V.1.9 and Remark V.1.9.1] and $((h'^*P)^2) > 0$, the intersection form is negative definite on the classes in $N^1(T)_{\mathbb{R}}$ orthogonal to h'^*P. It remains to show that the classes of the components of h'^*N are linearly independent in $N^1(T)_{\mathbb{R}}$. Suppose that $G_+ \equiv G_-$, where $G_+, G_- \geq 0$ are $\mathbb{R}$-linear combinations of components of h'^*N without common components. Then $(G_+^2) = (G_+ \cdot G_-) \geq 0$. If $G_+ \neq 0$, then G_+ is not numerically trivial, since $(G_+ \cdot H_T) > 0$ for an ample divisor H_T on T, and G_+ is orthogonal to h'^*P, so $(G_+^2) < 0$, a contradiction. Thus $G_+ = 0$, and similarly $G_- = 0$. Hence the intersection matrix of the components of h'^*N is negative definite, and by the uniqueness in [Laz04, Theorem 2.3.19], applied to $lh'^*(-g^*K_Z)$, the decomposition $h'^*(-g^*K_Z) = h'^*P + h'^*N$ is the Zariski decomposition.

Let $t \in T \setminus \operatorname{Supp} h'^*N$. For any positive integer m such that mlh'^*N is a divisor, by [Laz04, Proposition 2.3.21], multiplication by the canonical section of mlh'^*N is a bijection

$$H^0(T, \mathcal{O}_T(mlh'^*P)) \to H^0(T, \mathcal{O}_T(mlh'^*(-g^*K_Z))).$$

Since the canonical section of mlh'^*N does not vanish at t, $s(mlh'^*P, t) = s(mlh'^*(-g^*K_Z), t)$. By Lemma 2.6(2), (9.1) holds.

Step 4. In this step we prove that $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$.

Recall that $x \in X$ is the very general point of **Step 1** and $b = f(x)$. Since N depends only on Z and b is a very general point of Z, we may assume that $b \notin g(\operatorname{Supp} N)$. Since $b \in Z_{\mathrm{reg}}$, g is an isomorphism over a neighborhood of b. Let $b' := g^{-1}(b)$. Then b' is a very general point of Z' and $b' \notin \operatorname{Supp} N$.

We first prove the following inequality. Let Y be a smooth projective surface, L a $\mathbb{Q}$-divisor on Y, and $y \in Y$ a point such that $\epsilon_{\mathrm{mov}}(L, y) > 0$. Let $\Gamma_0 \subset Y$ be an integral curve through y. Then

$$(L \cdot \Gamma_0) \geq \epsilon_{\mathrm{mov}}(L, y) \operatorname{mult}_y \Gamma_0. \tag{9.2}$$

Let m be a positive integer such that mL is a divisor and $s := s(mL, y) \geq 0$. The tangent cone of Γ_0 at y is a finite union of lines in the tangent plane of Y at y. Let $\ell \in \mathfrak{m}_y$ be a local parameter whose differential does not vanish on any line of the tangent cone of Γ_0 at y. Since $H^0(Y, \mathcal{O}_Y(mL))$ generates s-jets at y, there exists a section of $\mathcal{O}_Y(mL)$ which is ℓ^s times a local generator modulo $\mathfrak{m}_y^{s+1}$. Let $G_0 \geq 0$ be the divisor of the section. Then $\operatorname{mult}_y G_0 = s$. If $s = 0$, then $y \notin \operatorname{Supp} G_0$. If $s \geq 1$, then the tangent cone of G_0 at y is the line defined by ℓ. In both cases $\Gamma_0 \not\subset \operatorname{Supp} G_0$. Let $\pi : \tilde{Y} \to Y$ be the blow-up of Y at y, and G_0' and Γ_0' the strict transforms of G_0 and Γ_0. By [Har77, Propositions V.3.2 and V.3.6],

$$(mL \cdot \Gamma_0) = (G_0 \cdot \Gamma_0) = (\pi^* G_0 \cdot \pi^* \Gamma_0) = (G_0' \cdot \Gamma_0') + s \operatorname{mult}_y \Gamma_0 \geq s \operatorname{mult}_y \Gamma_0.$$

Choosing a sequence of such m with $s(mL, y)/m \to \epsilon_{\mathrm{mov}}(L, y) > 0$ and dividing by m, we obtain (9.2).

Let $C' \subset Z'$ be an integral curve through b'. Then $C := g(C')$ is an integral curve through b and C' is the strict transform of C. Since $C \cap Z^\circ$ is an open subset of C containing b, there exists an affine open subset $C_0 \subset C \cap Z^\circ$ containing b. By Definition-Lemma 9.4, $X \times_Z C_0 \to C_0$ is a Brauer–Severi scheme of relative dimension $d-2$ over the integral affine curve C_0. By Theorem 2.22, $X \times_Z C_0 \cong \mathbb{P}^{d-2} \times C_0$ over C_0. Let $p \in \mathbb{P}^{d-2}$ be the point such that x corresponds to (p, b), and let $\sigma_C : C_0 \to X$ be the section of f over C_0 corresponding to $c \to (p, c)$. Then $\sigma_C(b) = x$. Let Γ be the closure of $\sigma_C(C_0)$ in X, with the reduced structure. Then Γ is an integral curve through x, and $f|_\Gamma : \Gamma \to C$ is birational and an isomorphism from a neighborhood of x in Γ onto a neighborhood of b in C.

Let z_1, z_2 and $v_1, \ldots, v_{d-2}$ be as in **Step 1**. For any j, $\sigma_C^* v_j \in \mathcal{O}_{C,b}$, and we choose $a_j \in \mathcal{O}_{Z,b}$ whose image in $\mathcal{O}_{C,b}$ is $\sigma_C^* v_j$. Let $v_j' := v_j - f^* a_j$. Then v_j' vanishes on $\sigma_C(C_0)$ near x, so $v_j' \in \mathcal{I}_{\Gamma,x}$. Since the vertical differential of $f^* a_j$ is zero, the vertical differentials of $v_1', \ldots, v_{d-2}'$ form a basis. Let (T, h, t) be a surface multisection constructed from $(\Gamma; v_1', \ldots, v_{d-2}')$. Then T, h, and t satisfy the conditions of Lemma 9.5, and $\Gamma \subset h(T)$. Since g is an isomorphism over a neighborhood of $b = f(h(t))$, the rational map $g^{-1} \circ f \circ h : T \dashrightarrow Z'$ is defined near t. Let $\mu : \tilde{T} \to T$ be a resolution of singularities of the closure of the graph of $g^{-1} \circ f \circ h$ in $T \times Z'$ which is an isomorphism over the open subset of T where $g^{-1} \circ f \circ h$ is defined (by Hironaka's resolution of singularities, which is an isomorphism over the smooth locus of the closure of the graph), and let $h' : \tilde{T} \to Z'$ be the induced morphism. We replace T by $\tilde{T}$, h by $h \circ \mu$, and t by $\mu^{-1}(t)$. The new triple is in general not constructed from $(\Gamma; v_1', \ldots, v_{d-2}')$, but T is a smooth projective surface, $g \circ h' = f \circ h$, h' is generically finite and surjective, and T, h, t still satisfy the conditions of Lemma 9.5, since μ is an isomorphism near t. Let $\Gamma_T \subset T$ be the strict transform of Γ. Since h is an isomorphism near t onto a neighborhood of x in $h(T)$, $h'|_{\Gamma_T} : \Gamma_T \to C'$ is birational and an isomorphism from a neighborhood of t in Γ_T onto a neighborhood of b'

in C'. Thus $h'_*\Gamma_T = C'$. Since the multiplicity at a point of a curve on a smooth surface is the multiplicity of the local ring of the curve at the point [Har77, Exercise V.3.4(d)], the local isomorphism implies the first equality below, and the projection formula implies the second:

$$\operatorname{mult}_t \Gamma_T = \operatorname{mult}_{b'} C', \qquad (h'^*P \cdot \Gamma_T) = (P \cdot C'). \tag{9.3}$$

By Lemma 9.5 and $(f \circ h)^*(-K_Z) = h'^*(-g^*K_Z)$, $\epsilon_{\mathrm{mov}}(h'^*(-g^*K_Z), t) \geq 1$. Since $h'(t) = b' \notin \operatorname{Supp} N$, $t \notin \operatorname{Supp} h'^*N$. By (9.1), $\epsilon_{\mathrm{mov}}(h'^*P, t) \geq 1$. Applying (9.2) with $Y = T$, $y = t$, $L = h'^*P$, and $\Gamma_0 = \Gamma_T$, and using (9.3), we obtain

$$(P \cdot C') = (h'^*P \cdot \Gamma_T) \geq \operatorname{mult}_t \Gamma_T = \operatorname{mult}_{b'} C'.$$

Since C' is an arbitrary integral curve through b', [Laz04, Proposition 5.1.5], applied to $l'P$ for a positive integer l' such that $l'P$ is a divisor, implies that $\epsilon(P, b') \geq 1$. Since P is big and b' is a very general point of Z', Lemma 2.9(4) implies that $\epsilon_{\mathrm{mov}}(P, b') = \epsilon(P, b') \geq 1$. By (9.1) applied to $h' = \mathrm{id}_{Z'}$ and $t = b'$, $\epsilon_{\mathrm{mov}}(-g^*K_Z, b') \geq 1$.

Finally, let l be as in **Step 2**. For any positive integer m divisible by l, the projection formula and $g_*\mathcal{O}_{Z'} = \mathcal{O}_Z$ imply that g^* induces an isomorphism

$$H^0(Z, \mathcal{O}_Z(-mK_Z)) \to H^0(Z', \mathcal{O}_{Z'}(-mg^*K_Z)).$$

Since g is an isomorphism near b', $s(-mK_Z, b) = s(-mg^*K_Z, b')$. By Lemma 2.6(2), and since b is a smooth point of Z,

$$\epsilon_{\mathrm{mov}}(-K_Z) \geq \epsilon_{\mathrm{mov}}(-K_Z, b) = \epsilon_{\mathrm{mov}}(-g^*K_Z, b') \geq 1.$$

Together with **Step 1**, this proves Theorem 9.2. □

10. Curve slices and the CM divisor

In this section, we prove Theorem 10.2 and Theorem 10.3. Together they are used in Section 12 for the volume estimates.

10.1. **Curve slices.** In this subsection, we prove Theorem 10.2.

Definition 10.1. Notation and conditions as in Set-up 9.1, and let $K_{X/Z} := K_X - f^*K_Z$, which is $\mathbb{Q}$-Cartier by Lemma 9.3. We define the $\mathbb{Q}$-divisor class

$$\Lambda := -(d-1)^{1-d} f_*\big((-K_{X/Z})^{d-1} \cdot [X]\big)$$

on Z, where f_* denotes the proper pushforward of cycle classes.

The sign and the factor $(d-1)^{1-d}$ in Definition 10.1 are normalized so that the inequality of Theorem 10.2 below takes the form $((-K_Z - \Lambda) \cdot H) \geq 1$. By Theorem 10.3, Λ is numerically equivalent to a $\mathbb{Q}$-divisor ≥ 0 on Z supported in $Z \backslash Z^\circ$.

In Theorem 10.2 below, the key point is that the restriction of $-K_X$ to $f^{-1}(C)$ is nef for a very general member C of a very ample linear system on Z, which follows from Lemma 7.2. See Remark 7.1 for the difference from [Zhu18, Lemma 5.4].

Theorem 10.2. *Notation and conditions as in Set-up 9.1. Let H be a very ample Cartier divisor on Z, $C \in |H|$ a very general member, $T := f^{-1}(C)$, and $L := -K_X|_T$. Then C is a smooth curve contained in Z_{reg}, T is a normal projective variety with klt singularities, L is nef and big, $\epsilon(L, x) = d-1$ for a very general point $x \in T$, and*

$$(L^{d-1}) = ((-K_X)^{d-1} \cdot f^*H) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H) \geq (d-1)^{d-1}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we list the conditions on C, and we show that C is a smooth curve contained in Z_{reg} and that T is a normal projective variety with klt singularities.

Fix an integer $I > 0$ such that IK_X is Cartier. Then $L = I^{-1}(-IK_X)|_T$ is a well-defined $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T. By Lemma 2.6(3) applied to $-K_X$ and by Lemma 2.6(2), there exist proper closed subsets $\Sigma_1, \Sigma_2, \dots$ of X such that any smooth point $y \in X \setminus \bigcup_{j\geq 1} \Sigma_j$ satisfies

$$\limsup_{m\to+\infty,\ I|m} \frac{s(-mK_X, y)}{m} = \epsilon_{\mathrm{mov}}(-K_X, y) = \epsilon_{\mathrm{mov}}(-K_X) = d - 1. \tag{10.1}$$

Let $q \geq 2$ be an integer. The hypotheses of Theorem 2.19 with $a = d-1$ and $t = 1/q$ hold: f is a contraction with $0 < \dim Z = 2 < d = \dim X$, K_X is $\mathbb{Q}$-Cartier since X is terminal, the general fiber of f is isomorphic to $\mathbb{P}^{d-2} = \mathbb{P}^{a-1}$ by Definition-Lemma 9.4, $a \geq 2$ since $d \geq 3$, $0 < t \leq a$, and $\epsilon_{\mathrm{mov}}(-K_X) = d - 1 > a - t$. Thus there exist a $\mathbb{Q}$-divisor $D_q \geq 0$ on X and a non-empty open subset $U_q \subset Z$ such that $D_q \sim_{\mathbb{Q}} -qK_X$ and $(f^{-1}(U_q), D_q|_{f^{-1}(U_q)})$ is klt.

Let $\mathcal{V}$ be the countable set of irreducible components of X_{sing}, of Σ_j for $j \geq 1$, and of $\operatorname{Supp} D_q$ for $q \geq 2$. Any $V \in \mathcal{V}$ is a proper closed subset of X. Let $\mathcal{W}$ be the countable set consisting of the points of Z_{sing}, the irreducible components of $Z \setminus Z^\circ$ and of $Z \setminus U_q$ for $q \geq 2$, and the sets $f(V)$ for $V \in \mathcal{V}$ with $f(V) \neq Z$. Any member of $\mathcal{W}$ is a point or an irreducible curve on Z. Since C is very general, we may assume that the following conditions hold.

(1) C contains no member of $\mathcal{W}$.
(2) $C \cap Z_{\mathrm{reg}}$ and $T \cap X_{\mathrm{reg}}$ are smooth.
(3) T is a general member of the free linear system $f^*|H|$ on X, and $T \cap f^{-1}(U_q)$ is a general member of the restriction of $f^*|H|$ to $f^{-1}(U_q)$ for any $q \geq 2$, in the sense of [KM98, Lemma 5.17].

Since $|H|$ is base point free, a general member of $|H|$ contains no given point and no given irreducible curve of Z, so (1) is a countable intersection of non-empty open conditions on $|H|$. Condition (2) holds for a general C by [Har77, Corollary III.10.9, Remark III.10.9.2], applied to the base point free linear systems $|H|$ on Z_{reg} and $f^*|H|$ on X_{reg}. Condition (3) is a countable intersection of non-empty open conditions by definition.

By (1), $C \cap Z_{\mathrm{sing}} = \emptyset$, so $C \subset Z_{\mathrm{reg}}$, and C is smooth by (2). Since $C \geq 0$ is an ample Cartier divisor on the normal projective surface Z, C is connected [Har77, Corollary III.7.9]. Thus C is a smooth projective curve. Since C is irreducible and contains no irreducible component of $Z \setminus Z^\circ$ or of $Z \setminus U_q$, we have $C \cap Z^\circ \neq \emptyset$ and $C \cap U_q \neq \emptyset$ for any $q \geq 2$.

Next, $T = f^*C \geq 0$ is a Cartier divisor on X, and $T \not\supset V$ for any $V \in \mathcal{V}$: if $f(V) = Z$, then $f(V) \not\subset C$, and otherwise $f(V) \not\subset C$ by (1). Since X is terminal, X has rational singularities [KM98, Theorem 5.22], hence X is Cohen–Macaulay [KM98, Theorem 5.10]. Thus the Cartier divisor T is Cohen–Macaulay [Mat89, Theorem 17.3(ii)], and in particular T satisfies S_2. By [KM98, Corollary 5.18], $\operatorname{codim}(X_{\mathrm{sing}}, X) \geq 3$. Since T contains no irreducible component of X_{sing}, we have $\operatorname{codim}(T \cap X_{\mathrm{sing}}, T) \geq 2$. Since $T \cap X_{\mathrm{reg}}$ is smooth by (2), T is normal by Serre's criterion [Mat89, Theorem 23.8]. The fibers of f are connected because $f_*\mathcal{O}_X = \mathcal{O}_Z$, so the fibers of $f|_T : T \to C$ are connected. Since C is connected, T is connected. A connected normal scheme is integral, so T is a normal projective variety and $\dim T = d - 1$. On the smooth variety $T \cap X_{\mathrm{reg}}$ we have $IK_T|_{T\cap X_{\mathrm{reg}}} \sim I(K_X + T)|_{T\cap X_{\mathrm{reg}}}$ by adjunction, and $T \setminus X_{\mathrm{reg}}$ has codimension at least 2 in T, so [Har77, Proposition II.6.5(b)] implies

$$K_T \sim_{\mathbb{Q}} (K_X + T)|_T. \tag{10.2}$$

In particular K_T is $\mathbb{Q}$-Cartier. Since X is terminal, [KM98, Lemma 5.17(1)] and (3) imply that T is klt.

Since T is integral with $\dim T = d - 1$ and $T \not\supset V$ for any $V \in \mathcal{V}$, T is not contained in Σ_j for any $j \geq 1$, and $T \not\subset \operatorname{Supp} D_q$ for any $q \geq 2$: if T were contained in an irreducible component V of Σ_j or of $\operatorname{Supp} D_q$, then $\dim V \leq d - 1 = \dim T$ would imply $T = V$.

Finally, let $z \in C \cap Z^\circ$, and let $F_z := f^{-1}(z)$. By Definition-Lemma 9.4, $f^{-1}(Z^\circ) \to Z^\circ$ is a Brauer–Severi scheme of relative dimension $d - 2$, so $F_z \cong \mathbb{P}^{d-2}$. Since Brauer–Severi schemes are smooth and $z \in Z_{\mathrm{reg}}$, X is smooth along F_z and F_z is a fiber of a smooth morphism onto a

smooth surface. Thus $K_{F_z} \sim K_X|_{F_z}$ by adjunction, and since $(f^*K_Z)|_{F_z} \sim_{\mathbb{Q}} 0$,

$$-K_{X/Z}|_{F_z} \sim_{\mathbb{Q}} -K_X|_{F_z} \sim -K_{F_z}, \qquad \mathcal{O}_{F_z}(-K_{F_z}) \cong \mathcal{O}_{\mathbb{P}^{d-2}}(d-1). \tag{10.3}$$

Step 2. In this step we fix a point $x \in T$ and show that $\epsilon_{\mathrm{mov}}(IL, x) \geq I(d-1)$.

By **Step 1**, $\Sigma_j \cap T$ is a proper closed subset of T for any $j \geq 1$, and $T \cap X_{\mathrm{reg}}$ and $T \cap f^{-1}(Z^\circ)$ are non-empty open subsets of T. Let $x \in T$ be a very general point. Then $x \in X_{\mathrm{reg}}$, $x \notin \Sigma_j$ for any $j \geq 1$, and $f(x) \in C \cap Z^\circ$. Moreover, x is a smooth point of T by **Step 1**(2), and x satisfies (10.1).

Let m be a positive integer such that $I \mid m$. Then $-mK_X$ is Cartier, and

$$\mathcal{O}_X(-mK_X)|_T = \mathcal{O}_T(mL).$$

Let $V_m := H^0(X, \mathcal{O}_X(-mK_X))$. By Lemma 2.6(5), applied to the Cartier divisor $-mK_X$ and to the normal closed subvariety $T \subset X$, the image $V_m|_T$ of V_m in $H^0(T, \mathcal{O}_T(mL))$ satisfies $s(V_m|_T, x) \geq s(-mK_X, x)$. Since $V_m|_T \subset H^0(T, \mathcal{O}_T(mL))$, Definition 2.4 implies $s(mL, x) \geq s(V_m|_T, x)$, where $s(mL, x)$ is computed on T. Thus $s(mL, x) \geq s(-mK_X, x)$. Since IL is a Cartier divisor on T, writing $m = kI$ and using (10.1), we obtain

$$\epsilon_{\mathrm{mov}}(IL, x) = \limsup_{k \to +\infty} \frac{s(kIL, x)}{k} \geq I \limsup_{m \to +\infty,\ I \mid m} \frac{s(-mK_X, x)}{m} = I(d-1). \tag{10.4}$$

Step 3. In this step we show that L is nef.

We check the hypotheses of Lemma 7.2 for $f|_T : T \to C$ and L. By **Step 1**, T is a normal projective variety and C is a smooth projective curve. The morphism $f|_T$ is surjective and the fibers of $f|_T$ are fibers of f, which are connected. Let $T \to C' \to C$ be the Stein factorization of $f|_T$. Since T is normal, C' is a smooth projective curve, and $C' \to C$ is a finite bijective morphism between smooth curves, hence an isomorphism (in characteristic zero a finite bijective morphism has degree one, and a finite birational morphism onto a smooth curve is an isomorphism). Thus $f|_T$ is a contraction.

(1) Since $-K_X$ is f-ample, L is ample over C. Since $T = f^*C$ and $\mathcal{O}_Z(C) \cong \mathcal{O}_Z(H)$, we have $T|_T \sim (f|_T)^*(H|_C)$. By (10.2),

$$K_T + L \sim_{\mathbb{Q}} (K_X + T)|_T - K_X|_T \sim_{\mathbb{Q}} (f|_T)^*(H|_C).$$

(2) By Definition-Lemma 9.4, $f^{-1}(Z^\circ) \to Z^\circ$ is a Brauer–Severi scheme, and Brauer–Severi schemes are smooth. Since $T = f^{-1}(C)$ is the scheme-theoretic preimage of C, the morphism $f|_T$ is smooth over the non-empty open subset $C \cap Z^\circ$ of C.

(3) Let $q \geq 2$ be an integer. By **Step 1**, $T \not\subset \operatorname{Supp} D_q$, so $D_q|_T \geq 0$ is a $\mathbb{Q}$-divisor on T with $D_q|_T \sim_{\mathbb{Q}} -qK_X|_T = qL$. Since $(f^{-1}(U_q), D_q|_{f^{-1}(U_q)})$ is klt, **Step 1**(3) and [KM98, Lemma 5.17(1)] (since T is general, the restriction to T of each component of D_q is a reduced divisor, so the coefficients of $D_q|_T$ are those of D_q) imply that

$$\big((f|_T)^{-1}(C \cap U_q), D_q|_{(f|_T)^{-1}(C \cap U_q)}\big)$$

is klt. Since $C \cap U_q \neq \emptyset$, $(T, D_q|_T)$ is klt over a non-empty open subset of C.

Thus L is nef by Lemma 7.2.

Step 4. In this step we show that L is big, that $\epsilon(L, x) = d-1$, and that $(L^{d-1}) \geq (d-1)^{d-1}$.

Recall that x is the very general point of T fixed in **Step 2**. By (10.4),

$$\epsilon_{\mathrm{mov}}(IL) \geq \epsilon_{\mathrm{mov}}(IL, x) \geq I(d-1) > 0.$$

Since IL is Cartier, IL is big by Lemma 2.6(4), so L is big. Since IL is nef and big by **Step 3** and x is a very general point of T, Lemma 2.9(4) and (10.4) imply

$$\epsilon(IL, x) = \epsilon_{\mathrm{mov}}(IL, x) \geq I(d-1).$$

By Definition 2.7, $\epsilon(IL, x) = I\epsilon(L, x)$, so $\epsilon(L, x) \geq d-1$.

Conversely, let $F_{f(x)} := f^{-1}(f(x))$. Since $f(x) \in C \cap Z^\circ$, **Step 1** implies $F_{f(x)} \cong \mathbb{P}^{d-2}$. Since $d \geq 3$, there exists a line $\ell \subset F_{f(x)} \subset T$ passing through x, and $\operatorname{mult}_x \ell = 1$. By (10.3), $(L \cdot \ell) = (-K_{F_{f(x)}} \cdot \ell) = d-1$. By Lemma 2.9(1),

$$\epsilon(L, x) \leq \frac{(L \cdot \ell)}{\operatorname{mult}_x \ell} = d - 1.$$

Thus $\epsilon(L, x) = d-1$. Since $\dim T = d-1$, Lemma 2.9(2) implies

$$(L^{d-1}) \geq \epsilon(L, x)^{d-1} = (d-1)^{d-1}.$$

Step 5. In this step we show that

$$(L^{d-1}) = ((-K_X)^{d-1} \cdot f^*H) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H).$$

Since $T = f^*C \geq 0$ is a Cartier divisor linearly equivalent to f^*H, we have $[T] = f^*H \cdot [X]$, and the projection formula for the inclusion $T \subset X$ implies $(L^{d-1}) = ((-K_X)^{d-1} \cdot f^*H)$. In the computation below, all cycle classes have rational coefficients, and intersections with the $\mathbb{Q}$-Cartier divisors K_X, $K_{X/Z}$, and K_Z are defined through Cartier multiples.

We first show that

$$f_*\big((-K_{X/Z})^{d-2} \cdot [X]\big) = (d-1)^{d-2}[Z]. \tag{10.5}$$

The left-hand side of (10.5) belongs to $A_2(Z)_\mathbb{Q} = \mathbb{Q}[Z]$, so it is equal to $c[Z]$ for some $c \in \mathbb{Q}$. Let $H' \in |H|$ be a general member. Since C contains no irreducible component of $Z \backslash Z^\circ$, the set $C \backslash Z^\circ$ is finite, and $H'|_C$ is a general member of the base point free linear system $|H|_C|$ on C. Thus $C \cap H'$ consists of $N := (H^2) > 0$ reduced points $z_1, \ldots, z_N \in C \cap Z^\circ$. Since $f|_T$ is smooth over $C \cap Z^\circ$ by **Step 3**, we have $f^*H \cdot f^*H' \cdot [X] = \sum_{i=1}^N [F_{z_i}]$, where $F_{z_i} = f^{-1}(z_i)$. By the projection formula [Ful98, Proposition 2.5(c)] and (10.3),

$$c(H^2) = \big((-K_{X/Z})^{d-2} \cdot f^*H \cdot f^*H'\big) = \sum_{i=1}^N \big((-K_{F_{z_i}})^{d-2}\big) = (H^2)(d-1)^{d-2}.$$

Thus $c = (d-1)^{d-2}$, which proves (10.5).

Now $-K_X = -K_{X/Z} + f^*(-K_Z)$. By the projection formula [Ful98, Proposition 2.5(c)],

$$f_*\big((-K_X)^{d-1} \cdot [X]\big) = \sum_{j=0}^{d-1} \binom{d-1}{j} (-K_Z)^j \cdot f_*\big((-K_{X/Z})^{d-1-j} \cdot [X]\big).$$

For $j \geq 2$, the class $f_*((-K_{X/Z})^{d-1-j} \cdot [X])$ belongs to $A_{j+1}(Z)_\mathbb{Q} = 0$, since $j+1 > 2 = \dim Z$. By Definition 10.1 and (10.5), the terms with $j = 0$ and $j = 1$ are equal to $-(d-1)^{d-1}\Lambda$ and $(d-1)^{d-1}(-K_Z) \cdot [Z]$, respectively. Thus

$$f_*\big((-K_X)^{d-1} \cdot [X]\big) = (d-1)^{d-1}\big((-K_Z) \cdot [Z] - \Lambda\big) \quad \text{in } A_1(Z)_\mathbb{Q}. \tag{10.6}$$

By the projection formula [Ful98, Proposition 2.5(c)] and (10.6),

$$((-K_X)^{d-1} \cdot f^*H) = \big(H \cdot f_*((-K_X)^{d-1} \cdot [X])\big) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H).$$

Combining **Step 4** and **Step 5**, we obtain

$$(L^{d-1}) = ((-K_X)^{d-1} \cdot f^*H) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H) \geq (d-1)^{d-1},$$

which finishes the proof. □

10.2. **The CM divisor.** In this subsection, we prove Theorem 10.3. Later we only use the last sentence of Theorem 10.3. Its first inequality follows from Theorem 10.2, and the second follows directly from [CP21, Theorem 7.1] applied to $T := f^{-1}(C) \to C$ for a very general $C \in |H|$, because T is normal and projective by Theorem 10.2, $-K_{T/C} = -K_{X/Z}|_T$ is $\mathbb{Q}$-Cartier and relatively ample over C, the general fiber $\mathbb{P}^{d-2}$ is K-semistable [BJ20, Corollary 7.17, Theorem B], and the degree of the CM line bundle of $T \to C$ is $(d-1)^{d-1}(\Lambda \cdot H)$. **Steps 1–4** below prove the first statement, that Λ is numerically equivalent to an effective $\mathbb{Q}$-divisor supported in $Z \backslash Z^\circ$, which implies the second inequality. The key steps are **Step 1** and **Step 3**.

Theorem 10.3. *Notation and conditions as in Set-up 9.1. Then there exists a $\mathbb{Q}$-divisor $\Lambda_{\mathrm{CM}} \geq 0$ on Z such that $\Lambda_{\mathrm{CM}} \equiv \Lambda$ and $\operatorname{Supp} \Lambda_{\mathrm{CM}} \subset Z \backslash Z^\circ$ (see Definition-Lemma 9.4 for Z°). Moreover, $1 \leq ((-K_Z - \Lambda) \cdot H) \leq (-K_Z \cdot H)$ for any very ample Cartier divisor H on Z.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct a line bundle Q on Z_{reg} and a rational section σ of Q which is regular and nowhere zero on $Z^\circ \cap Z_{\mathrm{reg}}$.

By Lemma 9.3, Z is $\mathbb{Q}$-factorial and no prime divisor on X is mapped by f to a point. Any irreducible component F of a fiber of f satisfies $\dim F \geq d-2$, and if $\dim F = d-1$, then F is a prime divisor on X mapped by f to a point. Thus $\dim f^{-1}(z) = d-2$ for any $z \in Z$. Since X is terminal, X has rational singularities, and hence X is Cohen–Macaulay [KM98, Theorems 5.22 and 5.10]. Therefore f is flat over Z_{reg} [Mat89, Theorem 23.1].

Let I be a positive integer such that IK_X and IK_Z are Cartier and $\mathcal{O}_X(-IK_{X/Z})$ is very ample over Z. Such I exists because X and Z are $\mathbb{Q}$-factorial and $-K_{X/Z}$ is f-ample. For any $z \in Z$, let $X_z := f^{-1}(z)$. By relative Serre vanishing [Har77, Theorem III.8.8(c)] and [Har77, Theorem III.12.11], there exists a positive integer q_0 such that

$$H^i\big(X_z, \mathcal{O}_X(-qIK_{X/Z})|_{X_z}\big) = 0$$

for any $z \in Z_{\mathrm{reg}}$, any integer $q \geq q_0$, and any $i > 0$, and such that for any $q \geq q_0$ the sheaf

$$V_q := f_*\mathcal{O}_X(-qIK_{X/Z})|_{Z_{\mathrm{reg}}}$$

is locally free on Z_{reg}. Moreover, let $q \geq q_0$ be an integer, let $u : S \to Z_{\mathrm{reg}}$ be a morphism from a locally noetherian scheme S, let $f_S : X_S := X \times_Z S \to S$ be the base change of f, and let $\mathcal{L}_S$ be the pullback of $\mathcal{O}_X(-qIK_{X/Z})$ to X_S. Then the natural map $u^*V_q \to f_{S*}\mathcal{L}_S$ is an isomorphism: we may assume that S is affine, hence noetherian, and by [Har77, Theorem III.12.11] applied to f_S (which is projective, with $\mathcal{L}_S$ flat over S since f is flat over Z_{reg}) and by [Har77, Proposition III.9.3], both sheaves are locally free, and at any point $s \in S$ the map is an isomorphism between their fibers, which are both

$$H^0(X_{u(s)}, \mathcal{O}_X(-qIK_{X/Z})|_{X_{u(s)}}) \otimes k(s).$$

Let

$$Q := \bigotimes_{j=0}^{d-1} \big(\det V_{q_0+j}\big)^{\otimes (-1)^{d-j}\binom{d-1}{j}}.$$

By [KM76, Theorem 4] (see also [CP21, Notation 3.4]), there exist line bundles $M_0, \dots, M_{d-1}$ on Z_{reg} such that

$$\det V_q \cong \bigotimes_{j=0}^{d-1} M_j^{\otimes \binom{q}{j}}$$

for any $q \geq q_0$. For $0 \leq i \leq d-1$,

$$\sum_{j=0}^{d-1} (-1)^{d-1-j} \binom{d-1}{j} \binom{q_0+j}{i} = \binom{q_0}{i-d+1},$$

which equals 1 for $i = d-1$ and 0 for $i \leq d-2$. Thus $Q \cong M_{d-1}^{-1}$, that is, Q is the inverse of the leading term of the Knudsen–Mumford expansion of $\mathcal{O}_X(-IK_{X/Z})|_{f^{-1}(Z_{\mathrm{reg}})}$.

Let $U := Z^\circ \cap Z_{\mathrm{reg}}$, which is a non-empty open subset of Z by Definition-Lemma 9.4. The morphism $f^{-1}(U) \to U$ is a Brauer–Severi scheme, but f is not assumed to have a rational section, so we cannot trivialize $Q|_U$ by a splitting of f. Instead, we use that PGL_{d-1} has no non-trivial character.

For any positive integer q, let

$$W_q := H^0(\mathbb{P}^{d-2}, \omega_{\mathbb{P}^{d-2}}^{\otimes(-qI)}).$$

The group $\mathrm{PGL}_{d-1} = \mathrm{Aut}(\mathbb{P}^{d-2})$ acts on $\omega_{\mathbb{P}^{d-2}}$ by pullback of forms, hence acts linearly on W_q and on the line $\det W_q$. Let $\chi_q : \mathrm{PGL}_{d-1} \to \mathbb{G}_m$ be the character of the action on $\det W_q$. Then χ_q is trivial: the composition of χ_q with the surjection $\mathrm{SL}_{d-1} \to \mathrm{PGL}_{d-1}$ is a character of SL_{d-1}, SL_{d-1} is generated by unipotent elements, and $\mathbb{G}_m$ contains no non-trivial unipotent element. Fix a non-zero vector $w_q \in \det W_q$ for any q.

Since $f^{-1}(U) \to U$ is a Brauer–Severi scheme, it is trivial locally in the étale topology [Gro68, 8.1 and Corollary 8.3]: there exist an étale surjective morphism $h : U' \to U$ and an isomorphism

$$\phi : \mathbb{P}^{d-2} \times U' \to f^{-1}(U) \times_U U'$$

over U'. Since f is smooth over U, $\mathcal{O}_X(K_{X/Z})|_{f^{-1}(U)}$ is the relative canonical sheaf of $f^{-1}(U) \to U$. Thus ϕ induces an isomorphism $W_q \otimes \mathcal{O}_{U'} \cong h^* V_q$ for any $q \geq q_0$, and the image of $w_q \otimes 1$ is a nowhere-zero section of $h^* \det V_q$. If ϕ' is another isomorphism, then $\phi^{-1} \circ \phi'$ is given by a morphism $\gamma : U' \to \mathrm{PGL}_{d-1}$, and the section defined by ϕ' is the section defined by ϕ multiplied by $\chi_q \circ \gamma = 1$. Hence the pullbacks of the section along both projections $U' \times_U U' \to U'$ coincide, and by étale descent the section descends to a nowhere-zero section σ_q of $\det V_q|_U$, which does not depend on h and ϕ. In particular, for any morphism $u : U'' \to U$ from a noetherian scheme U'', $u^*\sigma_q$ is the section descended from the image of $w_q \otimes 1$ under the isomorphism induced by any étale-local trivialization of $f^{-1}(U) \times_U U'' \to U''$.

Let σ be the section of $Q|_U$ corresponding to

$$\bigotimes_{j=0}^{d-1} \sigma_{q_0+j}^{\otimes(-1)^{d-j}\binom{d-1}{j}}.$$

Then σ is nowhere zero on U. Since U is dense in Z_{reg}, σ is a rational section of Q on Z_{reg}, and

$$\mathrm{Supp}\,\mathrm{div}(\sigma) \subset Z_{\mathrm{reg}} \backslash U = Z_{\mathrm{reg}} \backslash Z^\circ.$$

Replacing the vectors w_q by non-zero multiples multiplies σ by a non-zero constant, so $\mathrm{div}(\sigma)$ does not depend on the vectors w_q.

Step 2. In this step we compute the degree of Q on curves in Z_{reg} in terms of the CM line bundle.

First let $g : T \to C$ be a flat projective morphism with connected fibers from a normal projective variety T with $\dim T = d-1$ onto a smooth projective curve C. Assume that $IK_{T/C}$ is Cartier, $-K_{T/C}$ is g-ample, $\mathcal{O}_T(-IK_{T/C})$ is very ample over C, and

$$H^i\big(g^{-1}(c), \mathcal{O}_T(-qIK_{T/C})|_{g^{-1}(c)}\big) = 0$$

for any $c \in C$, any $q \geq q_0$, and any $i > 0$. Then for any $q \geq q_0$, $g_*\mathcal{O}_T(-qIK_{T/C})$ is locally free and commutes with base change [Har77, Theorem III.12.11], and we let

$$Q_g := \bigotimes_{j=0}^{d-1} \big(\det g_*\mathcal{O}_T(-(q_0+j)IK_{T/C})\big)^{\otimes(-1)^{d-j}\binom{d-1}{j}}.$$

Let λ_g be the CM line bundle of g [CP21, Definition 3.1]. Then $\deg \lambda_g = -((-K_{T/C})^{d-1})$ [CP21, Remark 3.2]. As in **Step 1**, [KM76, Theorem 4] implies that $Q_g \cong M_{d-1}(g)^{-1}$, where $M_{d-1}(g)$

is the leading term of the Knudsen–Mumford expansion of $\mathcal{O}_T(-IK_{T/C})$ [CP21, Notation 3.4]. By [CP21, Proposition 3.7(b)], $c_1(M_{d-1}(g)) = -I^{d-1}\lambda_g$. Thus

$$\deg Q_g = I^{d-1}\deg \lambda_g = -I^{d-1}\big((-K_{T/C})^{d-1}\big). \tag{10.7}$$

Now let $C \subset Z_{\mathrm{reg}}$ be a smooth projective curve such that $C \not\subset Z_{\mathrm{reg}}\backslash U$ and $T := f^{-1}(C)$ is normal and connected, and let $g := f|_T : T \to C$. Since T is connected and normal, T is integral. The morphism g is flat with connected fibers, because f is flat over Z_{reg} and has connected fibers. We show that $K_{T/C} = K_{X/Z}|_T$. Since $\dim X_{\mathrm{sing}} \le d-3$ [KM98, Corollary 5.18] and T is normal, the complement of $T^\circ := T_{\mathrm{reg}} \cap X_{\mathrm{reg}}$ in T is of codimension at least 2. On T°, the adjunction formula implies $K_T|_{T^\circ} = (K_X + T)|_{T^\circ}$, and on the smooth surface Z_{reg} it implies $K_C = (K_Z + C)|_C$. Since $T = f^*C$, we obtain $K_{T/C}|_{T^\circ} = K_{X/Z}|_{T^\circ}$, hence $K_{T/C} = K_{X/Z}|_T$ because T is normal. Thus $\mathcal{O}_T(-qIK_{T/C}) \cong \mathcal{O}_X(-qIK_{X/Z})|_T$ for any q. Since $\mathcal{O}_X(-IK_{X/Z})$ is very ample over Z, $\mathcal{O}_T(-IK_{T/C}) \cong \mathcal{O}_X(-IK_{X/Z})|_T$ is very ample over C. Hence g satisfies the conditions of the previous paragraph, $V_q|_C \cong g_*\mathcal{O}_T(-qIK_{T/C})$ for $q \ge q_0$, and $Q|_C \cong Q_g$. Since $C \not\subset Z_{\mathrm{reg}}\backslash U$, $\sigma|_C$ is a rational section of Q_g, and by the base change property of the sections σ_q in **Step 1**, $\sigma|_{C\cap U}$ is the section defined in the last two paragraphs of **Step 1** with $g^{-1}(C\cap U) \to C\cap U$ in place of $f^{-1}(U) \to U$. By (10.7),

$$\deg \operatorname{div}(\sigma|_C) = \deg Q_g = -I^{d-1}\big((-K_{T/C})^{d-1}\big). \tag{10.8}$$

Step 3. In this step we show that $\operatorname{div}(\sigma) \ge 0$.

To bound each coefficient of $\operatorname{div}(\sigma)$, and not only the intersection numbers $(\operatorname{div}(\sigma)\cdot H)$, we restrict to a curve C transverse to one component D of $\operatorname{div}(\sigma)$ at a general point p, and we use Theorem 2.22 to replace the fibers over $C\backslash\{p\}$ by a trivial family. The family $g_p : T_p \to C$ constructed below has only one fiber, over p, that may fail to be isomorphic to $\mathbb{P}^{d-2}$, and [CP21, Theorem 7.1] has no condition on special fibers.

Let D be a prime divisor on Z_{reg} which is a component of $\operatorname{div}(\sigma)$, and let c be the coefficient of D in $\operatorname{div}(\sigma)$. Then $D \subset Z_{\mathrm{reg}}\backslash Z^\circ$. Since D is Cartier on Z_{reg}, any irreducible component P of $f^{-1}(D)$ is a prime divisor on X, and $f(P) = D$ by Lemma 9.3. Let

$$V := \big(f^{-1}(D)_{\mathrm{red}}\big)_{\mathrm{sing}} \cup \big(X_{\mathrm{sing}} \cap f^{-1}(D)\big).$$

Then $\dim V \le d-2$, because $f^{-1}(D)_{\mathrm{red}}$ is reduced and $\dim X_{\mathrm{sing}} \le d-3$ [KM98, Corollary 5.18]. Any irreducible component of V dominating D has fibers of dimension at most $d-3$ over a dense open subset of D, and the components of V not dominating D are mapped to finitely many points. By generic smoothness [Har77, Corollary III.10.7], applied to the connected components of the smooth variety $(f^{-1}(D)_{\mathrm{red}}\backslash V) \cap f^{-1}(D_{\mathrm{reg}})$, there exists a dense open subset $D' \subset D_{\mathrm{reg}}$ over which $f^{-1}(D)_{\mathrm{red}}\backslash V \to D$ is smooth.

Let $p \in D'$ be a general point. Then p is not contained in any other component of $\operatorname{div}(\sigma)$, $\dim(f^{-1}(p)\cap V) \le d-3$, and $f^{-1}(D)_{\mathrm{red}}\backslash V \to D$ is smooth at any point of $f^{-1}(p)\backslash V$. Since $p \in Z_{\mathrm{reg}}$, $f^{-1}(p)$ is locally defined by two equations in X, so any irreducible component F of $f^{-1}(p)$ satisfies $\dim F \ge d-2$, hence $\dim F = d-2$ by **Step 1**. Thus $F \not\subset V$.

Let H' be a very ample Cartier divisor on Z, and let C be a general member of the linear system $|H' \otimes \mathfrak{m}_p|$ of members of $|H'|$ containing p. Since H' is very ample, the members of $|H' \otimes \mathfrak{m}_p|$ have local equations at p whose classes cover $\mathfrak{m}_p/\mathfrak{m}_p^2$. Thus a local equation t of C at p satisfies $t \notin \mathfrak{m}_p^2$ and $t|_D \notin \mathfrak{m}_{D,p}^2$, that is, C is smooth at p and $(C\cdot D)_p = 1$. The base locus of $|H' \otimes \mathfrak{m}_p|$ is $\{p\}$, so by Bertini [Har77, Corollary III.10.9 and Remark III.10.9.2], C is smooth on $Z_{\mathrm{reg}}\backslash\{p\}$, and $C \cap Z_{\mathrm{sing}} = \emptyset$ because Z_{sing} is finite and $p \notin Z_{\mathrm{sing}}$. By [Har77, Corollary III.7.9], C is connected. Thus C is a smooth projective curve contained in Z_{reg}. Moreover, $C \not\subset Z_{\mathrm{reg}}\backslash U$, and $C\backslash Z^\circ$ is a finite set containing p.

We show that $T := f^{-1}(C)$ is normal and connected. Let $x \in f^{-1}(p)\backslash V$. Then X and $f^{-1}(D)_{\mathrm{red}}$ are smooth at x, and $f^{-1}(D)_{\mathrm{red}} \to D$ is smooth at x. Since $(C\cdot D)_p = 1$, $t|_D$ is a uniformizer of D at p. Since $f^{-1}(D)_{\mathrm{red}} \to D$ is smooth at x, the pullback of $t|_D$ to $f^{-1}(D)_{\mathrm{red}}$

is not contained in the square of the maximal ideal of $f^{-1}(D)_{\mathrm{red}}$ at x. Hence $f^*t \notin \mathfrak{m}_{X,x}^2$, and $T = \{f^*t = 0\}$ is smooth at x. On $X_{\mathrm{reg}} \backslash f^{-1}(p)$, the linear system $f^*|H' \otimes \mathfrak{m}_p|$ is base point free, so $T \cap X_{\mathrm{reg}}$ is smooth outside $f^{-1}(p)$ by Bertini [Har77, Corollary III.10.9 and Remark III.10.9.2]. Thus

$$T_{\mathrm{sing}} \subset (T \cap X_{\mathrm{sing}}) \cup (f^{-1}(p) \cap V),$$

and both sets on the right-hand side are of dimension at most $d-3 = \dim T - 2$. Since T is a Cartier divisor on the Cohen–Macaulay variety $f^{-1}(Z_{\mathrm{reg}})$, T is Cohen–Macaulay. By Serre's criterion [Mat89, Theorem 23.8], T is normal. Since C is connected and f has connected fibers, T is connected. Let $g := f|_T : T \to C$. Near p, $\mathrm{div}(\sigma) = cD$, so $(C \cdot D)_p = 1$ implies that $\mathrm{ord}_p(\sigma|_C) = c$.

Since $C \backslash Z^\circ$ is a finite set containing p, and the complement of a non-empty finite subset of the smooth projective curve C is affine [Har77, Exercise IV.1.3], there exists an affine open neighborhood $C_1 \subset C$ of p such that $C_1 \backslash \{p\} \subset Z^\circ$. Then $C_1 \backslash \{p\}$ is a smooth affine curve contained in Z°, and $g^{-1}(C_1 \backslash \{p\}) \to C_1 \backslash \{p\}$ is a Brauer–Severi scheme of relative dimension $d-2$ by Definition-Lemma 9.4. By Theorem 2.22, there exists an isomorphism

$$\psi : \mathbb{P}^{d-2} \times (C_1 \backslash \{p\}) \to g^{-1}(C_1 \backslash \{p\})$$

over $C_1 \backslash \{p\}$. Let $g_p : T_p \to C$ be the morphism obtained by gluing $g^{-1}(C_1) \to C_1$ and

$$\mathbb{P}^{d-2} \times (C \backslash \{p\}) \to C \backslash \{p\}$$

along ψ. Both pieces are normal and integral, and $g^{-1}(C_1 \backslash \{p\})$ is dense in both pieces, so T_p is normal and integral. Since $\{C_1, C \backslash \{p\}\}$ is an open cover of C and ψ is an isomorphism over $C_1 \backslash \{p\}$, the morphism g_p is proper and flat with connected fibers, $IK_{T_p/C}$ is Cartier, and $-K_{T_p/C}$ is g_p-ample. Moreover, $\mathcal{L} := \mathcal{O}_{T_p}(-IK_{T_p/C})$ is very ample over C: over C_1, $\mathcal{L}$ is $\mathcal{O}_T(-IK_{T/C})|_{g^{-1}(C_1)}$, over $C \backslash \{p\}$, $\mathcal{L}$ is the pullback of $\mathcal{O}_{\mathbb{P}^{d-2}}((d-1)I)$, and both the surjectivity of $g_p^* g_{p*} \mathcal{L} \to \mathcal{L}$ and the condition that the induced morphism $T_p \to \mathbb{P}(g_{p*}\mathcal{L})$ over C is a closed immersion are local on C. Hence g_p is projective and T_p is projective. The fibers of g_p over $C \backslash \{p\}$ are isomorphic to $\mathbb{P}^{d-2}$. Thus g_p satisfies the conditions of [CP21, Notation 6.1] with zero boundary. Since

$$H^i(\mathbb{P}^{d-2}, \mathcal{O}_{\mathbb{P}^{d-2}}((d-1)qI)) = 0$$

for any $q > 0$ and $i > 0$, g_p also satisfies the conditions of the first paragraph of **Step 2**.

The moment polytope of $-K_{\mathbb{P}^{d-2}}$ is the simplex

$$\Big\{ u \in \mathbb{R}^{d-2} : u_i \geq -1 \text{ for any } i,\ \sum_{i=1}^{d-2} u_i \leq 1 \Big\},$$

whose vertices are $(-1, \dots, -1)$ and $(d-1)e_i - (1, \dots, 1)$ for $1 \leq i \leq d-2$, where e_i is the i-th unit vector. The average of the vertices is the origin, and the barycenter of a simplex is the average of the vertices, so $\mathbb{P}^{d-2}$ is K-semistable [BJ20, Corollary 7.17], that is, $\delta(\mathbb{P}^{d-2}) \geq 1$ [BJ20, Theorem B]. By [CP21, Theorem 7.1], $\deg \lambda_{g_p} \geq 0$.

Since $g_p^{-1}(C_1) = g^{-1}(C_1)$ and determinants of direct images are local on the base, $Q_{g_p}|_{C_1} \cong Q_g|_{C_1}$. Over $C \backslash \{p\}$, the family $\mathbb{P}^{d-2} \times (C \backslash \{p\})$ is trivial, and the vectors w_{q_0+j} define a nowhere-zero section of $Q_{g_p}|_{C \backslash \{p\}}$ as in **Step 1**. Over $C_1 \backslash \{p\} \subset U$, by **Step 2** the section $\sigma|_C$ is the section defined in the last two paragraphs of **Step 1** from the vectors w_{q_0+j}, with $g^{-1}(C_1 \backslash \{p\}) \to C_1 \backslash \{p\}$ in place of $f^{-1}(U) \to U$. That section does not depend on the trivialization, so it may be computed with $h = \mathrm{id}_{C_1 \backslash \{p\}}$ and $\phi = \psi$. Hence $\sigma|_{C_1}$ and the section over $C \backslash \{p\}$ coincide on $C_1 \backslash \{p\}$, and they glue to a rational section σ_p of Q_{g_p} which is nowhere zero on $C \backslash \{p\}$ and satisfies $\mathrm{ord}_p \sigma_p = \mathrm{ord}_p(\sigma|_C) = c$. By (10.7) for g_p,

$$c = \deg \mathrm{div}(\sigma_p) = \deg Q_{g_p} = I^{d-1} \deg \lambda_{g_p} \geq 0.$$

Since D is an arbitrary component of $\mathrm{div}(\sigma)$, $\mathrm{div}(\sigma) \geq 0$.

Step 4. In this step we construct Λ_{CM} and prove the inequalities.

Since Z_{sing} is finite, $\mathrm{div}(\sigma)$ extends uniquely to a divisor $\overline{\mathrm{div}(\sigma)}$ on Z, the closure of $\mathrm{div}(\sigma)$. By **Step 3**, $\overline{\mathrm{div}(\sigma)} \geq 0$, and by **Step 1**, $\operatorname{Supp} \overline{\mathrm{div}(\sigma)}$ is contained in the closure of $Z_{\mathrm{reg}} \backslash Z^\circ$, hence in $Z \backslash Z^\circ$. Let

$$\Lambda_{\mathrm{CM}} := \frac{1}{I^{d-1}(d-1)^{d-1}} \overline{\mathrm{div}(\sigma)}.$$

Then $\Lambda_{\mathrm{CM}} \geq 0$ and $\operatorname{Supp} \Lambda_{\mathrm{CM}} \subset Z \backslash Z^\circ$.

Let H be a very ample Cartier divisor on Z, $C \in |H|$ a very general member, $T := f^{-1}(C)$, and $g := f|_T$. By Theorem 10.2, C is a smooth curve contained in Z_{reg} and T is a normal projective variety. Since C is very general,

$$C \not\subset \operatorname{Supp} \overline{\mathrm{div}(\sigma)} \cup (Z \backslash U).$$

Since C is connected and f has connected fibers, T is connected. Since $C \subset Z_{\mathrm{reg}}$ and $C \in |H|$,

$$\deg \mathrm{div}(\sigma|_C) = (\overline{\mathrm{div}(\sigma)} \cdot H).$$

By **Step 2**, $K_{T/C} = K_{X/Z}|_T$. Since T is integral and is the subscheme of X defined by the Cartier divisor $f^*C \geq 0$, and $C \sim H$, we have $[T] = f^*H \cap [X]$. By the projection formula [Ful98, Proposition 2.5(c)] and Definition 10.1,

$$\big((-K_{T/C})^{d-1}\big) = \big((-K_{X/Z})^{d-1} \cdot f^*H\big) = \big(f_*((-K_{X/Z})^{d-1} \cdot [X]) \cdot H\big) = -(d-1)^{d-1}(\Lambda \cdot H).$$

By (10.8),

$$(\overline{\mathrm{div}(\sigma)} \cdot H) = I^{d-1}(d-1)^{d-1}(\Lambda \cdot H),$$

that is, $(\Lambda_{\mathrm{CM}} \cdot H) = (\Lambda \cdot H)$ for any very ample Cartier divisor H on Z.

We show that $\Lambda_{\mathrm{CM}} \equiv \Lambda$. Let Γ be a curve on Z. Since Z is $\mathbb{Q}$-factorial, there exists a positive integer m such that $m\Gamma$ is Cartier. Since any Cartier divisor on Z is the difference of two very ample Cartier divisors, there exist very ample Cartier divisors H_1 and H_2 on Z such that $m\Gamma \sim H_1 - H_2$. Thus

$$\big((\Lambda_{\mathrm{CM}} - \Lambda) \cdot \Gamma\big) = \frac{1}{m}\Big(\big((\Lambda_{\mathrm{CM}} - \Lambda) \cdot H_1\big) - \big((\Lambda_{\mathrm{CM}} - \Lambda) \cdot H_2\big)\Big) = 0.$$

Since Γ is arbitrary, $\Lambda_{\mathrm{CM}} \equiv \Lambda$.

Finally, let H be a very ample Cartier divisor on Z. By Theorem 10.2,

$$(d-1)^{d-1}((-K_Z - \Lambda) \cdot H) \geq (d-1)^{d-1},$$

so $((-K_Z - \Lambda) \cdot H) \geq 1$. Since $\Lambda_{\mathrm{CM}} \geq 0$ and H is very ample, $(\Lambda \cdot H) = (\Lambda_{\mathrm{CM}} \cdot H) \geq 0$, so $((-K_Z - \Lambda) \cdot H) \leq (-K_Z \cdot H)$. □

11. Ruled bases

In this section, we prove Theorem 11.1. It treats the case of Section 12 in which the K_Z-MMP ends with a ruling over a curve. On such bases, Proposition 12.2 does not bound $\mathrm{vol}(-K_X)$ in terms of d only, so we regard the composition of f with the ruling as a fibration over a curve whose general fibers are weak Fano varieties.

Theorem 11.1. *Notation and conditions as in Set-up 9.1. Assume that there exists a contraction $g : Z \to C$ onto a smooth projective curve C of genus at most one such that the general fiber of g is isomorphic to $\mathbb{P}^1$. Let $h := g \circ f$, let $X_{\bar{\eta}}$ be the geometric generic fiber of h, and let $v := ((-K_{X_{\bar{\eta}}})^{d-1})$. Then $X_{\bar{\eta}}$ is terminal, $-K_{X_{\bar{\eta}}}$ is nef and big, and $v \geq (d-1)^{d-1}$. Moreover,*

(1) if $v > (d-1)^{d-1}$, then X is birational over C to $C \times \mathbb{P}^{d-1}$, and
(2) if $v = (d-1)^{d-1}$, then $C \cong \mathbb{P}^1$ and $\mathrm{vol}(-K_X) \leq c_d$, where c_d is a positive real number depending only on d.

In particular, for any integer $d \geq 3$, the set of varieties X such that some $f : X \to Z$ and $g : Z \to C$ satisfy the conditions above is birationally bounded.

Remark 11.2. Parts (1) and (2) of Theorem 11.1 use different arguments. For part (1), Proposition 4.2, used in dimension $d-1$, assumes $v > (d-1)^{d-1}$. For part (2), **Step 3** of the proof uses $v = (d-1)^{d-1}$: the restrictions of $|-mK_X|$ to a very general fiber T of h then have volume equal to $\mathrm{vol}(-K_T)$. From this, **Step 3** constructs, for any rational number $q > 0$, a $\mathbb{Q}$-divisor $D_q \geq 0$ on X such that $D_q \sim_{\mathbb{Q}} -qK_X$ and (X, D_q) is klt over a non-empty open subset of C, and **Step 4** bounds $\mathrm{vol}(-K_X)$ by Lemma 7.3 and Lemma 7.4, using D_2 and a rational number α depending only on d [Bir21, Theorem 1.6].

Step 3 follows the first half of [Zhu18, proof of Lemma 5.4]. There, X is a Mori fiber space over a curve, and the equality of the volume of the restricted anti-canonical systems on a general fiber with the anti-canonical volume of the general fiber, together with [FKL16, Theorem A], implies that for any sufficiently large integer m, there exists a $\mathbb{Q}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{Q}} -mK_X$ and (X, D) is klt along a general fiber. In **Step 3** we use in addition [Blu21, Proposition 2.6(1)]. The second half of [Zhu18, proof of Lemma 5.4] uses $\rho(X) = 2$ and the length of extremal rays to show that $-K_X$ is nef. In Theorem 11.1, $\rho(X/Z) = 1$ and $\rho(Z) \geq 2$ since Z admits the contraction g onto a curve, so $\rho(X) \geq 3$. Thus **Step 4** bounds $\mathrm{vol}(-K_X)$ directly, without the nefness of $-K_X$.

Proof of Theorem 11.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that a very general fiber T of h is a normal terminal projective variety such that $K_T = K_X|_T$ and $\epsilon_{\mathrm{mov}}(-K_T) \geq d-1$.

Since f and g are contractions,

$$h_*\mathcal{O}_X = g_*f_*\mathcal{O}_X = g_*\mathcal{O}_Z = \mathcal{O}_C,$$

so h is a contraction. Let I be a positive integer such that $-IK_X$ is Cartier, and let $x_0 \in X$ be a very general point. By Lemma 2.6(3),

$$\epsilon_{\mathrm{mov}}(-K_X, x_0) = \epsilon_{\mathrm{mov}}(-K_X) = d-1.$$

Thus, by Lemma 2.6(2), there exist positive integers $m_1 < m_2 < \cdots$ divisible by I such that the integers $s_j := s(-m_jK_X, x_0)$ satisfy $s_j \geq 0$ and

$$\lim_{j\to+\infty} \frac{s_j}{m_j} = d-1.$$

For any j, let U_j be the set of smooth points $y \in X$ such that $s(-m_jK_X, y) \geq s_j$. By Lemma 2.6(3), U_j is open in X, and $x_0 \in U_j$.

Let $c \in C$ be a very general point and $T := h^*c$. Since h is proper and surjective, $h(U_j)$ contains a non-empty open subset of C for any j, so $T \cap U_j \neq \emptyset$ for any j. By generic smoothness [Har77, Corollary III.10.7], applied to $h|_{X_{\mathrm{reg}}}$, the scheme $T \cap X_{\mathrm{reg}}$ is smooth. Since X is terminal, X has rational singularities [KM98, Theorem 5.22], so X is Cohen–Macaulay [KM98, Theorem 5.10], and $\mathrm{codim}(X_{\mathrm{sing}}, X) \geq 3$ [KM98, Corollary 5.18]. Hence the Cartier divisor T is Cohen–Macaulay, and $\mathrm{codim}(T \cap X_{\mathrm{sing}}, T) \geq 2$ since c is general. Thus T is regular in codimension one, and T is normal by Serre's criterion. Since h is a contraction, T is connected, so T is a normal projective variety of dimension $d-1$. Since $\mathcal{O}_T(T) \cong \mathcal{O}_T$, adjunction implies $K_T = K_X|_T$. In particular, $-IK_T = (-IK_X)|_T$ is Cartier.

Let $\sigma : X' \to X$ be a log resolution, and let $E_1, \ldots, E_r$ be the σ-exceptional prime divisors, so that

$$K_{X'} = \sigma^*K_X + \sum_{i=1}^{r}(a(E_i, X, 0) - 1)E_i \quad \text{and} \quad a(E_i, X, 0) > 1 \text{ for any } i.$$

Since c is general, by generic smoothness, $T' := (h \circ \sigma)^*c$ is smooth, T' intersects $\sum_i E_i$ transversally, and $\mathrm{codim}(\sigma(E_i) \cap T, T) \geq 2$ for any i. Thus $\sigma|_{T'} : T' \to T$ is a log resolution whose exceptional prime divisors are the components of $E_1|_{T'}, \ldots, E_r|_{T'}$, and adjunction implies

$$K_{T'} = (\sigma|_{T'})^*K_T + \sum_{i=1}^{r}(a(E_i, X, 0) - 1)E_i|_{T'}.$$

Hence any $\sigma|_{T'}$-exceptional prime divisor has log discrepancy greater than 1 with respect to T. Since $\sigma|_{T'}$ is a log resolution of T, [KM98, Corollary 2.32(2)] with $\Delta = 0$ implies that the discrepancy of T is the minimum of 1 and the discrepancies of the $\sigma|_{T'}$-exceptional prime divisors, which is positive. Thus T is terminal.

For any positive integer m divisible by I, let W_m be the image of the restriction map

$$H^0(X, \mathcal{O}_X(-mK_X)) \to H^0(T, \mathcal{O}_T(-mK_T)).$$

Any point $y \in T \cap U_j$ is a smooth point of X and of T. By Lemma 2.6(5), applied to the Cartier divisor $-m_jK_X$ and $V = H^0(X, \mathcal{O}_X(-m_jK_X))$, we have $s(W_{m_j}, y) \geq s_j$ for any $y \in T \cap U_j$. Let $x \in T$ be a very general point. Then $x \in T \cap U_j$ for any j, so

$$s(-m_jK_T, x) \geq s(W_{m_j}, x) \geq s_j \quad \text{for any } j.$$

By Lemma 2.6(2), applied on T,

$$\epsilon_{\mathrm{mov}}(-K_T) \geq \epsilon_{\mathrm{mov}}(-K_T, x) \geq \lim_{j\to+\infty} \frac{s_j}{m_j} = d - 1.$$

Step 2. In this step, we show that $-K_T$ is nef and big, that $v = ((-K_T)^{d-1}) \geq (d-1)^{d-1}$, and that $X_{\bar{\eta}}$ is terminal and $-K_{X_{\bar{\eta}}}$ is nef and big.

Since c is general, $g^{-1}(c) \cong \mathbb{P}^1$, and $T = f^{-1}(g^{-1}(c))$. The fibers of $f|_T : T \to g^{-1}(c)$ are fibers of f, hence connected, so $f|_T$ is a contraction by the Stein factorization. Since Z° is a non-empty open subset of Z (Definition-Lemma 9.4) and c is general, $g^{-1}(c) \cap Z^\circ$ is a non-empty open subset of $g^{-1}(c)$ over which $f|_T$ is a Brauer–Severi scheme of relative dimension $d-2$. In particular, $f|_T$ is smooth over $g^{-1}(c) \cap Z^\circ$, and the general fiber of $f|_T$ is $\mathbb{P}^{d-2}$.

Let $q \geq 2$ be an integer. Since $\epsilon_{\mathrm{mov}}(-K_T) \geq d - 1 > d - 1 - 1/q$, Theorem 2.19, applied to $f|_T$ and $t = 1/q$, implies that there exists a $\mathbb{Q}$-divisor $D_{T,q} \geq 0$ on T such that $D_{T,q} \sim_{\mathbb{Q}} -qK_T$ and $(T, D_{T,q})$ is klt over a non-empty open subset of $g^{-1}(c)$. Moreover, $-K_T = -K_X|_T$ is ample over $g^{-1}(c)$ since $-K_X$ is ample over Z, and $K_T + (-K_T) = 0$. Thus Lemma 7.2, applied to $f|_T$, $L = -K_T$, and $A = 0$, implies that $-K_T$ is nef. By Lemma 2.6(4), $-K_T$ is big.

Recall that $x \in T$ is a very general point. Then x is a smooth point of T, and

$$\epsilon_{\mathrm{mov}}(-K_T, x) = \epsilon_{\mathrm{mov}}(-K_T) \geq d - 1$$

by Lemma 2.6(3). Since $-K_T$ is nef and big and T is normal, Lemma 2.9(4) implies $\epsilon(-K_T, x) = \epsilon_{\mathrm{mov}}(-K_T, x)$. Thus Lemma 2.9(2) implies

$$((-K_T)^{d-1}) \geq \epsilon(-K_T, x)^{d-1} \geq (d-1)^{d-1}.$$

We now pass from T to $X_{\bar{\eta}}$, which is a normal projective variety since $h_*\mathcal{O}_X = \mathcal{O}_C$, the characteristic is zero, and normality is preserved under separable field extensions. Since X is integral and C is a smooth curve, h is flat [Har77, Proposition III.9.7], and $K_{X_{\bar{\eta}}} = K_X|_{X_{\bar{\eta}}}$ [Har77, Proposition II.8.10]. By [Kol96, Proposition VI.2.9], applied to h, the sheaf $\mathcal{O}_X$, and the line bundle $\mathcal{O}_X(-IK_X)$, the intersection number $((-IK_X|_{X_t})^{d-1})$ on the fiber X_t is a locally constant function of $t \in C$, where t runs over all points of C, also the generic point. It is constant since C is connected. Since intersection numbers on the fibers are defined by Euler characteristics [Kol96, Definition VI.2.6], $((-IK_X|_{X_\eta})^{d-1})$ does not change under the base change from $\mathbb{C}(C)$ to $\overline{\mathbb{C}(C)}$ [Har77, Proposition III.9.3]. Thus $v = ((-K_T)^{d-1}) \geq (d-1)^{d-1}$.

Any curve $\Gamma \subset X_{\bar{\eta}}$ is defined over a finite extension of $\mathbb{C}(C)$. Thus, after a finite base change by a smooth projective curve $C' \to C$ and a restriction to a non-empty open subset of C', the curve Γ spreads out to a flat family of curves in the fibers of h, whose member Γ_1 over a very general point of C' is a curve in a very general fiber T_1 of h. Since **Steps 1** and 2 hold for T_1 in place of T, $-K_{T_1}$ is nef. By [Kol96, Proposition VI.2.9 and Definition VI.2.6], applied to the structure sheaf of the family of curves spread out from Γ and the pullback of $\mathcal{O}_X(-IK_X)$, as in the previous paragraph,

$$((-K_{X_{\bar{\eta}}}) \cdot \Gamma) = ((-K_{T_1}) \cdot \Gamma_1) \geq 0.$$

Thus $-K_{X_{\bar{\eta}}}$ is nef, and $-K_{X_{\bar{\eta}}}$ is big since $v > 0$ [Laz04, Theorem 2.2.16] (Remark 2.13).

Finally, the base change $\sigma_{\bar{\eta}}$ of σ to $\bar{\eta}$ is a log resolution of $X_{\bar{\eta}}$, whose exceptional prime divisors are the components of the base changes of the σ-exceptional prime divisors E_i dominating C, and each component of the base change of E_i has log discrepancy $a(E_i, X, 0) > 1$ with respect to $X_{\bar{\eta}}$. Thus [KM98, Corollary 2.32(2)], applied with $\Delta = 0$ to $\sigma_{\bar{\eta}}$, implies that the discrepancy of $X_{\bar{\eta}}$ is the minimum of 1 and the discrepancies of the $\sigma_{\bar{\eta}}$-exceptional prime divisors, which is positive. Thus $X_{\bar{\eta}}$ is terminal.

Step 3. In this step, we assume that $v = (d-1)^{d-1}$, and we show that for any rational number $q > 0$, there exists a $\mathbb{Q}$-divisor $D_q \geq 0$ on X such that $D_q \sim_{\mathbb{Q}} -qK_X$ and (X, D_q) is klt over a non-empty open subset of C.

Recall that $T = h^*c$ for a very general point $c \in C$, that $x \in T$ is a very general point, and that $s(W_{m_j}, x) \geq s_j$ for any j (**Step 1**). By **Step 2**, $((-K_T)^{d-1}) = (d-1)^{d-1}$. For any positive integer ℓ, let $\mathfrak{b}_\ell \subset \mathcal{O}_T$ be the base ideal of $W_{I\ell}$, where $\mathfrak{b}_\ell = 0$ if $W_{I\ell} = 0$. Since the multiplication of sections maps $W_{I\ell} \otimes W_{I\ell'}$ into $W_{I(\ell+\ell')}$, we have

$$\mathfrak{b}_\ell \mathfrak{b}_{\ell'} \subset \mathfrak{b}_{\ell+\ell'},$$

so $\mathfrak{b}_\bullet$ is a graded sequence of ideals [Blu21, Sections 2.5–2.6].

Write $m_j = I\ell_j$. Since x is a smooth point of T, $\dim T = d-1$, and W_{m_j} generates s_j-jets at x, we have

$$\dim W_{I\ell_j} \geq \binom{s_j + d - 1}{d-1} \geq \frac{s_j^{d-1}}{(d-1)!}.$$

Since $\lim_{j\to+\infty} s_j/(I\ell_j) = d-1$, we obtain

$$\limsup_{\ell\to+\infty} \frac{\dim W_{I\ell}}{\ell^{d-1}/(d-1)!} \geq \lim_{j\to+\infty} \frac{s_j^{d-1}}{\ell_j^{d-1}} = I^{d-1}(d-1)^{d-1}.$$

Since $W_{I\ell} \subset H^0(T, \mathcal{O}_T(-I\ell K_T))$ and $\mathrm{vol}(-IK_T) = I^{d-1}((-K_T)^{d-1}) = I^{d-1}(d-1)^{d-1}$ [Laz04, Corollary 1.4.41], equality holds.

Let E be a prime divisor over T, and let

$$\mathrm{mult}_E \mathfrak{b}_\bullet := \inf_\ell \frac{\mathrm{mult}_E \mathfrak{b}_\ell}{\ell},$$

where ℓ runs over the positive integers such that $\mathfrak{b}_\ell \neq 0$. Suppose that $\mathrm{mult}_E \mathfrak{b}_\bullet > 0$. Let λ be a rational number such that $0 < \lambda < \mathrm{mult}_E \mathfrak{b}_\bullet$, and let $\mu : \widetilde{T} \to T$ be a resolution such that E is a prime divisor on $\widetilde{T}$. For any ℓ such that $W_{I\ell} \neq 0$, we have $\mathrm{mult}_E \mathfrak{b}_\ell > \ell\lambda$, so $\mathrm{mult}_E \mathfrak{b}_\ell \geq \lceil \ell\lambda \rceil$. Thus pullback by μ embeds $W_{I\ell}$ into

$$H^0(\widetilde{T}, \mathcal{O}_{\widetilde{T}}(\ell\mu^*(-IK_T) - \lceil \ell\lambda \rceil E)) = H^0(\widetilde{T}, \mathcal{O}_{\widetilde{T}}(\lfloor \ell(\mu^*(-IK_T) - \lambda E) \rfloor)).$$

Since $\mu^*(-IK_T)$ is nef and big and $\lambda E \neq 0$, [FKL16, Theorem A] implies

$$\limsup_{\ell\to+\infty} \frac{\dim W_{I\ell}}{\ell^{d-1}/(d-1)!} \leq \mathrm{vol}(\mu^*(-IK_T) - \lambda E) < \mathrm{vol}(\mu^*(-IK_T)) = I^{d-1}(d-1)^{d-1},$$

where the last equality holds since $\mu_*\mathcal{O}_{\widetilde{T}} = \mathcal{O}_T$, a contradiction. Thus $\mathrm{mult}_E \mathfrak{b}_\bullet = 0$ for any prime divisor E over T. Since T is terminal, in particular klt, [Blu21, Proposition 2.6(1)] (see also [JM12, Corollary 2.16] for smooth varieties) implies

$$\sup_\ell \ell \,\mathrm{lct}(T; \mathfrak{b}_\ell) = \inf_E \frac{a(E, T, 0)}{\mathrm{mult}_E \mathfrak{b}_\bullet} = +\infty,$$

where E runs over the prime divisors over T, and $a(E, T, 0)/0 := +\infty$.

Let $q > 0$ be a rational number. Then there exists a positive integer ℓ_0 such that $\ell_0 \,\mathrm{lct}(T; \mathfrak{b}_{\ell_0}) > q/I$. For any positive integer p, since $\mathfrak{b}_{\ell_0}^p \subset \mathfrak{b}_{p\ell_0}$, we have

$$\mathrm{lct}(T; \mathfrak{b}_{p\ell_0}) \geq \mathrm{lct}(T; \mathfrak{b}_{\ell_0}^p) = \frac{\mathrm{lct}(T; \mathfrak{b}_{\ell_0})}{p}.$$

We choose p such that $m := Ip\ell_0 > q$, and let $\ell := p\ell_0$. Then

$$\frac{q}{m} < \min\{1, \mathrm{lct}(T; \mathfrak{b}_\ell)\}.$$

Let $\mu' : \widehat{T} \to T$ be a log resolution of $(T, \mathfrak{b}_\ell)$, and let $F \geq 0$ be the divisor on $\widehat{T}$ such that

$$\mathfrak{b}_\ell \mathcal{O}_{\widehat{T}} = \mathcal{O}_{\widehat{T}}(-F).$$

Then the pullback of W_m to $\widehat{T}$ is the sum of the fixed part F and a base point free linear system Λ on $\widehat{T}$, and $\mathrm{Supp}\, F \cup \mathrm{Exc}(\mu')$ is snc. Let H_T be the divisor of a general element of W_m. Then $\mu'^* H_T = F + M$, where M is a general member of Λ, so M is smooth and intersects $\mathrm{Supp}\, F \cup \mathrm{Exc}(\mu')$ transversally by [Har77, Corollary III.10.9 and Remark III.10.9.2], applied to Λ and to the restrictions of Λ to the strata of $\mathrm{Supp}\, F \cup \mathrm{Exc}(\mu')$. Thus

$$K_{\widehat{T}} + \frac{q}{m} M + \sum_G \Big(1 - a(G, T, 0) + \frac{q}{m} \mathrm{mult}_G F\Big) G = \mu'^* \Big(K_T + \frac{q}{m} H_T\Big),$$

where G runs over the prime divisors on $\widehat{T}$ that are μ'-exceptional or components of F. Since $q/m < \mathrm{lct}(T; \mathfrak{b}_\ell)$, any coefficient in the sum is less than 1, and $q/m < 1$. Thus $(T, (q/m)H_T)$ is klt.

Since W_m is the image of $H^0(X, \mathcal{O}_X(-mK_X))$ and H_T is general, there exists $H_X \in |-mK_X|$ such that $T \not\subset \mathrm{Supp}\, H_X$ and $H_X|_T = H_T$. Let $D_q := (q/m)H_X$. Then $D_q \geq 0$ and $D_q \sim_{\mathbb{Q}} -qK_X$. Since $\mathcal{O}_T(T) \cong \mathcal{O}_T$, we have $(K_X + T + D_q)|_T = K_T + (q/m)H_T$. By inversion of adjunction [KM98, Theorem 5.50(1)], $(X, T + D_q)$ is plt near T, so (X, D_q) is klt near T. Since $\mathrm{Nklt}(X, D_q)$ is closed and h is proper, $h(\mathrm{Nklt}(X, D_q))$ is a closed subset of C not containing c. Thus (X, D_q) is klt over the non-empty open subset $C \backslash h(\mathrm{Nklt}(X, D_q))$.

Step 4. In this step, we assume that $v = (d-1)^{d-1}$, and we show that $C \cong \mathbb{P}^1$ and $\mathrm{vol}(-K_X) \leq c_d$, where $c_d := 8d(d-1)^{d-1}/\alpha$ for a rational number $\alpha \in (0, 1]$ depending only on d.

By **Step 2**, $X_{\bar{\eta}}$ is terminal, hence 1-lc, and $-K_{X_{\bar{\eta}}}$ is nef and big. By [Bir21, Theorem 1.6], applied in dimension $d-1$ with $\epsilon = 1$ and $B = 0$ (Remark 2.13), there exists a rational number $\alpha \in (0, 1]$ depending only on d such that $(X_{\bar{\eta}}, \alpha\Delta)$ is lc for any $\mathbb{Q}$-divisor $\Delta \geq 0$ on $X_{\bar{\eta}}$ with $\Delta \sim_{\mathbb{Q}} -K_{X_{\bar{\eta}}}$. Let $D := D_2/2$, where D_2 is the $\mathbb{Q}$-divisor of **Step 3** for $q = 2$. Then $D \geq 0$, $D \sim_{\mathbb{Q}} -K_X$, and $(X, 2D)$ is klt over a non-empty open subset of C.

Recall that $T = h^*c$, where $c \in C$ is the very general point of **Step 1**. Thus Lemma 7.3, applied to h and the ample Cartier divisor $H = c$ of degree 1 on C, implies

$$a\alpha \leq 4 \deg(-K_C)$$

for any rational number $a > 0$ such that $-K_X - aT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. By Lemma 2.6(4), $-K_X$ is big. Since the big cone is open, $-K_X - aT$ is big for some rational number $a > 0$, so $-K_X - aT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. Hence $\deg(-K_C) > 0$, so C has genus zero and $C \cong \mathbb{P}^1$. Since $\deg(-K_{\mathbb{P}^1}) = 2$, we have $a \leq 8/\alpha$ for any such a. Let τ be the supremum of the rational numbers $a \geq 0$ such that $-K_X - aT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. Then $\tau \leq 8/\alpha$.

By **Step 1**, T is a normal Cartier prime divisor on X with $\mathcal{O}_T(T) \cong \mathcal{O}_T$, so $H^0(T, \mathcal{O}_T(T)) \neq 0$. By **Step 2**, $-K_T$ is nef, so $\mathrm{vol}(-K_T) = ((-K_T)^{d-1}) = (d-1)^{d-1}$ [Laz04, Corollary 1.4.41]. Thus Lemma 7.4, applied to $L = -K_X$ and T, implies

$$\mathrm{vol}(-K_X) \leq d\tau \,\mathrm{vol}(-K_X|_T) = d\tau \,\mathrm{vol}(-K_T) \leq \frac{8d(d-1)^{d-1}}{\alpha} = c_d.$$

Step 5. In this step, we assume that $v > (d-1)^{d-1}$, and we show that X is birational over C to $C \times \mathbb{P}^{d-1}$.

Let X_η be the generic fiber of h. By **Step 2**, $X_{\bar{\eta}}$ is terminal and $-K_{X_{\bar{\eta}}}$ is nef and big. Since $-IK_{X_{\bar{\eta}}}$ is Cartier and $-IK_{X_{\bar{\eta}}} - K_{X_{\bar{\eta}}}$ is nef and big, the base point free theorem [KM98,

Theorem 3.3] (Remark 2.13) implies that $|-bIK_{X_{\bar{\eta}}}|$ is base point free for any sufficiently large integer b. By flat base change [Har77, Proposition III.9.3],

$$H^0(X_{\bar{\eta}}, \mathcal{O}_{X_{\bar{\eta}}}(-bIK_{X_{\bar{\eta}}})) = H^0(X_\eta, \mathcal{O}_{X_\eta}(-bIK_{X_\eta})) \otimes_{\mathbb{C}(C)} \overline{\mathbb{C}(C)},$$

so the base locus of $|-bIK_{X_\eta}|$ is empty after base change to $\overline{\mathbb{C}(C)}$, hence empty. Thus $-K_{X_\eta}$ is semi-ample. Let $\phi : X_\eta \to X'_\eta$ be the contraction defined by $-K_{X_\eta}$ [Laz04, Theorem 2.1.27], whose proof works over any field, so that $-bIK_{X_\eta} = \phi^* A$ for some positive integer b and some ample Cartier divisor A on X'_η. Since $-K_{X_\eta}$ is big, ϕ is birational, so $-bIK_{X'_\eta} = \phi_*(-bIK_{X_\eta}) \sim A$. Thus $-K_{X'_\eta}$ is an ample $\mathbb{Q}$-Cartier divisor and $K_{X_\eta} = \phi^* K_{X'_\eta}$. Let $X'_{\bar{\eta}}$ be the base change of X'_η to $\overline{\mathbb{C}(C)}$. Then the base change $X_{\bar{\eta}} \to X'_{\bar{\eta}}$ of ϕ is birational and crepant, so $X'_{\bar{\eta}}$ is canonical [KM98, Lemma 2.30], and $((-K_{X'_{\bar{\eta}}})^{d-1}) = v > (d-1)^{d-1}$.

For any positive integer m divisible by I, since $-mK_{X_\eta} = \phi^*(-mK_{X'_\eta})$ and ϕ is birational, pullback of rational functions by ϕ induces an isomorphism

$$H^0(X'_\eta, \mathcal{O}_{X'_\eta}(-mK_{X'_\eta})) \to H^0(X_\eta, \mathcal{O}_{X_\eta}(-mK_{X_\eta})).$$

Let $U' \subset X'_\eta$ be the largest open subset over which ϕ is an isomorphism, and let $U \subset X$ be an open subset such that $U \cap X_\eta = \phi^{-1}(U')$. Since ϕ is birational, $U \neq \emptyset$.

By **Step 2**, $X_{\bar{\eta}}$ is klt, hence dlt, and $-K_{X_{\bar{\eta}}}$ is nef and big. Thus [HM07, Corollary 1.3(1) and Corollary 1.5(2)], applied with S a point and $\Delta = 0$ (Remark 2.13), implies that $X_{\bar{\eta}}$ is rationally connected, and so is a resolution of $X_{\bar{\eta}}$ [Kol96, Proposition IV.3.3]. Recall the integers m_j and s_j and the non-empty open subsets $U_j \subset X$ of **Step 1**. For any j, $-m_j K_X$ is Cartier, and $H^0(X, \mathcal{O}_X(-m_j K_X))$ generates s_j-jets at any point of U_j. Thus Lemma 4.1, applied to h, the integers m_j and s_j, and U, implies that there exists a $\mathbb{C}(C)$-rational point p of X_η such that $p \in U$, p is a smooth point of X_η, and $H^0(X_\eta, \mathcal{O}_{X_\eta}(-m_j K_{X_\eta}))$ generates s_j-jets at p for any j. Let $x' := \phi(p)$. Then x' is a $\mathbb{C}(C)$-rational point in the smooth locus of X'_η. Since ϕ is an isomorphism near p, $H^0(X'_\eta, \mathcal{O}_{X'_\eta}(-m_j K_{X'_\eta}))$ generates s_j-jets at x' for any j. By flat base change [Har77, Proposition III.9.3],

$$H^0(X'_{\bar{\eta}}, \mathcal{O}_{X'_{\bar{\eta}}}(-m_j K_{X'_{\bar{\eta}}}))$$

generates s_j-jets at the point $x'_{\bar{\eta}}$ of $X'_{\bar{\eta}}$ over x'. Thus, by Lemma 2.6(2) (Remark 2.13),

$$\epsilon_{\mathrm{mov}}(-K_{X'_{\bar{\eta}}}, x'_{\bar{\eta}}) \geq \lim_{j \to +\infty} \frac{s_j}{m_j} = d - 1.$$

Since $-K_{X'_{\bar{\eta}}}$ is ample, Lemma 2.9(3) implies $\epsilon(-K_{X'_{\bar{\eta}}}, x'_{\bar{\eta}}) \geq d - 1$.

Thus Proposition 4.2, applied to X'_η and x', implies that X'_η is birational over $\mathbb{C}(C)$ to $\mathbb{P}^{d-1}_{\mathbb{C}(C)}$. Since ϕ is birational, X_η is birational over $\mathbb{C}(C)$ to $\mathbb{P}^{d-1}_{\mathbb{C}(C)}$. Hence the function field of X is purely transcendental of degree $d-1$ over $\mathbb{C}(C)$, so X is birational over C to $C \times \mathbb{P}^{d-1}$.

Step 6. In this step, we show that the set $\mathfrak{X}$ of varieties X such that some $f : X \to Z$ and $g : Z \to C$ satisfy the conditions of Theorem 11.1 is birationally bounded.

Let $\mathfrak{X}_1$ be the set of varieties $X \in \mathfrak{X}$ such that $v = (d-1)^{d-1}$. Let $X \in \mathfrak{X}_1$. Since $d \geq 3$, we have $\epsilon_{\mathrm{mov}}(-K_X) = d - 1 > d/2$. Moreover, K_X is $\mathbb{Q}$-Cartier, and $\mathrm{vol}(-K_X) \leq c_d$ by **Step 4**. Since c_d depends only on d, Lemma 7.6, applied to the positive real number c_d, implies that $\mathfrak{X}_1$ is birationally bounded.

Let $\mathfrak{X}_2$ (resp. $\mathfrak{X}_3$) be the set of varieties $X \in \mathfrak{X}$ such that $v > (d-1)^{d-1}$ and $C \cong \mathbb{P}^1$ (resp. C has genus one). By **Step 2**, $\mathfrak{X} = \mathfrak{X}_1 \cup \mathfrak{X}_2 \cup \mathfrak{X}_3$. By **Step 5**, any $X \in \mathfrak{X}_2$ is birational to $\mathbb{P}^1 \times \mathbb{P}^{d-1}$, hence to $\mathbb{P}^d$, and any $X \in \mathfrak{X}_3$ is birational to $C \times \mathbb{P}^{d-1}$ for a smooth projective curve C of genus one. The set $\{\mathbb{P}^d\}$ is bounded, and the set of varieties $C \times \mathbb{P}^{d-1}$, where C runs over the smooth projective curves of genus one, is bounded by Lemma 2.15(2). Thus Lemma 2.15(1) implies that $\mathfrak{X}_2$ and $\mathfrak{X}_3$ are birationally bounded, and that $\mathfrak{X} = \mathfrak{X}_1 \cup \mathfrak{X}_2 \cup \mathfrak{X}_3$ is birationally bounded. □

12. Mori fiber spaces over surfaces

In this section, we prove Theorem 12.1. The key step is Proposition 12.2, which bounds $\mathrm{vol}(-K_X)$ in terms of the ratio $(-K_Z \cdot H)^2/(H^2)$ for any very ample Cartier divisor H on Z, whether or not f has a rational section. We then run a K_Z-MMP. If it ends with a contraction onto a curve, we use Theorem 11.1. Otherwise it ends with a surface Z'' such that $-K_{Z''}$ is ample, and we choose H so that the ratio is bounded in terms of d only.

Theorem 12.1. *Let $d \geq 3$ be an integer. Then there exist positive real numbers δ_d and W_d depending only on d with the following property. Let $f : X \to Z$ satisfy the conditions of Set-up 9.1 for this d. Then Z is δ_d-lc, and one of the following holds.*

(1) There exists a contraction $g : Z \to C$ onto a smooth projective curve C of genus at most one such that the general fiber of g is isomorphic to $\mathbb{P}^1$.

(2) There exists a birational morphism $g : Z \to Z''$ onto a δ_d-lc projective surface Z'' such that $-K_{Z''}$ is ample, $((-K_{Z''})^2) \leq W_d$, and $\mathrm{vol}(-K_X) \leq 8d(d-1)^d(W_d+1)$.

In particular, the set of varieties X such that some $f : X \to Z$ satisfies the conditions of Set-up 9.1 for this d is birationally bounded.

Restricting to $T = f^{-1}(C)$ for a very general member $C \in |H|$ splits the bound into two factors: the supremum of the rational numbers $s \geq 0$ such that $-K_X - sT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor ≥ 0, which is bounded by Lemma 7.3, and $\mathrm{vol}(-K_X|_T)$, which is bounded by Theorems 10.2 and 10.3.

Proposition 12.2. *Notation and conditions as in Set-up 9.1. Let H be a very ample Cartier divisor on Z. Then*

$$\mathrm{vol}(-K_X) \leq 8d(d-1)^d \frac{(-K_Z \cdot H)^2}{(H^2)}.$$

Proof. We may assume that $-K_X$ is big. Otherwise $\mathrm{vol}(-K_X) = 0$, while $(H^2) > 0$ and

$$(-K_Z \cdot H) \geq ((-K_Z - \Lambda) \cdot H) \geq 1$$

by Theorems 10.2 and 10.3, where Λ is the $\mathbb{Q}$-divisor class on Z of Definition 10.1, so the right-hand side of the inequality is positive.

Let $C \in |H|$ be a very general member and $T := f^{-1}(C)$. By Theorem 10.2, C is a smooth curve contained in Z_{reg}, T is a normal projective variety, $-K_X|_T$ is nef and big, and

$$((-K_X|_T)^{d-1}) = (d-1)^{d-1}((-K_Z - \Lambda) \cdot H).$$

Since $-K_X|_T$ is nef, $\mathrm{vol}(-K_X|_T) = ((-K_X|_T)^{d-1})$ [Laz04, Corollary 1.4.41]. Thus the inequality $((-K_Z - \Lambda) \cdot H) \leq (-K_Z \cdot H)$ of Theorem 10.3 implies

$$\mathrm{vol}(-K_X|_T) \leq (d-1)^{d-1}(-K_Z \cdot H). \tag{12.1}$$

The divisor $T = f^*C$ is Cartier, and T is a prime divisor since T is a normal variety. Moreover, $T \sim f^*H$, and $\mathcal{O}_T(T)$ is the pullback of $\mathcal{O}_C(H|_C)$ under $f|_T$. Since $H|_C$ is very ample, $H^0(C, \mathcal{O}_C(H|_C)) \neq 0$, hence $H^0(T, \mathcal{O}_T(T)) \neq 0$. Let τ_H be the supremum of the rational numbers $s \geq 0$ such that $-K_X - sT$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. Lemma 7.4, applied to $-K_X$ and T, implies

$$\mathrm{vol}(-K_X) \leq d\tau_H \,\mathrm{vol}(-K_X|_T). \tag{12.2}$$

It remains to bound τ_H. We use Lemma 7.3 with $b = 2$ and $\alpha := 1/(2(d-1))$. Here Z is $\mathbb{Q}$-factorial by Lemma 9.3, and K_X is $\mathbb{Q}$-Cartier since X is terminal. We check conditions (1) and (2) of Lemma 7.3.

For condition (1), the general fiber of f is isomorphic to $\mathbb{P}^{d-2}$ by Definition-Lemma 9.4, $0 < \dim Z = 2 < d$, and

$$\epsilon_{\mathrm{mov}}(-K_X) = d - 1 > d - \frac{3}{2}.$$

Thus Theorem 2.19 applied with $a = d-1$ and $t = 1/2$ implies that there exists a $\mathbb{Q}$-divisor $D_2 \geq 0$ on X such that $D_2 \sim_{\mathbb{Q}} -2K_X$ and (X, D_2) is klt over a non-empty open subset of Z. Let $D := D_2/2$. Then $D \sim_{\mathbb{Q}} -K_X$ and $(X, 2D)$ is klt over a non-empty open subset of Z.

For condition (2), Set-up 9.1(3) implies $X_{\bar{\eta}} \cong \mathbb{P}^{d-2}_{\mathbb{C}(Z)}$. Let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor on $X_{\bar{\eta}}$ such that $\Delta \sim_{\mathbb{Q}} -K_{X_{\bar{\eta}}}$, and let $x \in X_{\bar{\eta}}$ be a closed point. Then Δ is $\mathbb{Q}$-linearly equivalent to $d-1$ times a hyperplane. Since $d-2 \geq 1$, there exists a line $\ell \subset X_{\bar{\eta}}$ passing through x such that $\ell \not\subset \operatorname{Supp} \Delta$, so

$$\operatorname{mult}_x \Delta \leq (\Delta \cdot \ell) = d-1,$$

since the intersection multiplicity at x of each component of Δ with ℓ is at least the multiplicity of that component at x [Ful98, Corollary 12.4]. Hence $\operatorname{mult}_x(\alpha\Delta) \leq 1/2 < 1$, and $(X_{\bar{\eta}}, \alpha\Delta)$ is terminal near x by [Zhu18, Lemma 2.13] and Remark 2.13. Since x is an arbitrary closed point, $(X_{\bar{\eta}}, \alpha\Delta)$ is klt, in particular lc.

Thus Lemma 7.3 implies $s\alpha(H^2) \leq 4(-K_Z \cdot H)$ for any rational number $s > 0$ such that $-K_X - sf^*H$ is $\mathbb{Q}$-linearly equivalent to a $\mathbb{Q}$-divisor $D' \geq 0$. Since $T \sim f^*H$ and $(-K_Z \cdot H) > 0$, taking the supremum over s implies

$$\tau_H \leq \frac{4(-K_Z \cdot H)}{\alpha(H^2)} = \frac{8(d-1)(-K_Z \cdot H)}{(H^2)}. \tag{12.3}$$

Combining (12.1), (12.2), and (12.3), we obtain

$$\operatorname{vol}(-K_X) \leq d \cdot \frac{8(d-1)(-K_Z \cdot H)}{(H^2)} \cdot (d-1)^{d-1}(-K_Z \cdot H) = 8d(d-1)^d \frac{(-K_Z \cdot H)^2}{(H^2)}.$$

□

To bound the ratio in Proposition 12.2 uniformly, we control the singularities of Z uniformly: [Bir23, Corollary 1.2] implies that Z is δ_d-lc for a number δ_d depending only on d, and [Bir21, Theorem 2.11] then bounds the anti-canonical volume of the output of a K_Z-MMP. The MMP is used only to construct a contraction of Z onto a curve or a polarization on Z itself, so we do not carry Set-up 9.1 or the moving Seshadri constant along the MMP.

Proof of Theorem 12.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we define δ_d and W_d and show that Z is δ_d-lc.

Let $\delta_d > 0$ be the number given by [Bir23, Corollary 1.2] for relative dimension $d-2$ and $\epsilon = 1$. Then δ_d depends only on d. The morphism f is a Fano fibration in the sense of [Bir23], that is, f is a contraction between normal varieties, X is lc, and $-K_X$ is ample over Z (see the definition before [Bir23, Corollary 1.2]). Since X is terminal, $a(F, X, 0) \geq 1$ for any prime divisor F over X [KM98, Definition 2.34], that is, X is 1-lc. Moreover, $\dim X - \dim Z = d-2$, and K_Z is $\mathbb{Q}$-Cartier by Lemma 9.3. Thus Z is δ_d-lc by [Bir23, Corollary 1.2].

By [Bir21, Theorem 2.11] in dimension two, there exists a positive real number W_d depending only on δ_d, hence only on d, such that $\operatorname{vol}(-K_V) \leq W_d$ for any δ_d-lc projective surface V such that $-K_V$ is nef and big. Here the hypothesis of [Bir21, Theorem 2.11], namely [Bir21, Theorem 1.1] in dimension one, holds because a projective curve V' such that (V', B') is ϵ-lc and $-(K_{V'} + B')$ is nef and big for some boundary B' satisfies $\deg K_{V'} < 0$, hence $V' \cong \mathbb{P}^1$, so the set of all such curves V' is the bounded set $\{\mathbb{P}^1\}$.

Step 2. In this step, we run a K_Z-MMP and show that the output is δ_d-lc.

By Theorem 9.2, $-K_Z$ is big. Fix an ample Cartier divisor H_0 on Z. Since $-K_Z$ is big, $(K_Z \cdot H_0) < 0$ [KM98, Lemma 2.60], while any pseudo-effective divisor has non-negative intersection number with H_0. Thus K_Z is not pseudo-effective. By Lemma 9.3, Z is $\mathbb{Q}$-factorial. By [Fuj12, Theorems 1.1 and 3.3], a K_Z-MMP consists of finitely many birational extremal contractions

$$Z = Z_0 \to Z_1 \to \cdots \to Z_n =: Z''$$

between $\mathbb{Q}$-factorial normal projective surfaces, followed by a contraction $h : Z'' \to C$ onto a normal projective variety C with $\dim C \leq 1$ (this C is not the member $C \in |H|$ in the proof of Proposition 12.2) such that $\rho(Z''/C) = 1$ and $-K_{Z''}$ is ample over C. Here $h_*\mathcal{O}_{Z''} = \mathcal{O}_C$ by [Fuj12, Theorem 3.2(3)(ii)]. Let $g : Z \to Z''$ be the composite morphism, which is birational.

For any $0 \leq i < n$, the morphism $Z_i \to Z_{i+1}$ is a K_{Z_i}-negative divisorial contraction, so [KM98, Lemma 3.38], applied with $Y = Z_{i+1}$ and f' the identity, implies $a(F, Z_i, 0) \leq a(F, Z_{i+1}, 0)$ for any prime divisor F over Z_{i+1} which is exceptional over Z_{i+1}. For a prime divisor F on Z_{i+1}, both sides are equal to 1. Since Z is δ_d-lc by **Step 1**, induction on i shows that Z'' is δ_d-lc.

Step 3. In this step, we assume that $\dim C = 1$ and show that (1) holds.

The curve C is normal and projective, hence smooth. Since Z'' is a normal surface, Z''_{sing} is finite. By generic smoothness [Har77, Corollary III.10.7] applied to $h|_{Z''_{\mathrm{reg}}}$, a general fiber G of h is smooth and contained in Z''_{reg}, and G is connected since $h_*\mathcal{O}_{Z''} = \mathcal{O}_C$. Since $(G^2) = 0$ and $-K_{Z''}$ is ample over C, the adjunction formula [Har77, Proposition V.1.5], applied on a resolution of Z'' which is an isomorphism over Z''_{reg}, implies

$$2p_a(G) - 2 = ((K_{Z''} + G) \cdot G) = (K_{Z''} \cdot G) < 0.$$

Thus $p_a(G) = 0$ and $G \cong \mathbb{P}^1$. Since Z'' is normal, g is an isomorphism over the complement of a finite subset of Z'', and a general fiber of h is disjoint from the finite subset. Hence the general fiber of $h \circ g$ is isomorphic to $\mathbb{P}^1$. Moreover, $g_*\mathcal{O}_Z = \mathcal{O}_{Z''}$ since g is birational and Z'' is normal, so $(h \circ g)_*\mathcal{O}_Z = \mathcal{O}_C$, and $h \circ g$ is a contraction.

We show that the genus of C is at most one. By Theorem 9.2, $\epsilon_{\mathrm{mov}}(-K_Z) \geq 1$. By Proposition 6.1, Z is birational to a projective surface Y with canonical singularities such that $-K_Y$ is ample, or to $E \times \mathbb{P}^1$ for some smooth projective curve E of genus one.

Suppose that Z is birational to Y. Then $(Y, 0)$ is a projective klt pair such that $-K_Y$ is nef and big, and $h \circ g$ induces a dominant rational map $Y \dashrightarrow C$. Since $(Y, 0)$ is klt, hence dlt, [HM07, Corollary 1.3(1) and Corollary 1.5(2)], applied with S a point and $\Delta = 0$, implies that Y is rationally connected [Kol96, Definition IV.3.2]. Since $Y \dashrightarrow C$ is dominant, an irreducible rational curve through two general points of Y with distinct images in C gives a non-constant rational map $\mathbb{P}^1 \dashrightarrow C$, which extends to a non-constant morphism $\mathbb{P}^1 \to C$ [Har77, Proposition I.6.8]. The Hurwitz formula [Har77, Corollary IV.2.4] implies $C \cong \mathbb{P}^1$.

Suppose that Z is birational to $E \times \mathbb{P}^1$, and let $\phi : E \times \mathbb{P}^1 \dashrightarrow C$ be the dominant rational map induced by $h \circ g$. If ϕ is non-constant on $\{e\} \times \mathbb{P}^1$ for a general point $e \in E$, then the Hurwitz formula applied to a non-constant morphism $\mathbb{P}^1 \to C$ implies $C \cong \mathbb{P}^1$. Otherwise, ϕ is constant on $\{e\} \times \mathbb{P}^1$ for a general point $e \in E$. Since ϕ is dominant, the restriction of ϕ to $E \times \{p\}$ for a general point $p \in \mathbb{P}^1$ is a non-constant rational map $E \dashrightarrow C$, which extends to a finite morphism $E \to C$ [Har77, Proposition I.6.8]. The Hurwitz formula [Har77, Corollary IV.2.4] implies that the genus of C is at most the genus of E, which is one.

Thus the contraction $h \circ g : Z \to C$ satisfies (1).

Step 4. In this step, we assume that $\dim C = 0$ and choose a very ample divisor H on Z with bounded ratio $(-K_Z \cdot H)^2/(H^2)$, which implies (2).

Since C is a point, $-K_{Z''}$ is ample. By **Step 2**, Z'' is a δ_d-lc projective surface, so the definition of W_d in **Step 1** implies

$$w := ((-K_{Z''})^2) = \mathrm{vol}(-K_{Z''}) \leq W_d.$$

Let $M := g^*(-K_{Z''})$. Then M is a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Z and $(M^2) = w$. The $\mathbb{Q}$-divisor $K_Z - g^*K_{Z''}$ is g-exceptional, so the projection formula implies $((K_Z - g^*K_{Z''}) \cdot M) = 0$. Hence

$$(-K_Z \cdot M) = (-g^*K_{Z''} \cdot M) - ((K_Z - g^*K_{Z''}) \cdot M) = (M^2) = w.$$

Recall that H_0 is the ample Cartier divisor on Z fixed in **Step 2**. For any rational number $u > 0$, the $\mathbb{Q}$-divisor $H_u := M + uH_0$ is ample [Laz04, Corollary 1.4.10], and

$$\frac{(-K_Z \cdot H_u)^2}{(H_u^2)} = \frac{(w + u(-K_Z \cdot H_0))^2}{w + 2u(M \cdot H_0) + u^2(H_0^2)}.$$

Since $w > 0$, the right-hand side tends to w as u tends to 0. Since $w \leq W_d < W_d + 1$, we may fix a rational number $u > 0$ such that

$$\frac{(-K_Z \cdot H_u)^2}{(H_u^2)} \leq W_d + 1.$$

Let m_0 be a positive integer such that $m_0 H_u$ is a Cartier divisor. By [Har77, Theorem II.7.6], applied to $\mathcal{O}_Z(m_0 H_u)$, there exists a positive integer m divisible by m_0 such that $H := mH_u$ is a very ample Cartier divisor. Then

$$\frac{(-K_Z \cdot H)^2}{(H^2)} = \frac{(-K_Z \cdot H_u)^2}{(H_u^2)} \leq W_d + 1,$$

and Proposition 12.2 implies $\mathrm{vol}(-K_X) \leq 8d(d-1)^d(W_d+1)$. The numbers u and m depend on Z and g, but the bound depends only on d. Thus $g : Z \to Z''$ satisfies (2).

Step 5. In this step, we show that the set of varieties X in the theorem is birationally bounded.

Let $\mathfrak{X}_1$ (resp. $\mathfrak{X}_2$) be the set of varieties X such that some $f : X \to Z$ satisfies the conditions of Set-up 9.1 for d and (1) (resp. (2)) holds. By **Steps 2, 3, and 4**, any such X belongs to $\mathfrak{X}_1 \cup \mathfrak{X}_2$. By Theorem 11.1, $\mathfrak{X}_1$ is birationally bounded. Any $X \in \mathfrak{X}_2$ is a normal projective variety of dimension d such that K_X is $\mathbb{Q}$-Cartier, $\mathrm{vol}(-K_X) \leq 8d(d-1)^d(W_d+1)$, and

$$\epsilon_{\mathrm{mov}}(-K_X) = d - 1 > \frac{d}{2},$$

where the inequality holds as $d \geq 3$. Thus Lemma 7.6 applied with $v = 8d(d-1)^d(W_d+1)$ implies that $\mathfrak{X}_2$ is birationally bounded. By Lemma 2.15(1), $\mathfrak{X}_1 \cup \mathfrak{X}_2$ is birationally bounded. □

13. Proof of the main theorem

In this section, we prove Theorem 1.2.

Proof of Theorem 1.2. Let $\mathfrak{X}$ be the set of normal projective varieties X of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d - 1$. We consider the following sets of projective varieties.

- $\mathfrak{Y}_1$ is the set of terminal projective varieties X'' of dimension d such that $-K_{X''}$ is ample.
- $\mathfrak{Y}_2$ is the set of terminal projective varieties Y of dimension d such that $\epsilon_{\mathrm{mov}}(-K_Y) \geq d-1$ and there exists a Mori fiber space from Y onto a smooth projective curve.
- If $d \geq 3$, then $\mathfrak{Y}_3$ is the set of varieties Y such that some $f : Y \to Z$ satisfies the conditions of Set-up 9.1 for d, with X replaced by Y. If $d = 2$, then $\mathfrak{Y}_3 := \emptyset$.

We show that any $X \in \mathfrak{X}$ is birational to a member of $\mathfrak{Y}_1 \cup \mathfrak{Y}_2 \cup \mathfrak{Y}_3$.

Let $X \in \mathfrak{X}$. By Construction-Theorem 3.1, there exist a terminal modification $X' \to X$, a composition $X' \dashrightarrow Y$ of finitely many steps of a $K_{X'}$-MMP, and a Mori fiber space $f : Y \to Z$ such that Y is terminal and Construction-Theorem 3.1(1)–(3) hold. Since $X' \to X$ and $X' \dashrightarrow Y$ are birational maps, X is birational to Y. In particular, $\dim Y = d$. Since X' is projective and any step of an MMP preserves projectivity, Y is projective, and Z is a normal projective variety as the base of a Mori fiber space.

Case 1. $\epsilon_{\mathrm{mov}}(-K_Y) > d - 1$ or $\dim Z = 0$. By Construction-Theorem 3.1(2), X is birational to a member of $\mathfrak{Y}_1$.

Suppose that we are not in Case 1. By Construction-Theorem 3.1(1),

$$\epsilon_{\mathrm{mov}}(-K_Y) \geq \epsilon_{\mathrm{mov}}(-K_X) \geq d - 1.$$

Thus $\epsilon_{\mathrm{mov}}(-K_Y) = d - 1$ and $\dim Z > 0$. By Construction-Theorem 3.1(3)(i), $\dim Z \in \{1, 2\}$. Hence we are in one of the following two cases.

Case 2. $\epsilon_{\mathrm{mov}}(-K_Y) = d-1$ and $\dim Z = 1$. Since Z is a normal projective curve, Z is a smooth projective curve. Since Y is a terminal projective variety of dimension d and $f : Y \to Z$ is a Mori fiber space, $Y \in \mathfrak{Y}_2$. Thus X is birational to a member of $\mathfrak{Y}_2$.

Case 3. $\epsilon_{\mathrm{mov}}(-K_Y) = d-1$ and $\dim Z = 2$. By Construction-Theorem 3.1(3)(ii), $d \geq 3$ and the geometric generic fiber of f is isomorphic to $\mathbb{P}^{d-2}$. Since Y is terminal of dimension d and f is a Mori fiber space onto the normal projective surface Z, the conditions (1)–(4) of Set-up 9.1 hold for $f : Y \to Z$, with X replaced by Y. Thus $Y \in \mathfrak{Y}_3$, and X is birational to a member of $\mathfrak{Y}_3$.

The three cases above are exhaustive. It remains to show that $\mathfrak{Y}_1$, $\mathfrak{Y}_2$, and $\mathfrak{Y}_3$ are birationally bounded. Since an ample divisor is nef and big, $\mathfrak{Y}_1$ is bounded by Theorem 2.16, hence birationally bounded by Lemma 2.15(1). The set $\mathfrak{Y}_2$ is birationally bounded by Theorem 5.1. If $d \geq 3$, then $\mathfrak{Y}_3$ is birationally bounded by Theorem 12.1. By Lemma 2.15(1), $\mathfrak{Y}_1 \cup \mathfrak{Y}_2 \cup \mathfrak{Y}_3$ is birationally bounded. Since any member of $\mathfrak{X}$ is birational to a member of $\mathfrak{Y}_1 \cup \mathfrak{Y}_2 \cup \mathfrak{Y}_3$, $\mathfrak{X}$ is birationally bounded by Lemma 2.15(1). □

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